

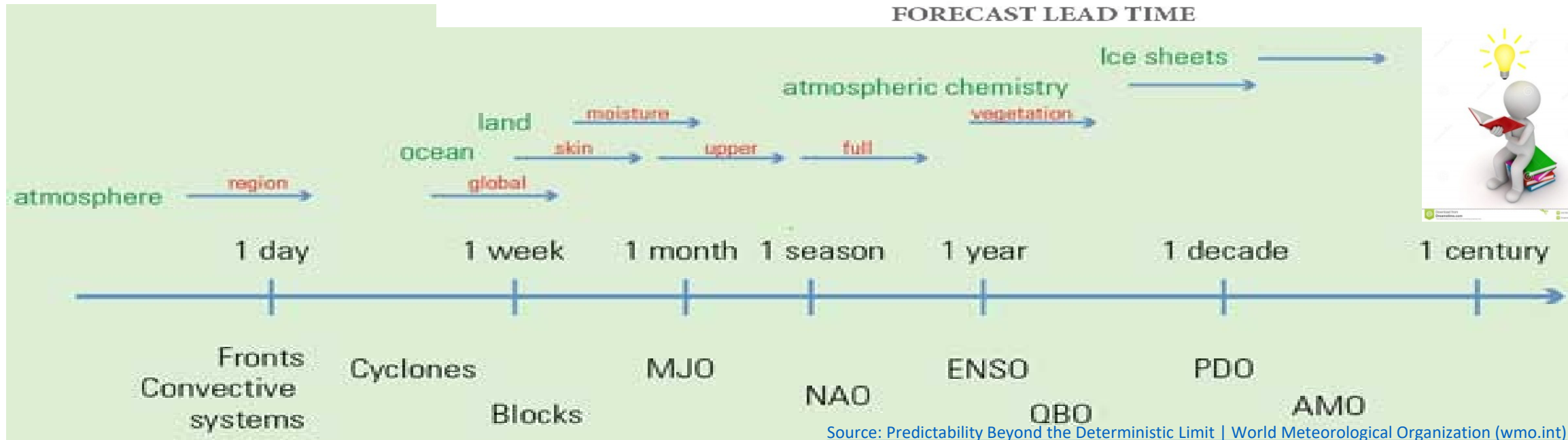
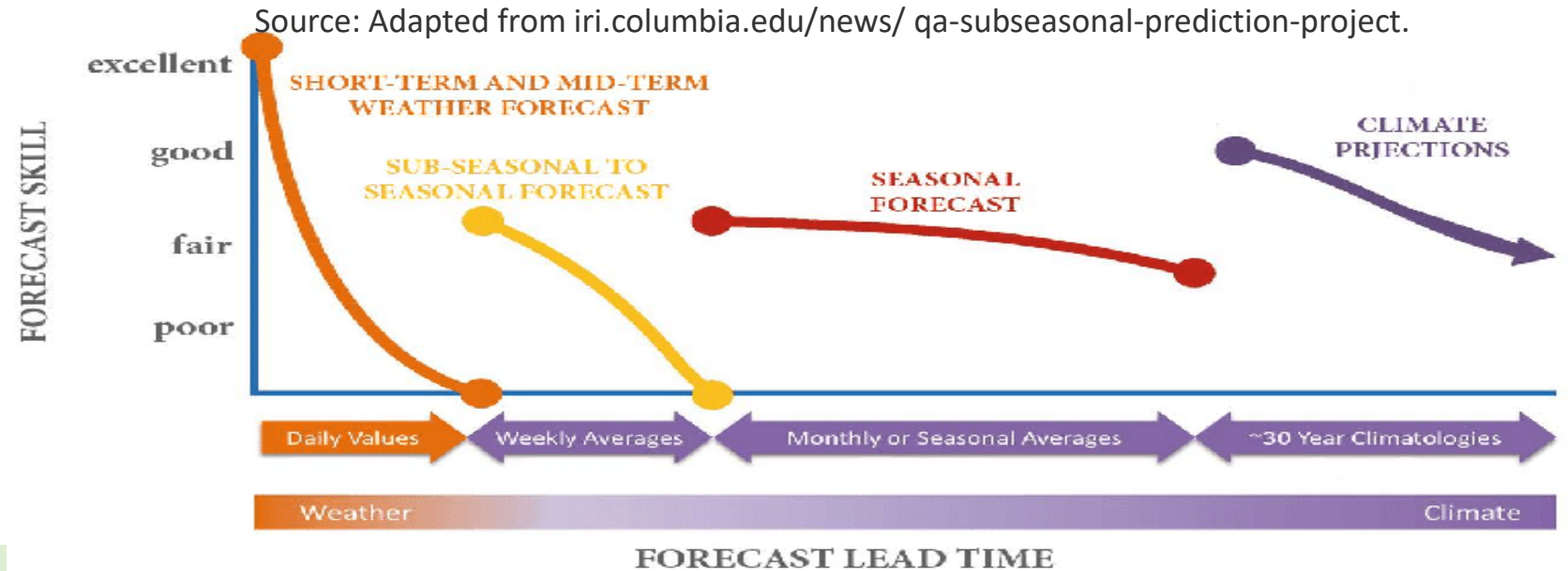
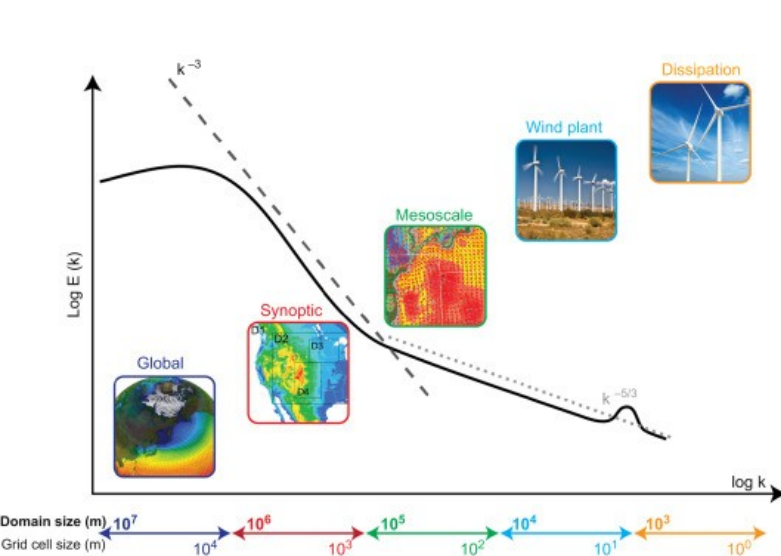


Características dos Modelos MPAS e FV3 Importantes Para as Demandas do MONAN

Instituto Nacional de Pesquisas Espaciais
Cachoeira paulista

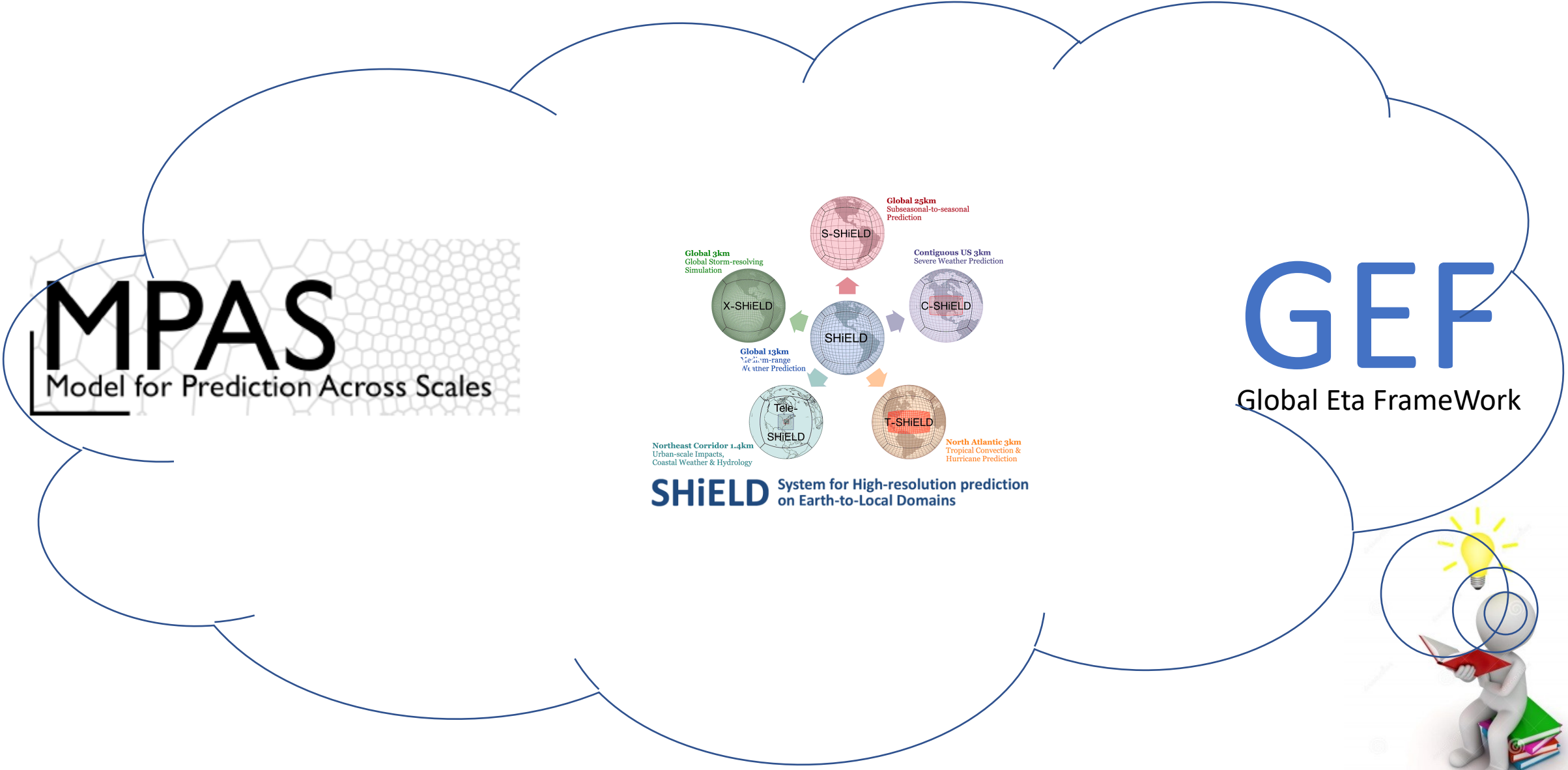


As Demandas do Modelo MONAN





Considerações Técnicas Sobre os Núcleos Dinâmicos que podem ser adotados pelo Modelos MONAN





Considerações Técnicas dos Núcleos Dinâmico dos Modelos



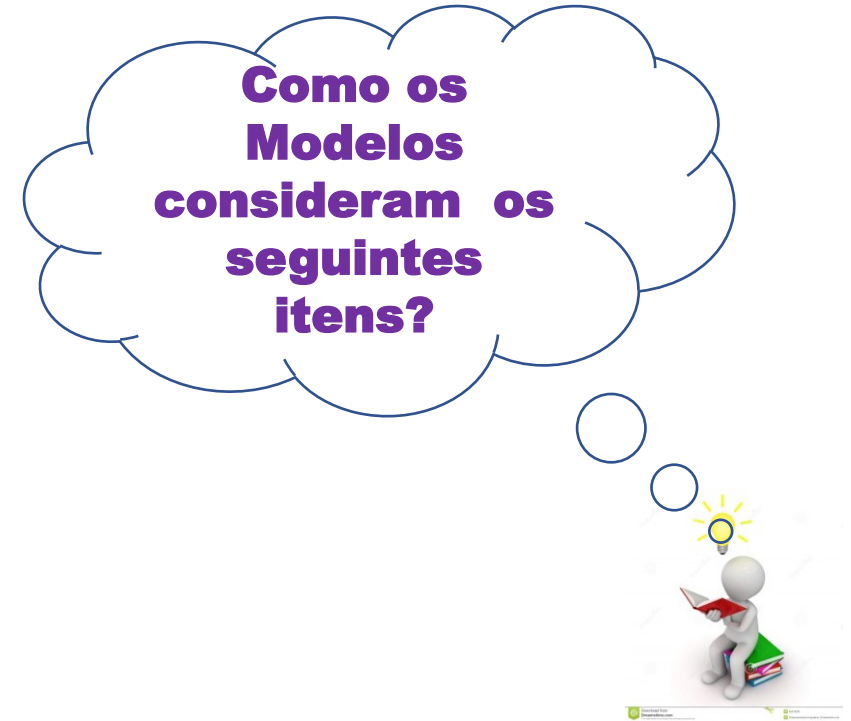
Formulação /Setup	MPAS	FV3-Shied	GEF
Dinâmica	Euleriana	Euleriana/ Lagrangian	Euleriana
Compressibilidade	compressible	compressible	compressible
Aproximação Vertical	nonhydrostatic	nonhydrostatic	hydrostatic
Grade	malhas de Voronoi não estruturadas (Grade-C)	cubed-sphere grid, in a version with uniform Jacobians (UJ) (Grade-C/ Grade-D)	cubed-sphere grid, in a version with uniform Jacobians (UJ) (Grade-B)
Coordenada vertical	terrain following coordinate) and a hybrid coordinate	Coordenada Lagrangiana	Coordenada Eta
Time Scheme	split-explicit time integration scheme employs a 3rd-order Runge-Kutta method	Forward e Backward dependendo da variável prognostica	Similar Ao modelo Eta Regional
Transporte de escalares	Volume finito polinomio de 3rd and 4th-order	Esquema Semi-Lagrangiano	Esquema Semi-Lagrangiano



Características Gerais que Atendem as Demandas do MONAN:



- 1.1 Grade com refinamento variável.
- 1.2 Regionalização com fronteira aberta.
- 1.3 Dinâmica não-hidrostática.
- 1.4 Transporte de escalares com manutenção de positividade e monotonicidade.
- 1.5 Suítes físicas avançadas, etc.





Uma visão geral da estrutura das malhas (grades)



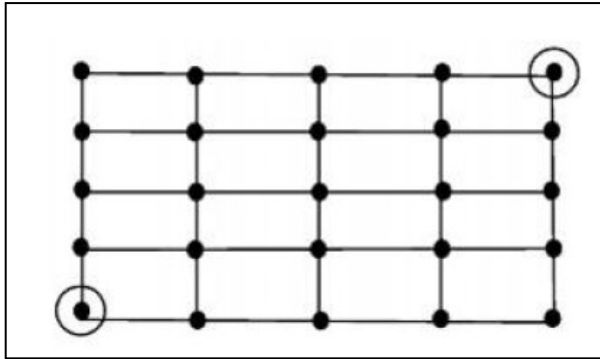
Uma visão geral da estrutura das malhas

Tipo de Malha Numérica



ESTRUTURADA

(consistes de retângulos 2D)

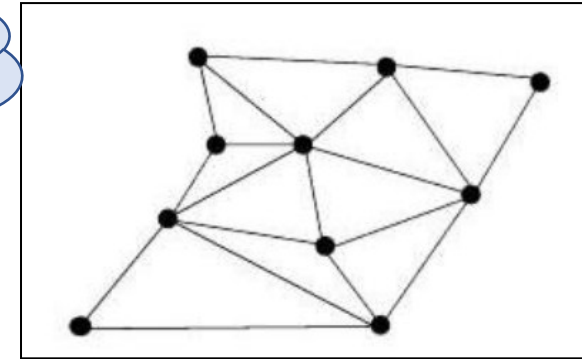


Como
Discretizar?



NÃO-ESTRUTURADA

(consistes de polígonos não homogêneos 2D)



Vantagens:

- A indexação sozinha, mostra uma conectividade limpa,
- Fácil de manipular os índices e armazenar na computação

Desvantagens:

- Restrições à ortogonalidade e razão de aspecto.
- Difícil envolver geometrias complexas.
- Menos eficiente para malhas localmente refinadas.

Vantagens:

- Informações de conectividade para cada célula precisam ser armazenadas
- Difícil de armazenar e manipular dados na computação

Desvantagens:

- Mais fácil envolver forma complexas.
- Muito eficiente para malhas localmente refinadas



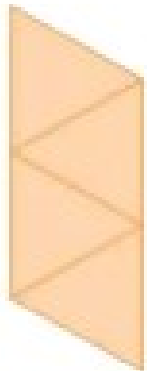
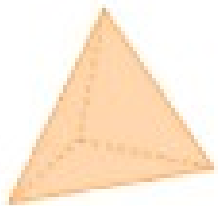
Uma visão geral da estrutura das malhas

Tipo de Malha Numérica

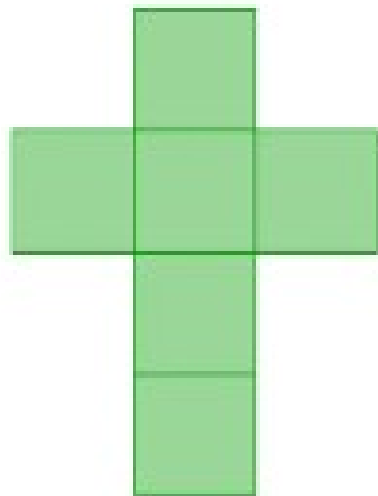
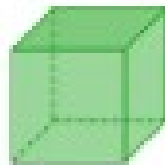


Planificações dos sólidos platônicos

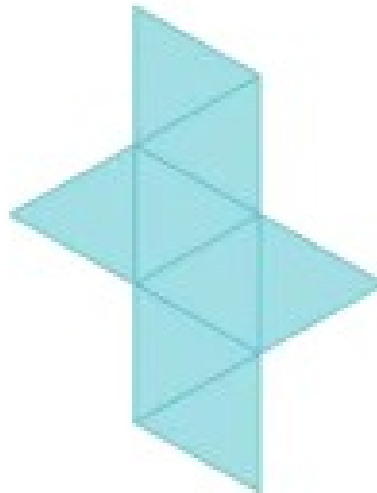
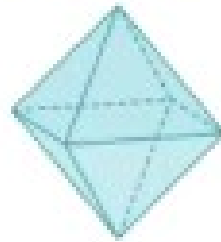
Tetraedro



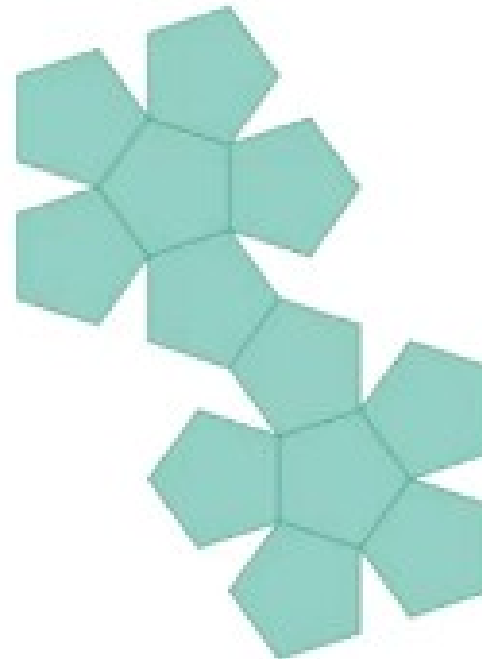
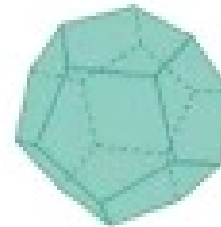
Cubo



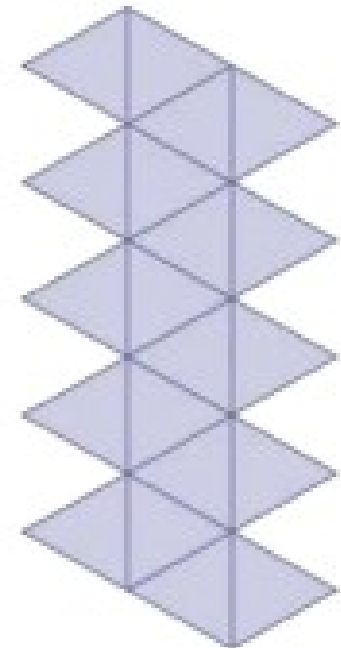
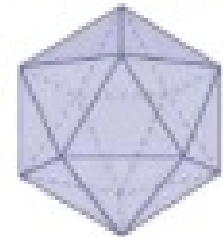
Octaedro



Dodecaedro



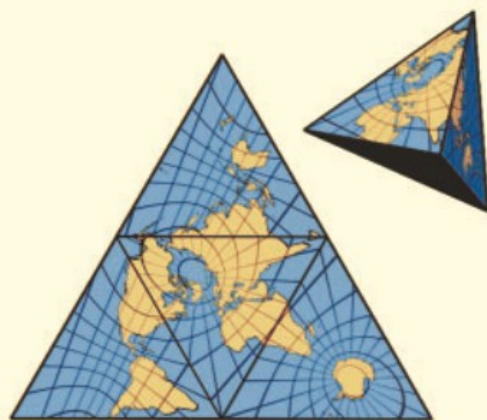
Icosaedro



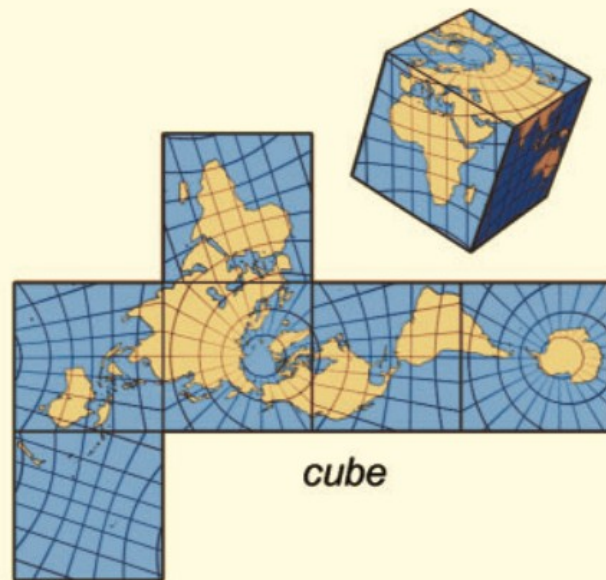


Uma visão geral da estrutura das malhas

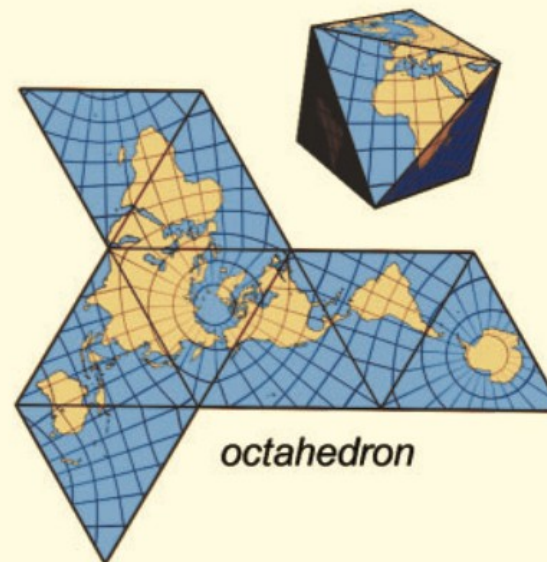
Tipo de Malha Numérica



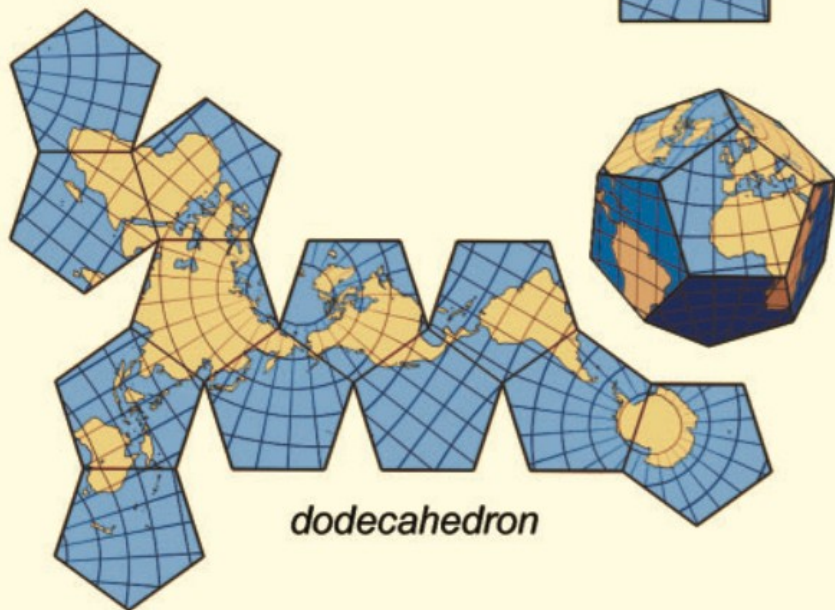
tetrahedron



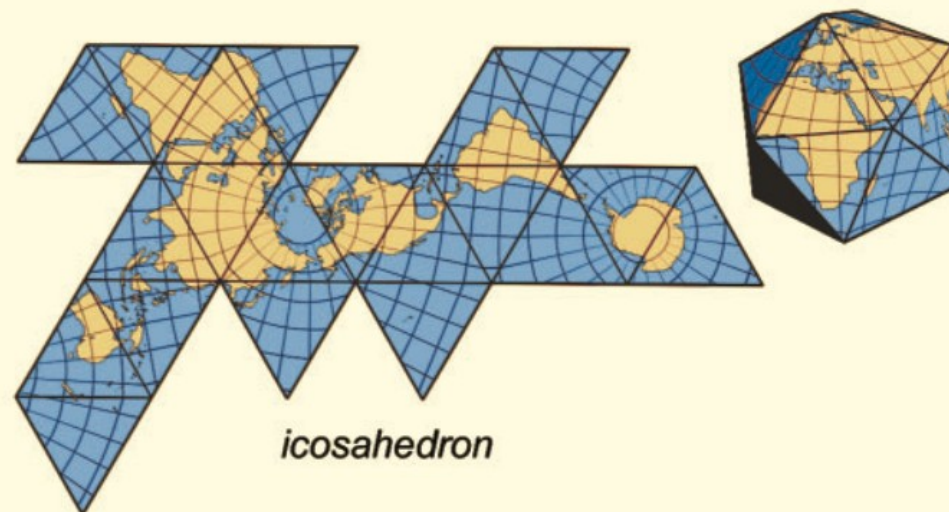
cube



octahedron



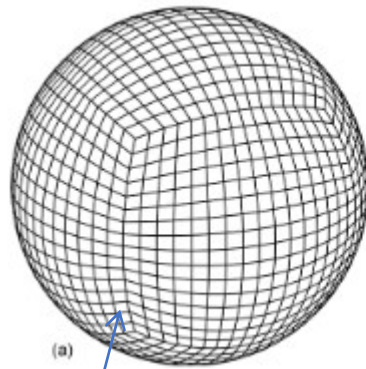
dodecahedron



icosahedron

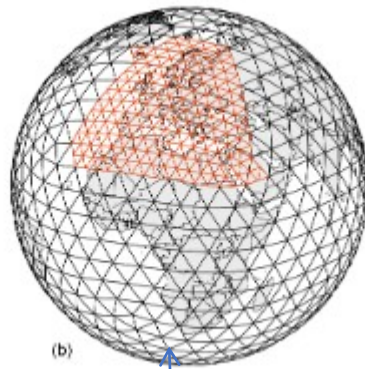


Uma visão geral da estrutura das malhas MPAS



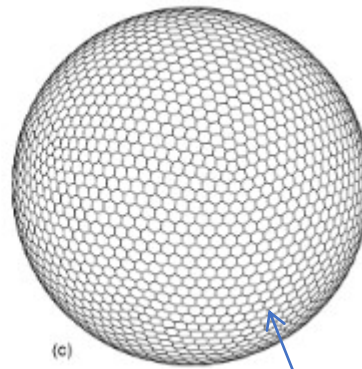
(a)

a) Esfera cubada



(b)

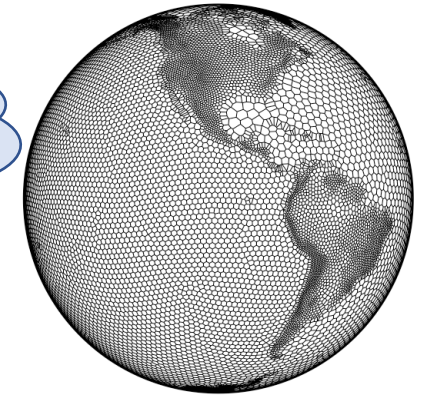
b) Icosaédrica triangular



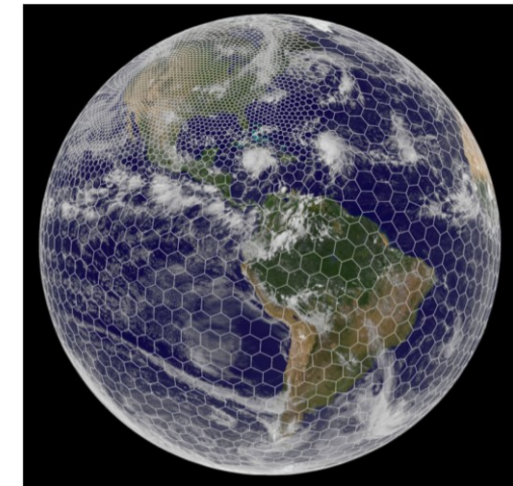
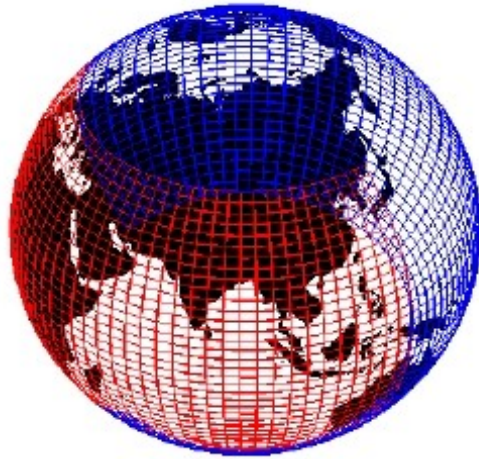
(c)

c) Icosaédrica hexagonal

Qual é a
Melhor Malha?



d) Tesselação
centroidal de
Voronoi (CVTs)





Considerações para a escolha das Malhas

Onda Espúrias (Aplica-se filtragem (Difusão))

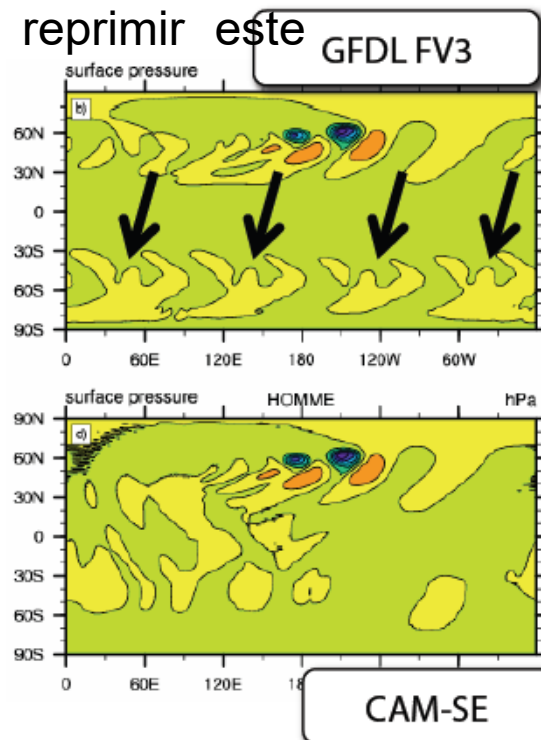
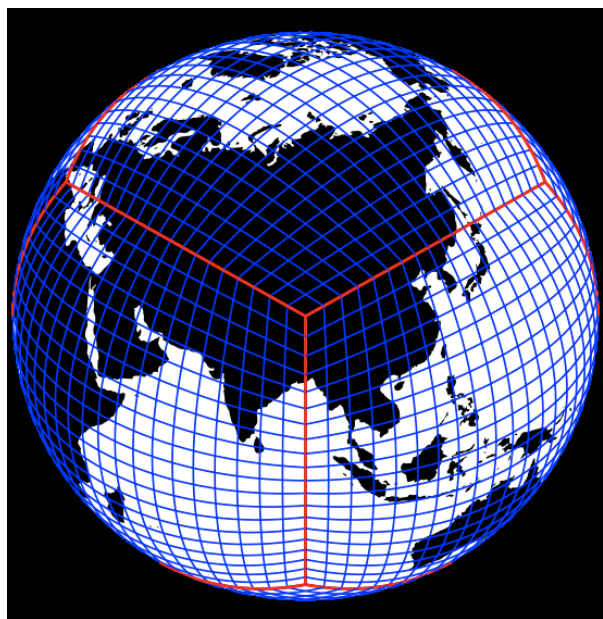


Métodos Numéricos: Problemas

Tanto o modelo **GFDL FV3** (FVcubed) como o modelo **CAM-SE** (elemento espectral) são construídos na esfera em cubos. Isso leva a um **realce** do modo de onda **$k = 4$** .



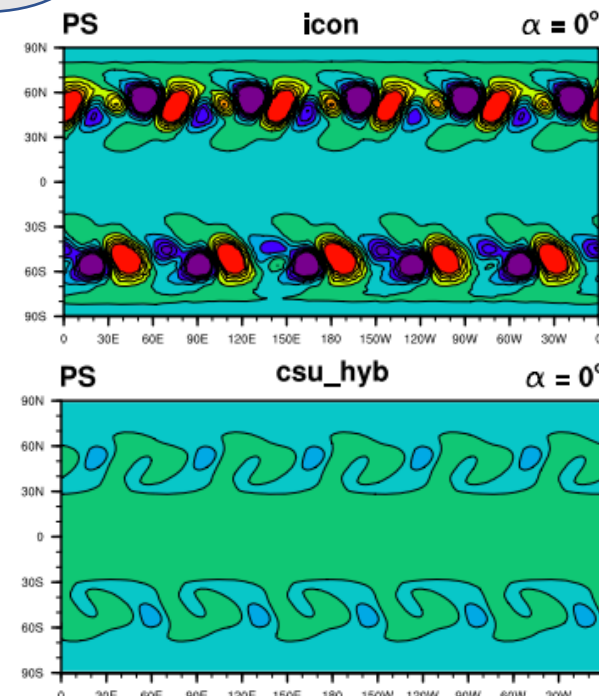
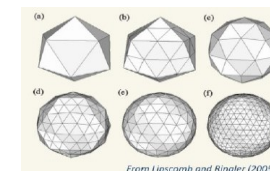
O uso de alta ordem de difusão numérica no CAM-SE é mais eficaz para reprimir este modo.



Ambos os modelos **MPAS**, **OLAM**, **ICON** e **CSU** são construídos em uma Malha icosaédrica (resultados da oficina de 2008). Isso leva a um **realce** do modo de onda **$k = 5$** .



Difusão Numérica
pode filtrar ondas
reais





1.1 Malhas do modelos adaptadas para todas as escalas

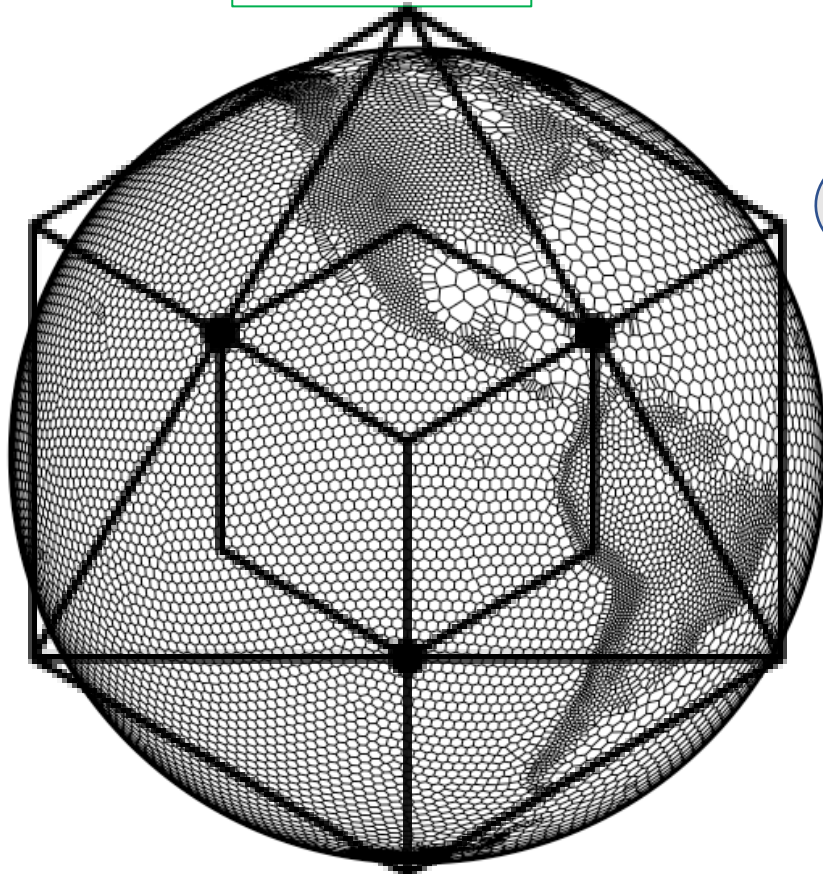
Malha
não estruturada

MPAS



Malha estruturada

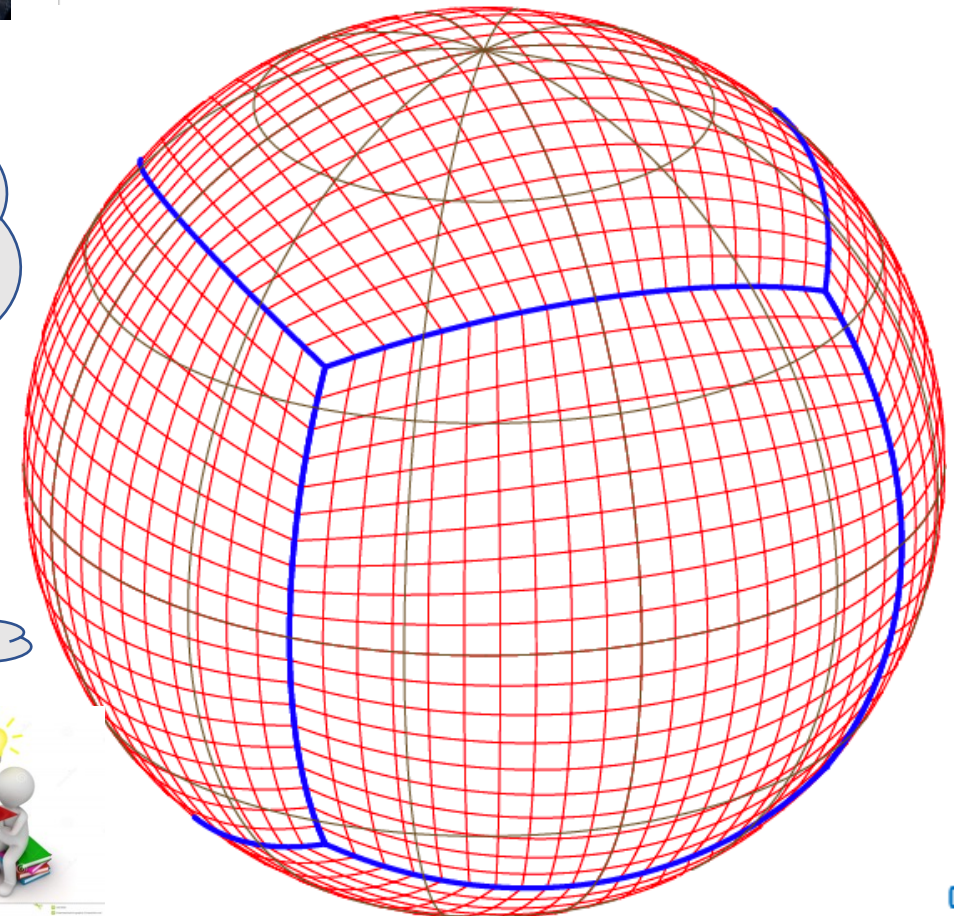
FV3 e GEF



Solução-
Tesselação
centroidal de
Voronoi (CVTs)



Porque?

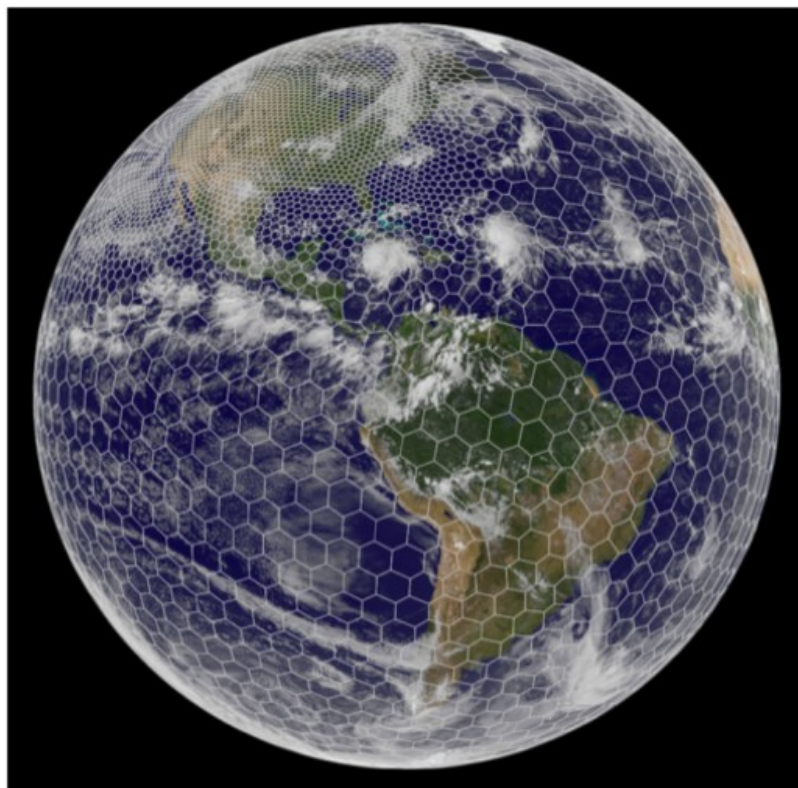




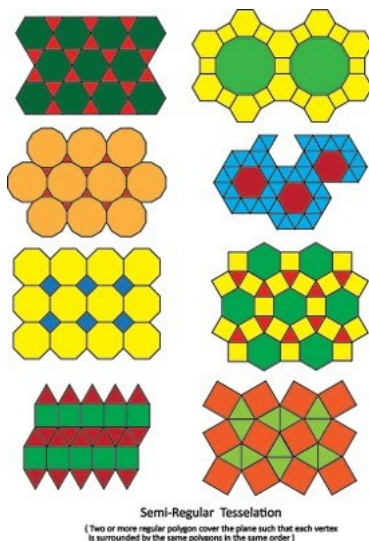
Solução (Estrutura das Malhas MPAS)



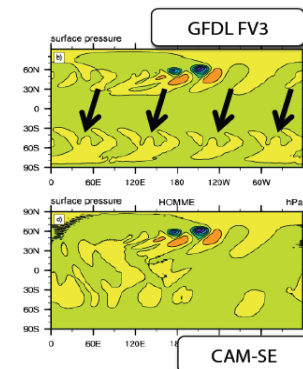
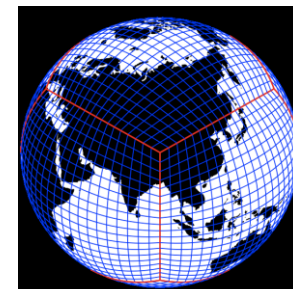
Uma característica definidora dos modelos MPAS é o uso de **Tesselação** centroidal de Voronoi (CVTs) com um deslocamento de grade C



– Quando se restringe a ficar na superfície de uma esfera, geralmente os chamamos de **Tesselação** centroidal de Voronoi na esfera (SCVTs)



Tesselação é o recobrimento de uma superfície bidimensional, tendo, como unidades básicas, polígonos congruentes (mesmo tamanho e a mesma forma) ou não, sem que existam espaços entre eles e de modo que a superfície total seja igual ao espaço particionado.



a) Adapta melhor a geometria esférica.

b) Pode ajudar a reduzir ondas espúrias devido ao formato da grade

c) Menos difusão Numérica



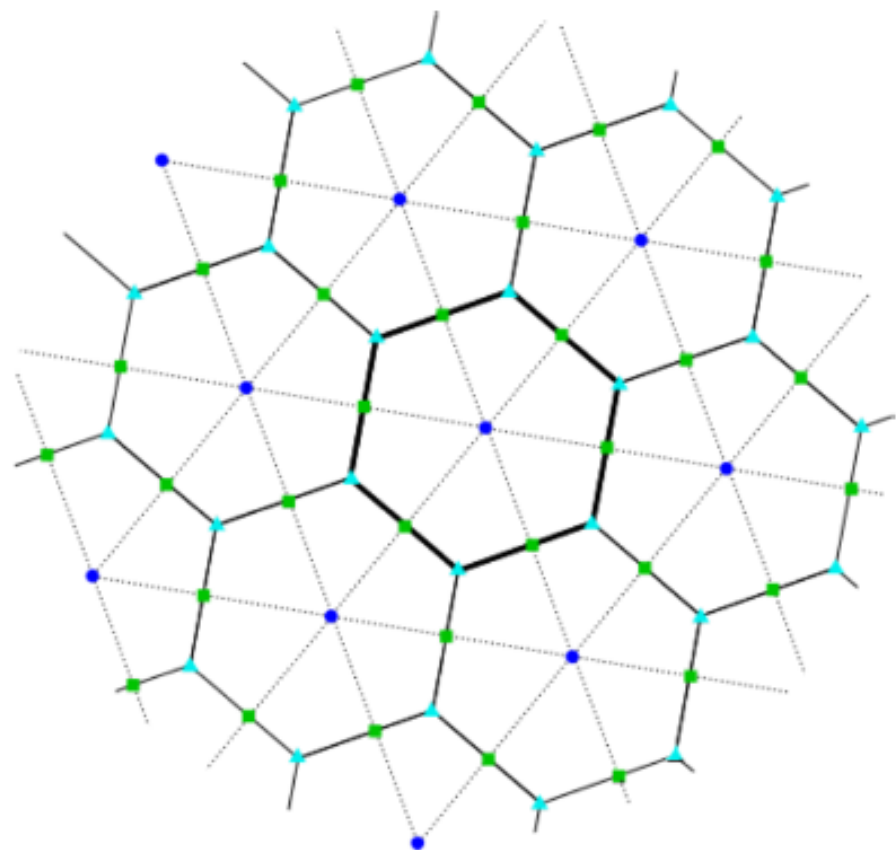


Uma visão geral da estrutura das malhas MPAS



Como o MPAS e você acompanham essa malha Voronoi não estruturada?

Esquemas para encontrar implicitamente os índices/identidades (os “IDs”) de elementos vizinhos da malha (ou seja, células, arestas, vértices) estão fadados ao fracasso...



Três tipos de elementos de malha são rastreados na representação de malha:

- Localização das células (círculos azuis) - os pontos geradores da malha de Voronoi
- Locais de vértices (triângulos ciano) - os cantos das células da malha primária
- Locais de borda (quadrados verdes) - os pontos onde as bordas da malha dupla que cruzam as bordas da malha primária

**Dificuldade no
Mapeamento da Malha**





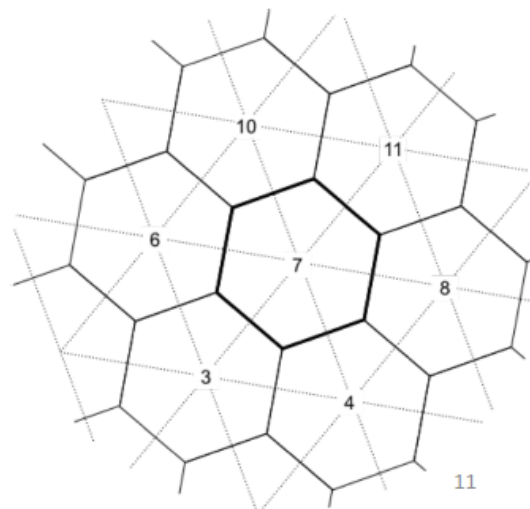
Uma visão geral da estrutura das malhas MPAS



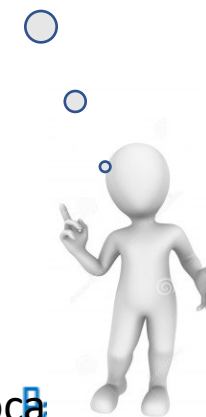
Os campos de conectividade explícita descrevem a estrutura de uma malha

- **nEdgesOnCell** (**nCells**) – o número de vizinhos para cada célula
- **cellsOnCell** (**maxEdges**, **nCells**) – os índices de células vizinhas para cada célula
- **edgeOnCell** (**maxEdges**, **nCells**) – os índices de bordas delimitadoras para cada célula
- **verticesOnCell** (**maxEdges**, **nCells**) – os índices de vértices de canto para cada célula
- **edgeOnVertex** (**vertexDegree**, **nVertices**) – os índices de arestas incidentes com cada vértice
- **verticesOnEdge** (**2**, **nEdges**) – os índices de vértices de extremidade para cada aresta
- **cellsOnVertex** (**vertexDegree**, **nVertices**) – os índices de células que se encontram em cada vértice
- **cellsOnEdge** (**2**, **nEdges**) – os índices de células separadas por cada aresta

```
nEdgesOnCell(7)=6  cellsOnCell(1,7)=8
                    cellsOnCell(2,7)=11
                    cellsOnCell(3,7)=10
                    cellsOnCell(4,7)=6
                    cellsOnCell(5,7)=3
                    cellsOnCell(6,7)=4
```



Mapeamento
da Malha



Na inicialização do modelo, todos os índices nessas matrizes são renumerados para um esquema de indexação local



Uma visão geral da estrutura das malhas MPAS



Um exemplo prático de campos de malha em uso

Um exemplo de uso de campos de malha: Calculando a média de um campo baseado em vértice, `vorticity(nVertLevels, nVertices)`, para células como `vortcell(nVertLevels, nCells)`:

`kiteAreasOnVertex(vertexDegree, nVertices)`

`cellsOnVertex(vertexDegree, nVertices)`

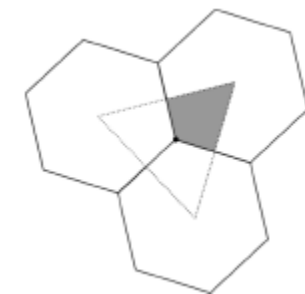
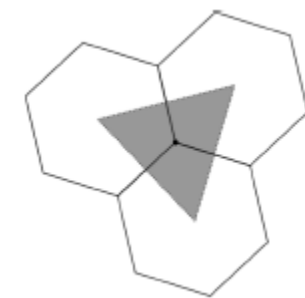
```
vortcell(:, :) = 0.0
do iVtx = 1, nVertices
  do j = 1, vertexDegree
    iCell = cellsOnVertex(j, iVtx)
    vortcell(:, iCell) = vortcell(:, iCell) + kiteAreasOnVertex(j, iVtx) * &
      vorticity(:, iVtx)
```

`end do`

`end do`

```
do iCell = 1, nCells
  vortcell(:, iCell) = vortcell(:, iCell) / areaCell(iCell)
end do
```

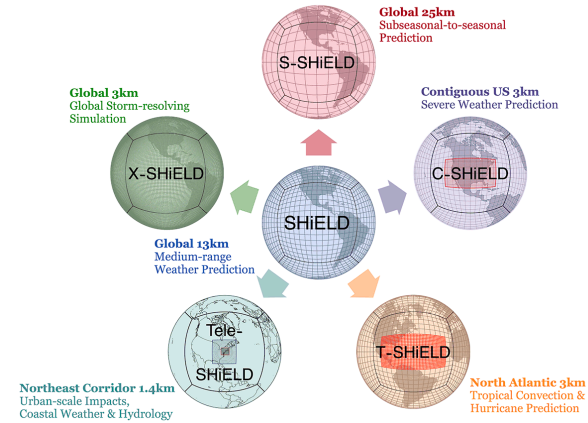
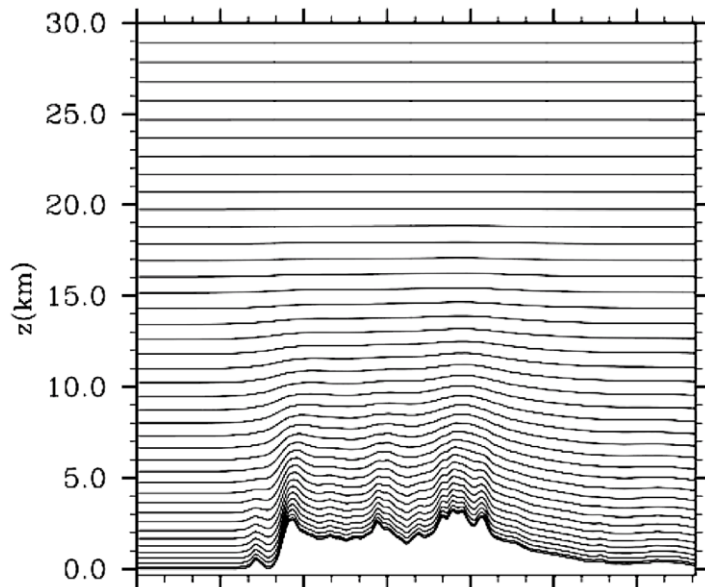
Operação com os índices
do mapeamento





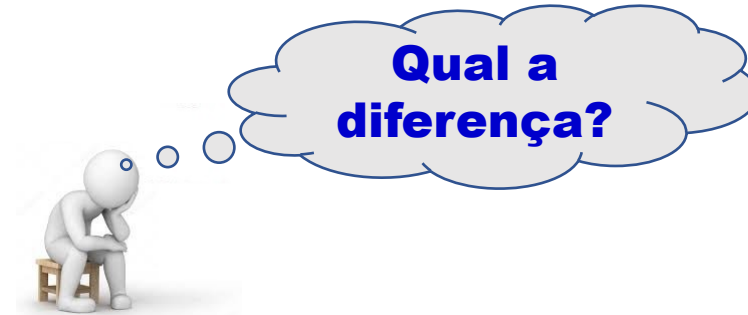
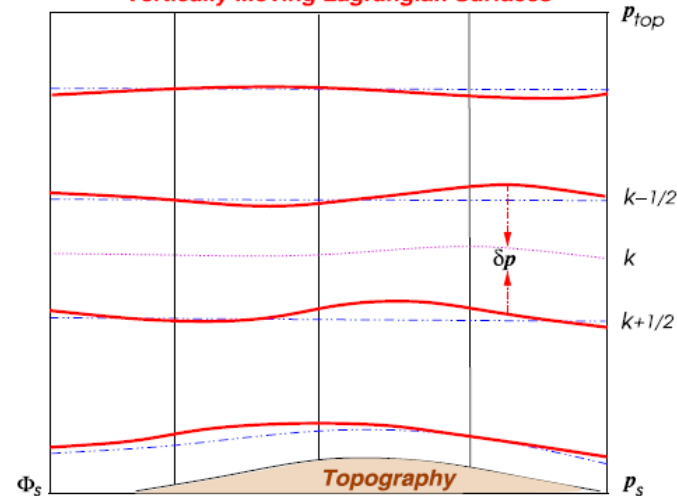
1.1 Malha Vertical

Grade Vertical



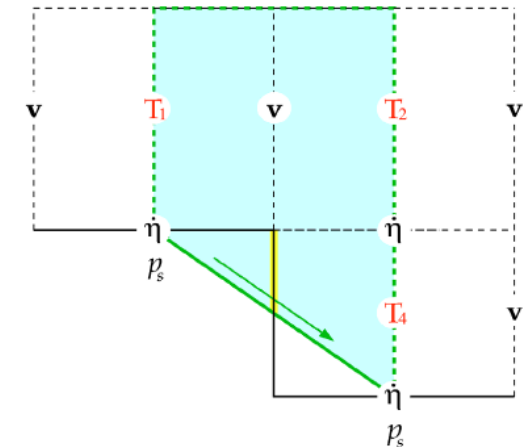
SHIELD System for High-resolution prediction on Earth-to-Local Domains

Vertically Moving Lagrangian Surfaces



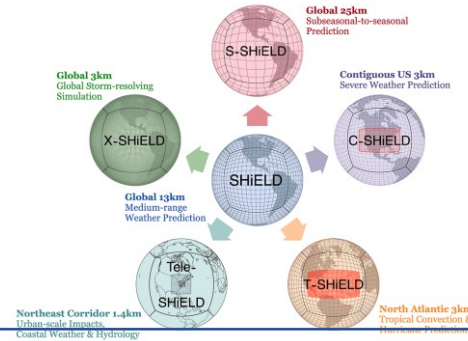
GEF

Global Eta FrameWork





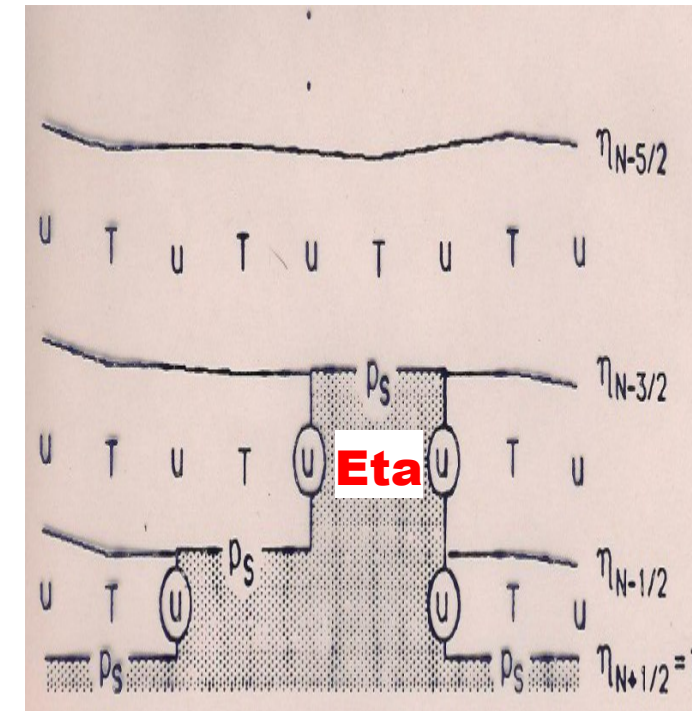
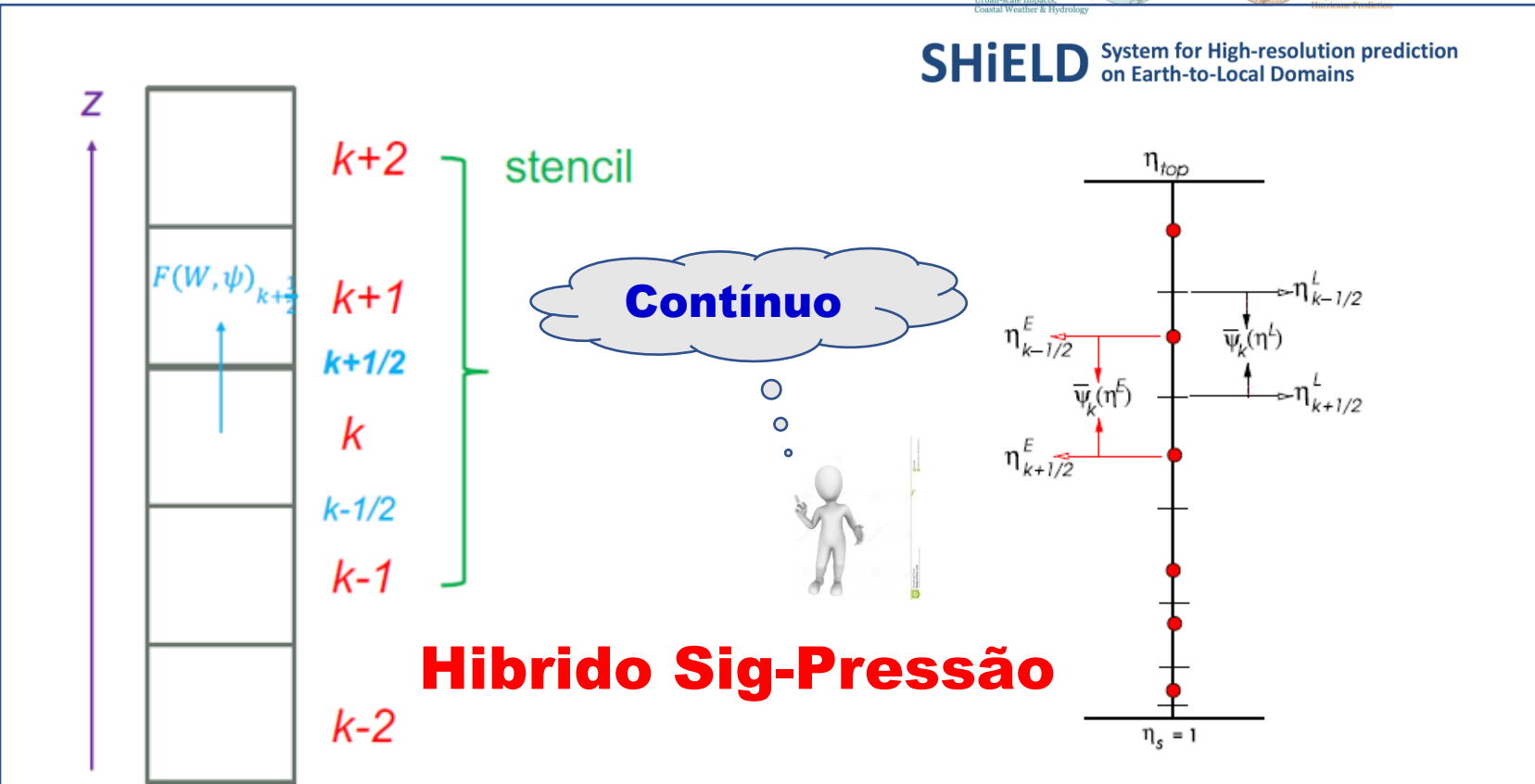
Uma visão geral da estrutura das malhas



Deslocamento das malhas das variáveis de momentum em relação às variáveis de massa

GEF

Global Eta FrameWork



Descontínuo

Paulo Yoshio

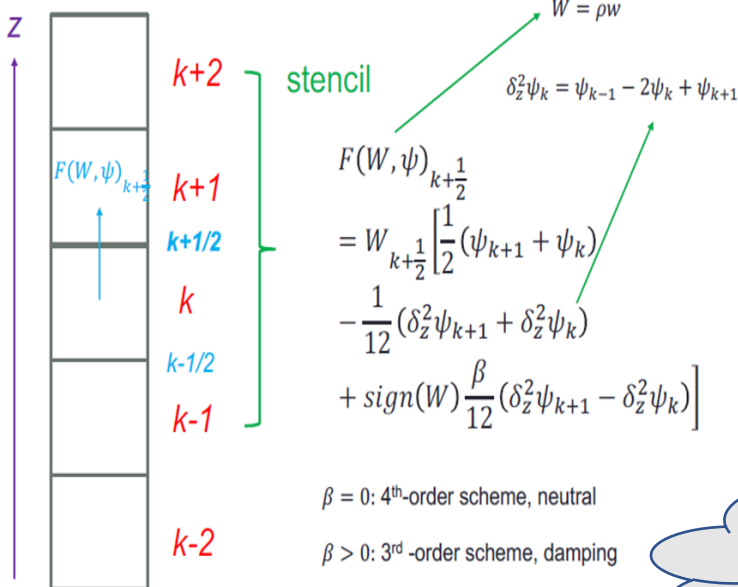


Uma visão geral da estrutura das malhas



3 (Volume Finito) Piecewise de 3 ou 4 ordem para a advection vertical

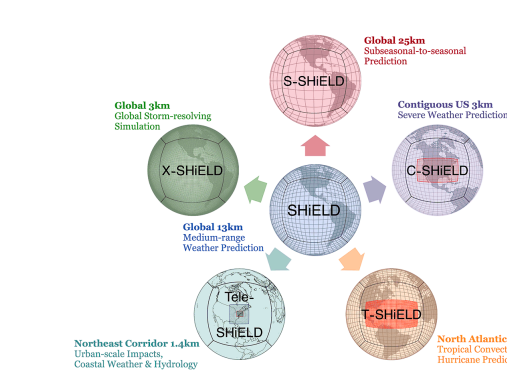
Vertical Discretization



$$\delta_z^2 \psi_k = \psi_{k-1} - 2\psi_k + \psi_{k+1}$$

$$F(W, \psi)_{k+1/2} = W_{k+1/2} \left[\frac{1}{2} (\psi_{k+1} + \psi_k) - \frac{1}{12} (\delta_z^2 \psi_{k+1} + \delta_z^2 \psi_k) + \text{sign}(W) \frac{\beta}{12} (\delta_z^2 \psi_{k+1} - \delta_z^2 \psi_k) \right]$$

$\beta = 0$: 4th-order scheme, neutral
 $\beta > 0$: 3rd-order scheme, damping



SHIELD System for High-resolution prediction on Earth-to-Local Domains

3 (Semi-Lagrangiano) Interpolação conservativa

Remapeamento nas coordenadas eulerianas de referência usando o método semi-lagrangiano conservativo de integração de célula 1-D (CISL) de Nair e Machenhauer [20].

Cuidados com a conservação de Massa



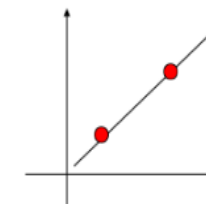
GEF

Global Eta FrameWork

3 (Volume Finito) Piecewise linear para a advection vertical

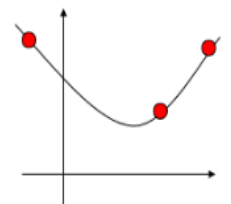
polinômio de ordem N=1

Linear Interpolation



Given any two points, there is one polynomial of order ≤ 1 that passes through the two points.

Quadratic Interpolation



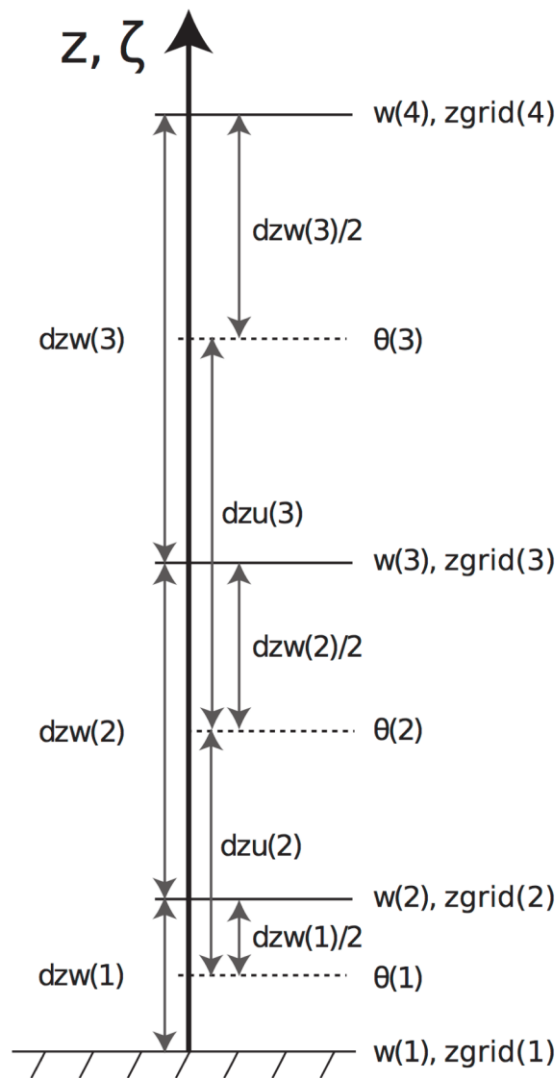
Given any three points there is one polynomial of order ≤ 2 that passes through the three points.



Uma visão geral da estrutura das malhas MPAS



Vertical grid



A grade vertical MPAS-Atmosphere também é deslocada:

- velocidades verticais em níveis w
- todos os outros campos em níveis θ

$zgrid$ fornece a altura geométrica nos níveis w

Os níveis θ estão nos pontos médios dos níveis w de colchetes

Para interpolar verticalmente o campo F dos níveis θ para os níveis w :

$$fzp(k) = 0.5 * dzw(k) / dzu(k)$$

$$fzm(k) = 0.5 * dzw(k-1) / dzu(k)$$

$$F_w(k) = fzm(k) * F_{\theta}(k) + fzp(k) * F_{\theta}(k-1)$$

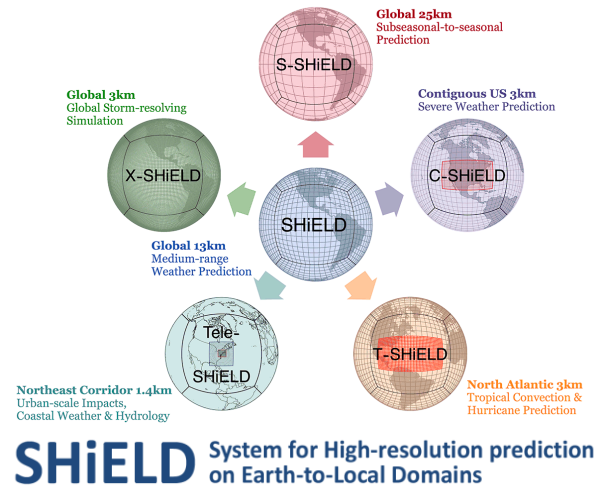
**Cuidados ao
trabalhar com os
índices**





1.1 Grade com Refinamento Variável

Grade com Refinamento Variável

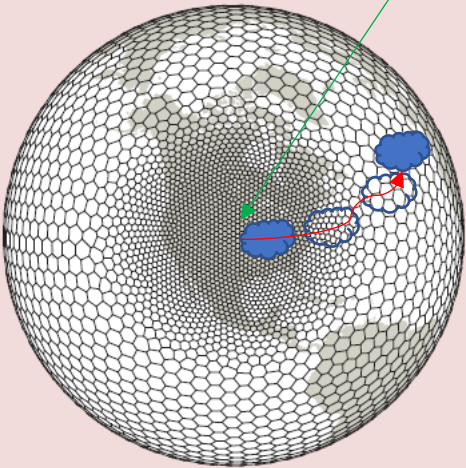


GEF
Global Eta FrameWork



1.1 Grade com Refinamento Variável

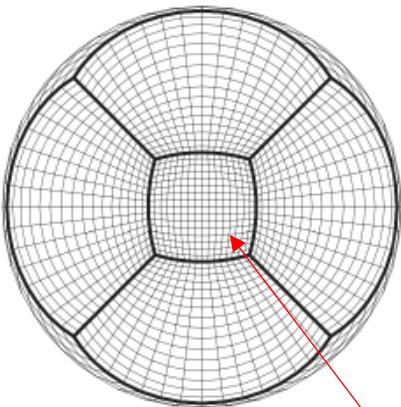
MPAS tem maior flexibilidade no refinamento de malha



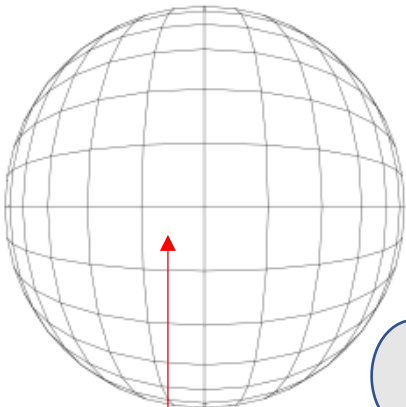
MPAS
Smooth grid refinement on a conformal mesh

- Increased accuracy and flexibility for variable resolution applications
- No abrupt mesh transitions.

FV3 Mesh Refinement



3x stretched-grid refinement from uniform

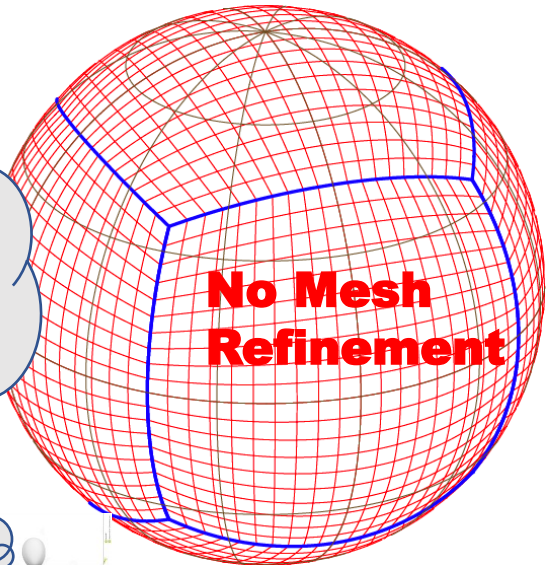


Opposite side of stretched-grid

Escolha pelo Refinamento com menor distorção

GEF

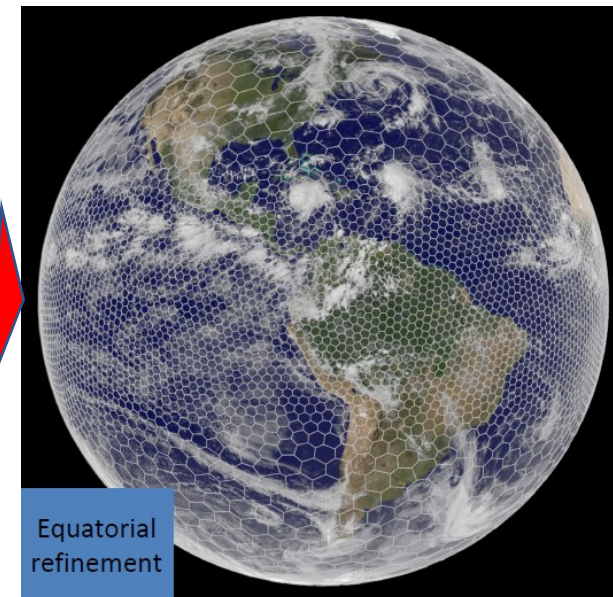
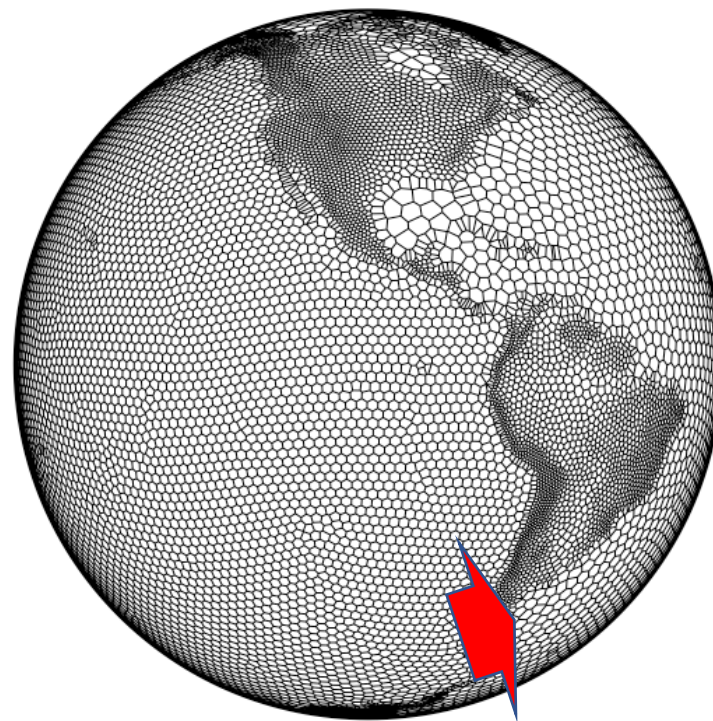
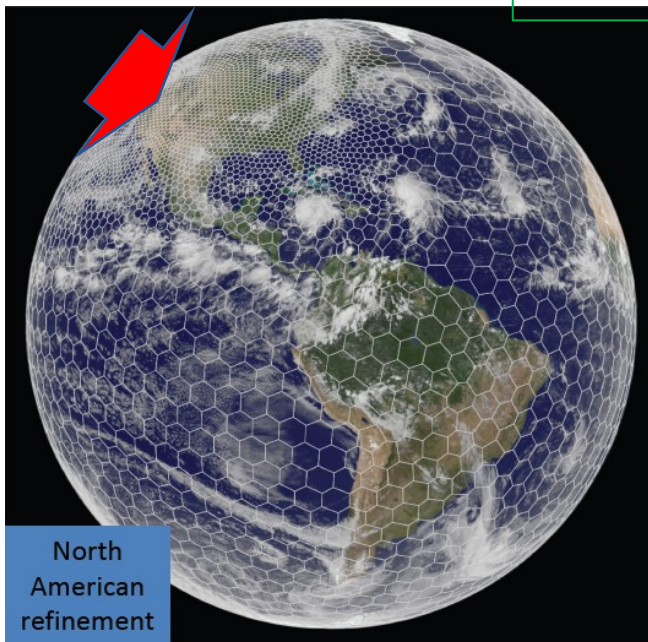
Grandes distorções podem não resolver alguns fenômenos



FV3 Resolução muito pobre no lado oposto da área com refinamento de malha

1.1 Grade com Refinamento Variável

MPAS
tem maior flexibilidade no refinamento de malha



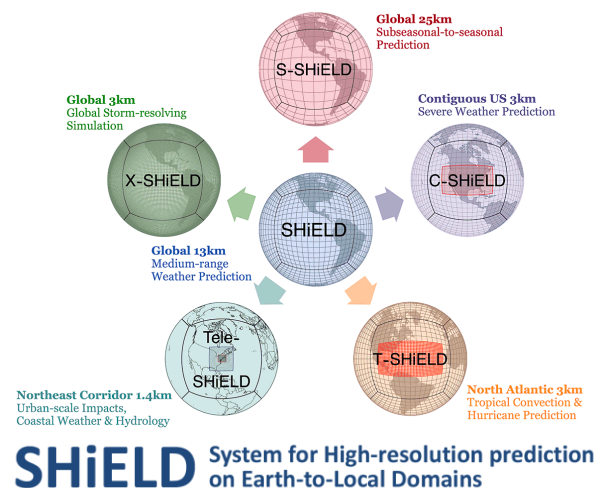
**Esta Malha
de Voronoi
Menor
Distorção**

**Esta Malha
de Voronoi
pode ser
considerada
a ideal para
o MONAN**



1.2 Regionalização com fronteira aberta

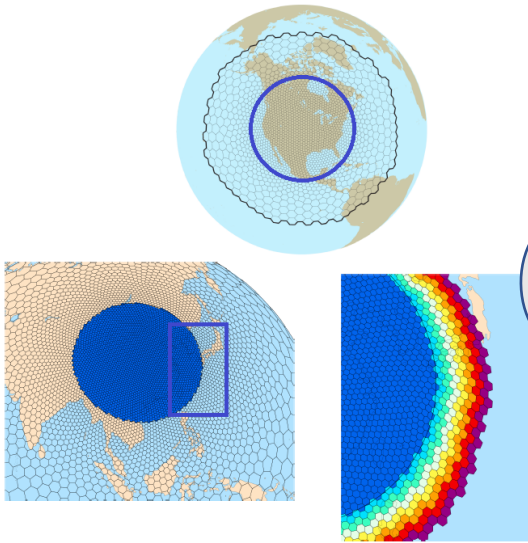
Regionalização Com Fronteira Aberta



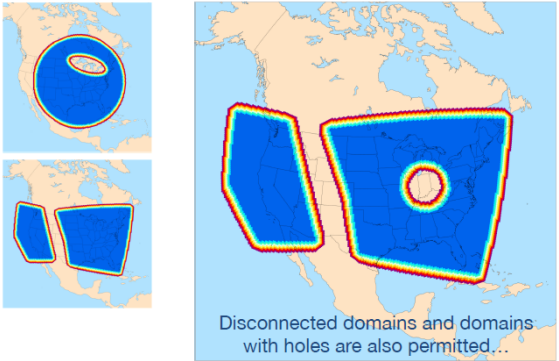
GEF
Global Eta FrameWork



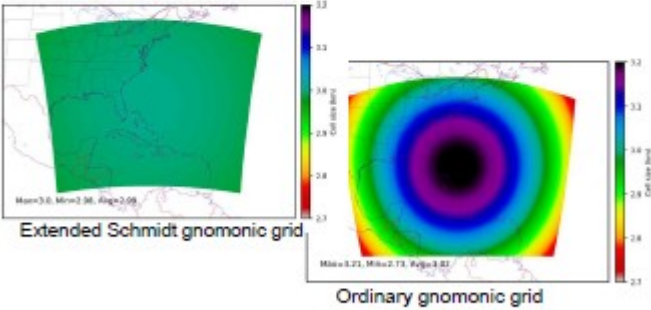
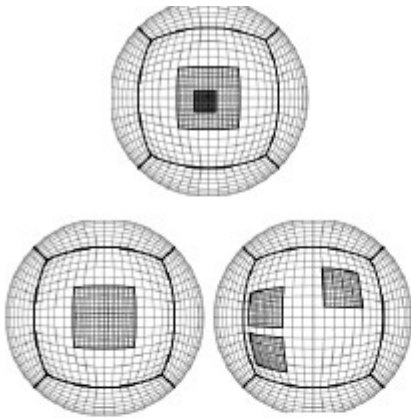
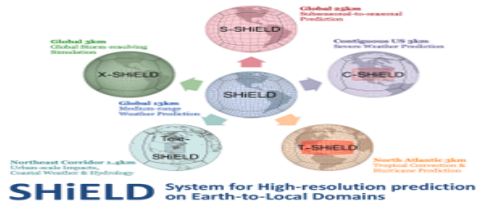
1.2 Regionalização com fronteira aberta



Flexibilidade
e em
configurar
Domínios
de fronteira
aberta



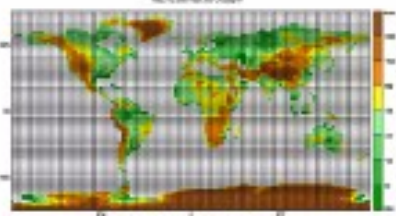
Disconnected domains and domains with holes are also permitted...



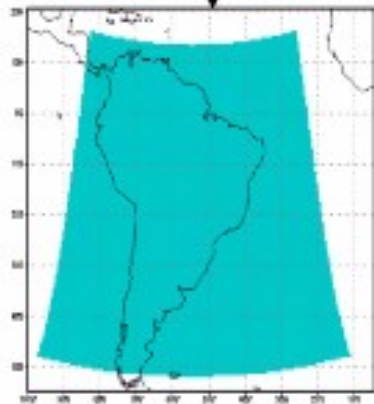
Regional-Domain Grid-cell Width
Courtesy Jim Purser and
Chan-Hoo Jeon (NCEP/EMC)



Modelo Eta global



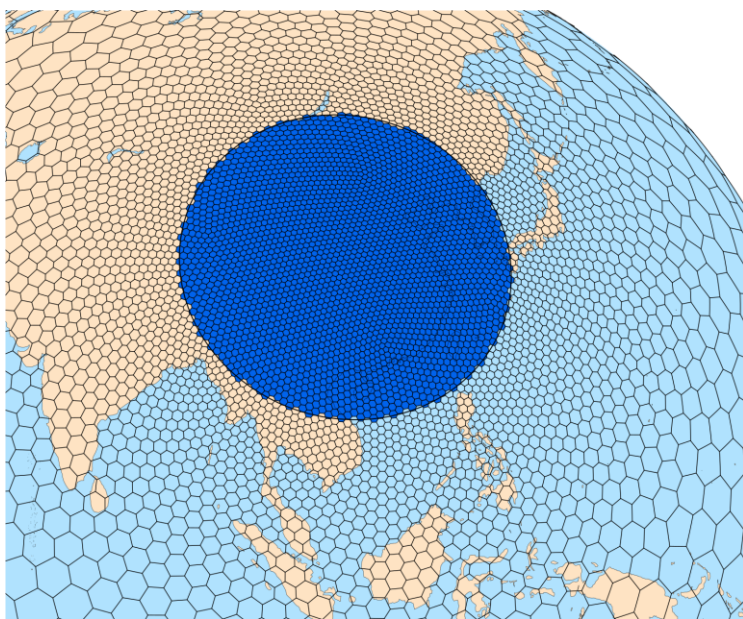
Eta CPTEC 40 km





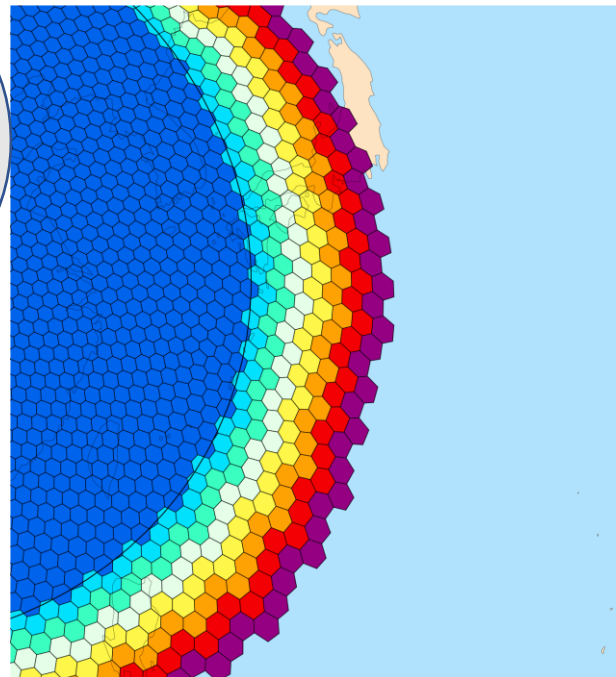
1.2 Regionalização com fronteira aberta

Comece com uma malha global, identifique o domínio regional desejado



A zona de relaxamento permite uma melhor downscale da condição de contorno global

Construir a zona de relaxamento
Construir a zona especificada
Recorte as células globais restantes





1.2 Regionalização com fronteira aberta

Especificação de Zona Fronteira Regional MPAS



Filtros na zona de relaxamento

$$\frac{\partial \psi}{\partial t} = RHS_{\psi} + F_1(\psi_{LS} - \psi) - F_2 \Delta x^2 \nabla^2 (\psi_{LS} - \psi)$$

Amortecimento de
Rayleigh para o valor
de grande escala (LS)

$$F_1 = \gamma_1 \frac{(i-1)}{m}$$

$$F_2 = \gamma_2 \frac{(i-1)}{m}$$

m é a largura da zona de relaxamento (em # células). Exemplo (à direita), m = 5 (padrão)

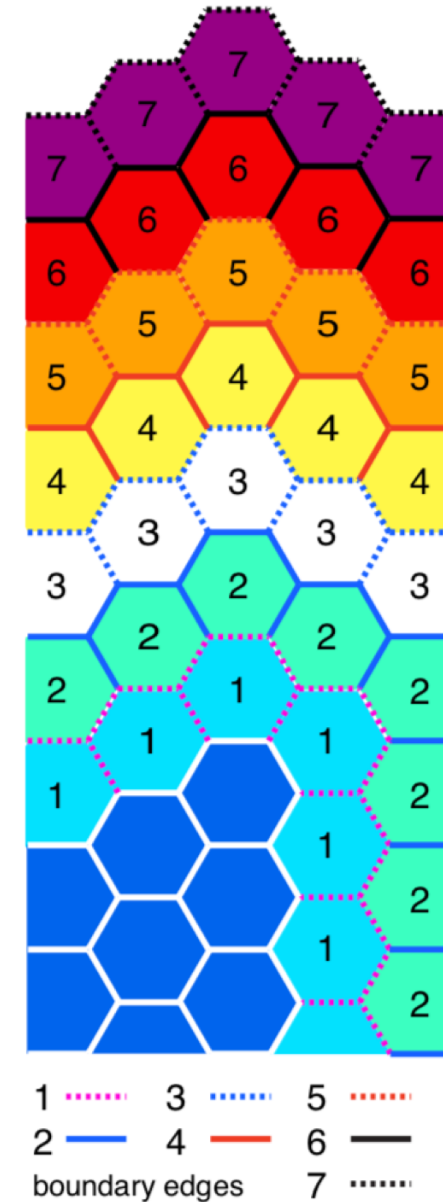
$$F_1 = \gamma_1(0, 0.2, 0.4, 0.6, 0.8)$$

$$F_2 = \gamma_2(0, 0.2, 0.4, 0.6, 0.8)$$

$$\gamma_1 = (0.06 \Delta x)^{-1}$$

$$\gamma_2 = (0.3 \Delta x)^{-1}$$

Esses valores (γ_1 e γ_2) são hardwire no MPAS





1.2 Regionalização com fronteira aberta

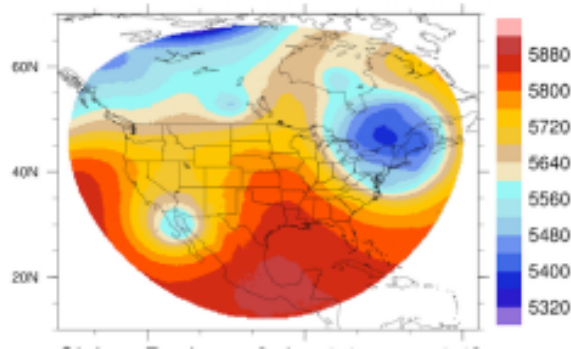
Regional MPAS



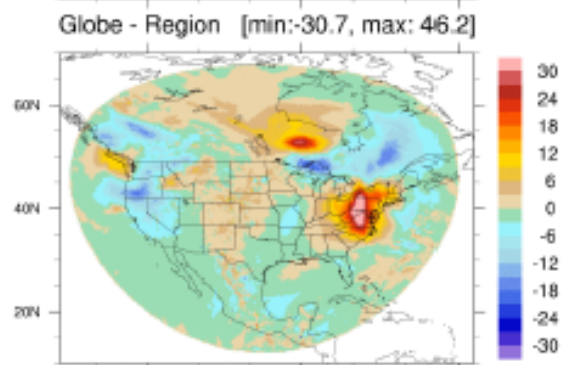
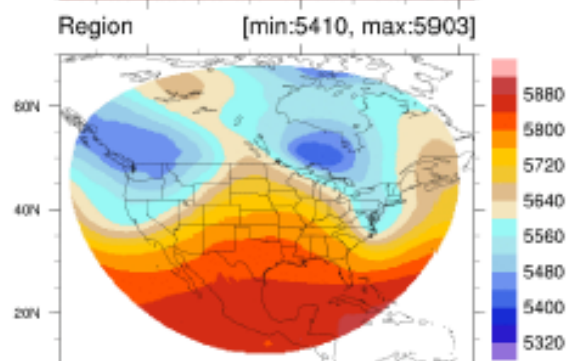
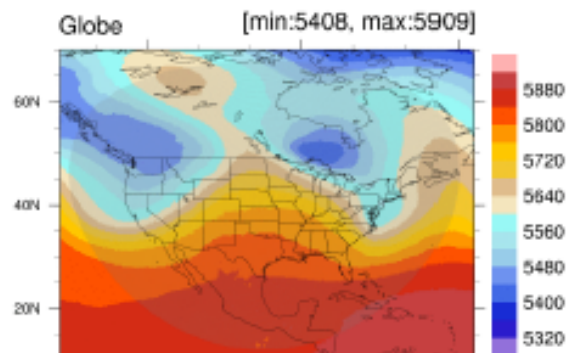
NCAR real-time forecasts
November 2016 - June 2017
Daily 10-day MPAS forecasts
00 UTC GFS analysis initialization



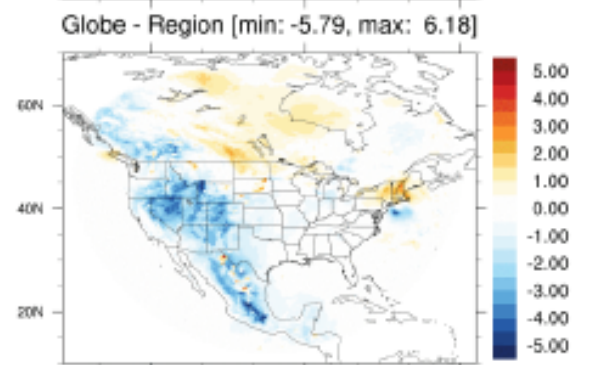
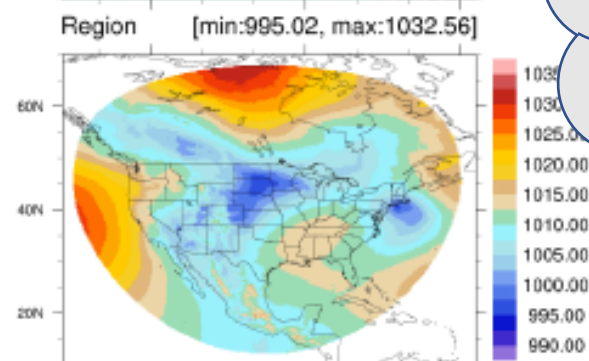
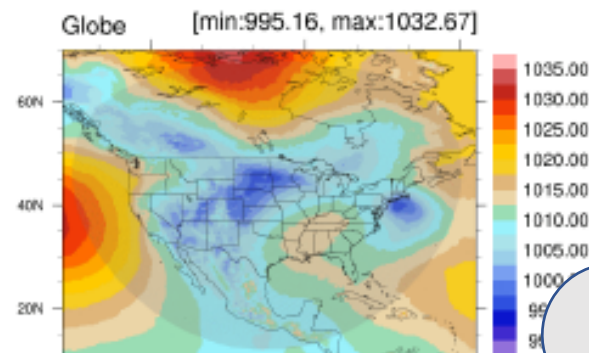
Regional initialization
500 hPa height field (m)
2017-05-09_00



FCST 120H at 2017-05-14_00 in height_500hPa [m]



FCST 120H at 2017-05-14_00 in mslp [hPa]



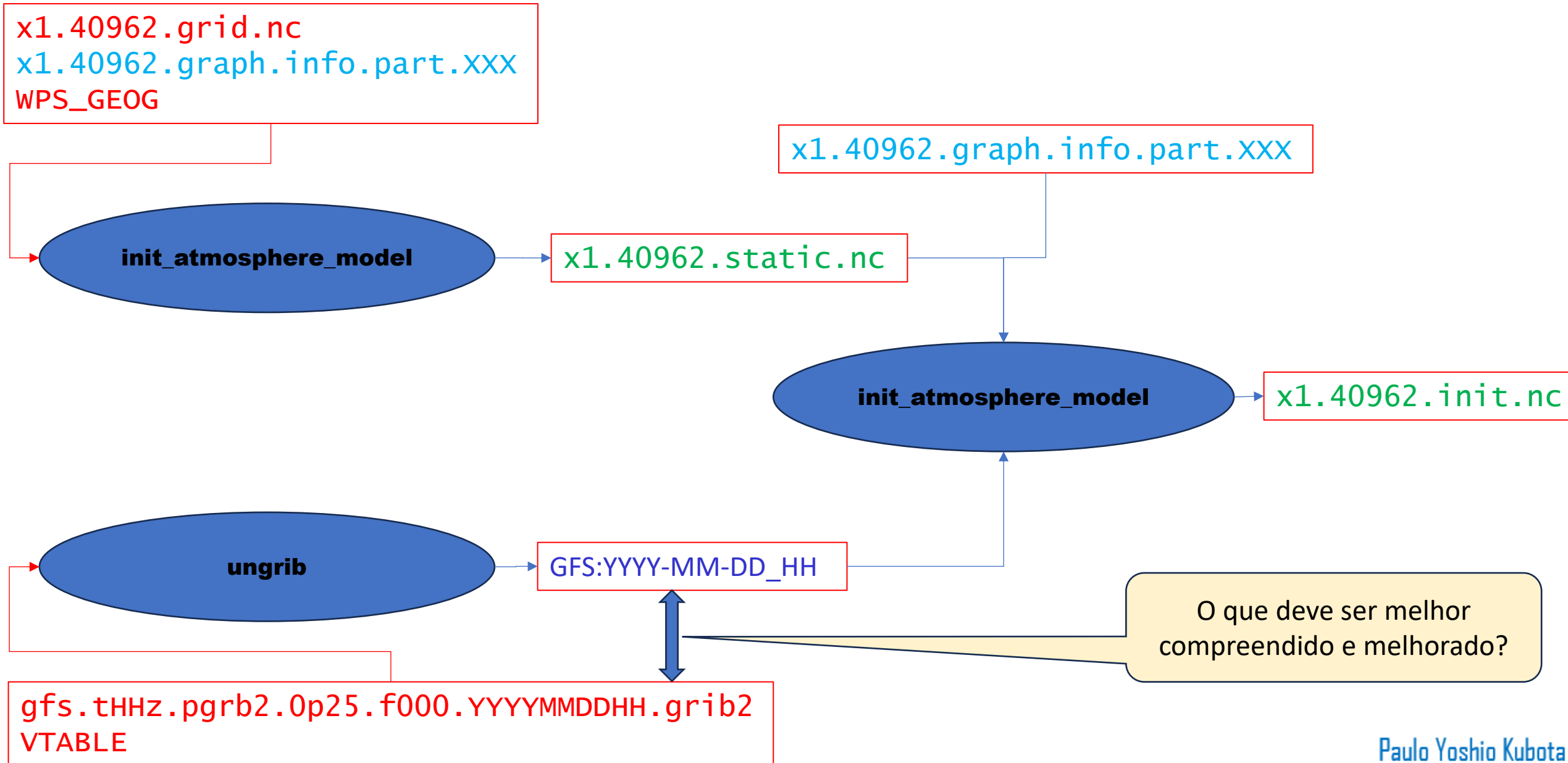
**Impacto da
regionalização**

**Global -
Regional**





Condição Inicial do modelo MONAN





Condição Inicial do modelo MONAN



O que se sabe até agora?

/oper/dados/ioper/tempo/GFS/.../gfs.t00z.pgrb2.0p25.**an1**.2024041300.grib2

/oper/dados/ioper/tempo/GFS/.../gfs.t00z.pgrb2.0p25.**f000**.2024041300.grib2

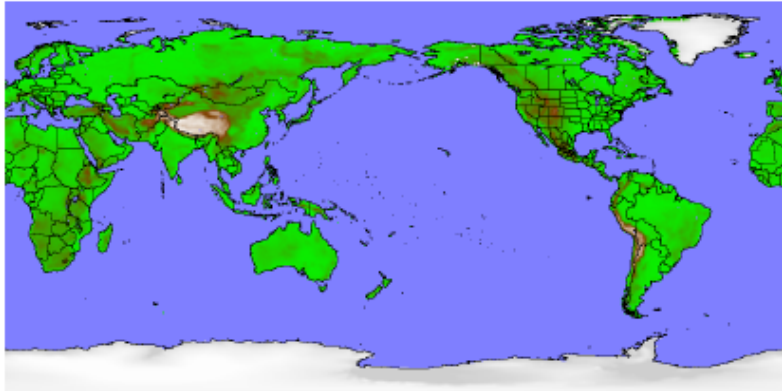


Condição Inicial do modelo MONAN

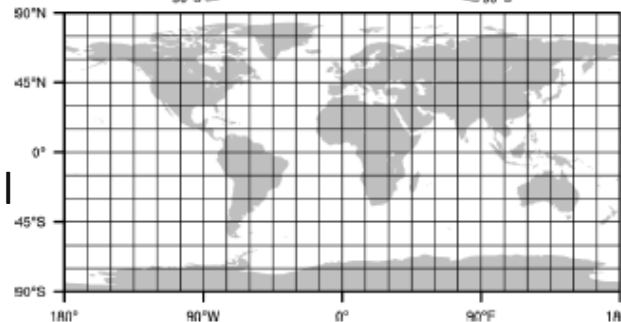
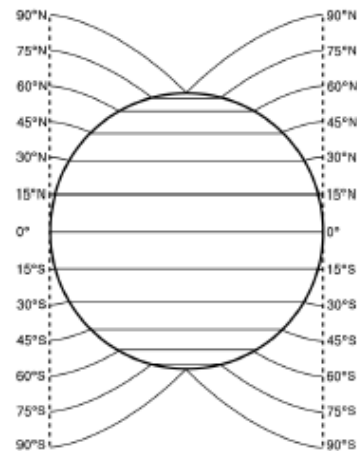


/oper/dados/ioper/tempo/GFS/.../gfs.t00z.pgrb2.0p25.**an1**.2024041300.**grib2**

/oper/dados/ioper/tempo/GFS/.../gfs.t00z.pgrb2.0p25.**f000**.2024041300.**grib2**



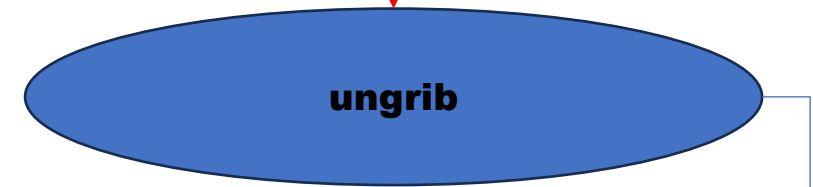
Cylindrical Equidistant



Necessário para domínios globais

- Pode ser usado para domínios regionais
- Pode ser usado em sua forma normal ou rotacionada

The WRF Users' Basic Tutorial



O que acontece neste passo?

GFS:YYYY-MM-DD_HH



Condição Inicial do modelo MONAN



GRIB1 Param	Level Type	From Level1	To Level2	metgrid Name	metgrid Units	metgrid Description	GRIB2 Discp	GRIB2 Catgy	GRIB2 Param	GRIB2 Level
11	100	*		TT	K	Temperature	0	0	0	100
33	100	*		UU	m s-1	U	0	2	2	100
34	100	*		VV	m s-1	V	0	2	3	100
52	100	*		RH	%	Relative Humidity	0	1	1	100
7	100	*		HGT	m	Height	0	3	5	100
11	105	2		TT	K	Temperature at 2 m	0	0	0	103
52	105	2		RH	%	Relative Humidity at 2 m	0	1	1	103
33	105	10		UU	m s-1	U at 10 m	0	2	2	103
34	105	10		VV	m s-1	V at 10 m	0	2	3	103
1	1	0		PSFC	Pa	Surface Pressure	0	3	0	1
130	102	0		PMSL	Pa	Sea-level Pressure	0	3	192	101
144	112	0	10	SM000010	fraction	Soil Moist 0-10 cm below grn layer (Up)	2	0	192	106
144	112	10	40	SM010040	fraction	Soil Moist 10-40 cm below grn layer	2	0	192	106
144	112	40	100	SM040100	fraction	Soil Moist 40-100 cm below grn layer	2	0	192	106
144	112	100	200	SM100200	fraction	Soil Moist 100-200 cm below gr layer	2	0	192	106
144	112	10	200	SM010200	fraction	Soil Moist 10-200 cm below gr layer	2	0	192	106
11	112	0	10	ST000010	K	T 0-10 cm below ground layer (Upper)	0	0	0	106
11	112	10	40	ST010040	K	T 10-40 cm below ground layer (Upper)	0	0	0	106
11	112	40	100	ST040100	K	T 40-100 cm below ground layer (Upper)	0	0	0	106
11	112	100	200	ST100200	K	T 100-200 cm below ground layer (Bottom)	0	0	0	106
85	112	0	10	ST000010	K	T 0-10 cm below ground layer (Upper)	2	0	2	106
85	112	10	40	ST010040	K	T 10-40 cm below ground layer (Upper)	2	0	2	106
85	112	40	100	ST040100	K	T 40-100 cm below ground layer (Upper)	2	0	2	106
85	112	100	200	ST100200	K	T 100-200 cm below ground layer (Bottom)	2	0	2	106
11	112	10	200	ST010200	K	T 10-200 cm below ground layer (Bottom)	0	0	0	106
91	1	0		SEAICE	proprtn	Ice flag	10	2	0	1
81	1	0		LANDSEA	proprtn	Land/Sea flag (1=land, 0 or 2=sea)	2	0	0	1
81	1	0		LANDN	proprtn		2	0	218	1
7	1	0		SOILHGT	m	Terrain field of source analysis	0	3	5	1
11	1	0		SKINTEMP	K	Skin temperature	0	0	0	1
65	1	0		SNOW	kg m-2	Water equivalent snow depth	0	1	13	1
	1	0		SNOWH	m	Physical Snow Depth	0	1		1
33	6	0		UMAXW	m s-1	U at max wind	0	2	2	6
34	6	0		VMAXW	m s-1	V at max wind	0	2	3	6
2	6	0		PMAXW	Pa	Pressure of max wind level	0	3	0	6
	6	0		PMAXWNN	Pa	PMAXW, used for nearest neighbor interp	0	3	0	6
2	6	0		TMAXW	K	Temperature at max wind level	0	0	0	6
7	6	0		HGTMAXW	m	Height of max wind level	0	3	5	6
33	7	0		UTROP	m s-1	U at tropopause	0	2	2	7
34	7	0		VTROP	m s-1	V at tropopause	0	2	3	7
2	7	0		PTROP	Pa	Pressure of tropopause	0	3	0	7
	7	0		PTROPNN	Pa	PTROP, used for nearest neighbor interp	0	3	0	7
2	7	0		TTROP	K	Temperature at tropopause	0	0	0	7
7	7	0		HGTTRIP	m	Height of tropopause	0	3	5	7

gfs.t00z.pgrb2.0p25.an1.2024041300.grib2

Estas variáveis não
está no arquivo de
análise

Estas variáveis são
necessárias para
domínios globais?

gfs.t00z.pgrb2.0p25.f000.2024041300.grib2



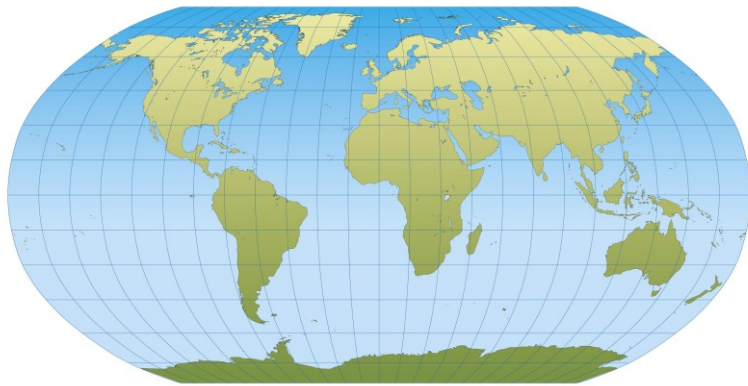
Condição Inicial do modelo MONAN



gdas.T00Z.sfcan1.netcdf.2024041300
gdas.T00Z.atman1.netcdf.2024041300

Eccodes
Ou
wgrib2api

gdas.T00Z.an1.grib2.2024041300



Chopping_BAM

GFS:YYYY-MM-DD_HH

ungrib

GRIB1 Param	Level Type	From Level1	To Level2	metgrid Name	metgrid Units	metgrid Description	GRIB2 Discp	GRIB2 Catgy	GRIB2 Param	GRIB2 Level
11	100	*		TT	K	Temperature	0	0	0	100
33	100	*		UU	m s-1	U	0	2	2	100
34	100	*		VV	m s-1	V	0	2	3	100
52	100	*		RH	%	Relative Humidity	0	1	1	100
7	100	*		HGT	m	Height	0	3	5	100
11	105	2		TT	K	Temperature at 2 m	0	0	0	103
52	105	2		RH	%	Relative Humidity at 2 m	0	1	1	103
33	105	10		UU	m s-1	U at 10 m	0	2	2	103
34	105	10		VV	m s-1	V at 10 m	0	2	3	103
1	1	0		PSFC	Pa	Surface Pressure	0	3	0	1
130	102	0		PMSL	Pa	Sea-level Pressure	0	3	192	101
144	112	0	10	SM000010	fraction	Soil Moist 0-10 cm below grn layer (Up)	2	0	192	106
144	112	10	40	SM010040	fraction	Soil Moist 10-40 cm below grn layer	2	0	192	106
144	112	40	100	SM040100	fraction	Soil Moist 40-100 cm below grn layer	2	0	192	106
144	112	100	200	SM100200	fraction	Soil Moist 100-200 cm below gr layer	2	0	192	106
144	112	10	200	SM010200	fraction	Soil Moist 10-200 cm below gr layer	2	0	192	106
11	112	0	10	ST000010	K	T 0-10 cm below ground layer (Upper)	0	0	0	106
11	112	10	40	ST010040	K	T 10-40 cm below ground layer (Upper)	0	0	0	106
11	112	40	100	ST040100	K	T 40-100 cm below ground layer (Upper)	0	0	0	106
11	112	100	200	ST100200	K	T 100-200 cm below ground layer (Bottom)	0	0	0	106
85	112	0	10	ST000010	K	T 0-10 cm below ground layer (Upper)	2	0	2	106
85	112	10	40	ST010040	K	T 10-40 cm below ground layer (Upper)	2	0	2	106
85	112	40	100	ST040100	K	T 40-100 cm below ground layer (Upper)	2	0	2	106
85	112	100	200	ST100200	K	T 100-200 cm below ground layer (Bottom)	2	0	2	106
11	112	10	200	ST010200	K	T 10-200 cm below ground layer (Bottom)	0	0	0	106
91	1	0		SEAICE	proprtn	Ice flag	10	2	0	1
81	1	0		LANDSEA	proprtn	Land/Sea flag (1=land, 0 or 2=sea)	2	0	0	1
81	1	0		LANDN	proprtn		2	0	218	1
7	1	0		SOILHGT	m	Terrain field of source analysis	0	3	5	1
11	1	0		SKINTEMP	K	Skin temperature	0	0	0	1
65	1	0		SNOW	kg m-2	Water equivalent snow depth	0	1	13	1
	1	0		SNOWH	m	Physical Snow Depth	0	1		1



“Existem **dois executáveis** que precisamos **inicializar** e **executar** uma simulação do MPAS-Atmosphere:

init_atmosphere_model

- Lida com todas as etapas do processamento de condições iniciais com dados reais
- Lida com o processamento de arquivos de atualização de SST e gelo marinho
- Lida com a criação de várias condições iniciais idealizadas
- Lida com a criação de condições de contorno lateral

atmosphere_model

- O próprio modelo, responsável por realizar a integração/simulação dadas quaisquer fontes de condições iniciais"



"Como o núcleo 'init_atmosphere' consegue combinar toda essa funcionalidade em um único programa?

1. A ideia chave é que init_atmosphere_model pode ser executado em estágios usando diferentes opções"

```
&nhyd_model
config_init_case = 7
...
/
&preproc_stages
config_static_interp = true
config_native_gwd_static = true
config_vertical_grid = false
config_met_interp = false
config_input_sst = false
config_frac_seaice = false
/
```

2 = *ideal baroclinic wave*
4 = *ideal squall line*
5 = *ideal supercell*
6 = *ideal mountain wave*
7 = *real-data initialization*
8 = *surface update file creation*
9 = *lateral boundary conditions*

Only used for real-data cases to control, e.g., whether we interpolate static fields, compute fields for GWDO scheme, interpolate meteorological data, etc.



"Em geral, existem dois arquivos que devem ser editados toda vez que o programa `init_atmosphere_model` é executado:"

`namelist.init_atmosphere`

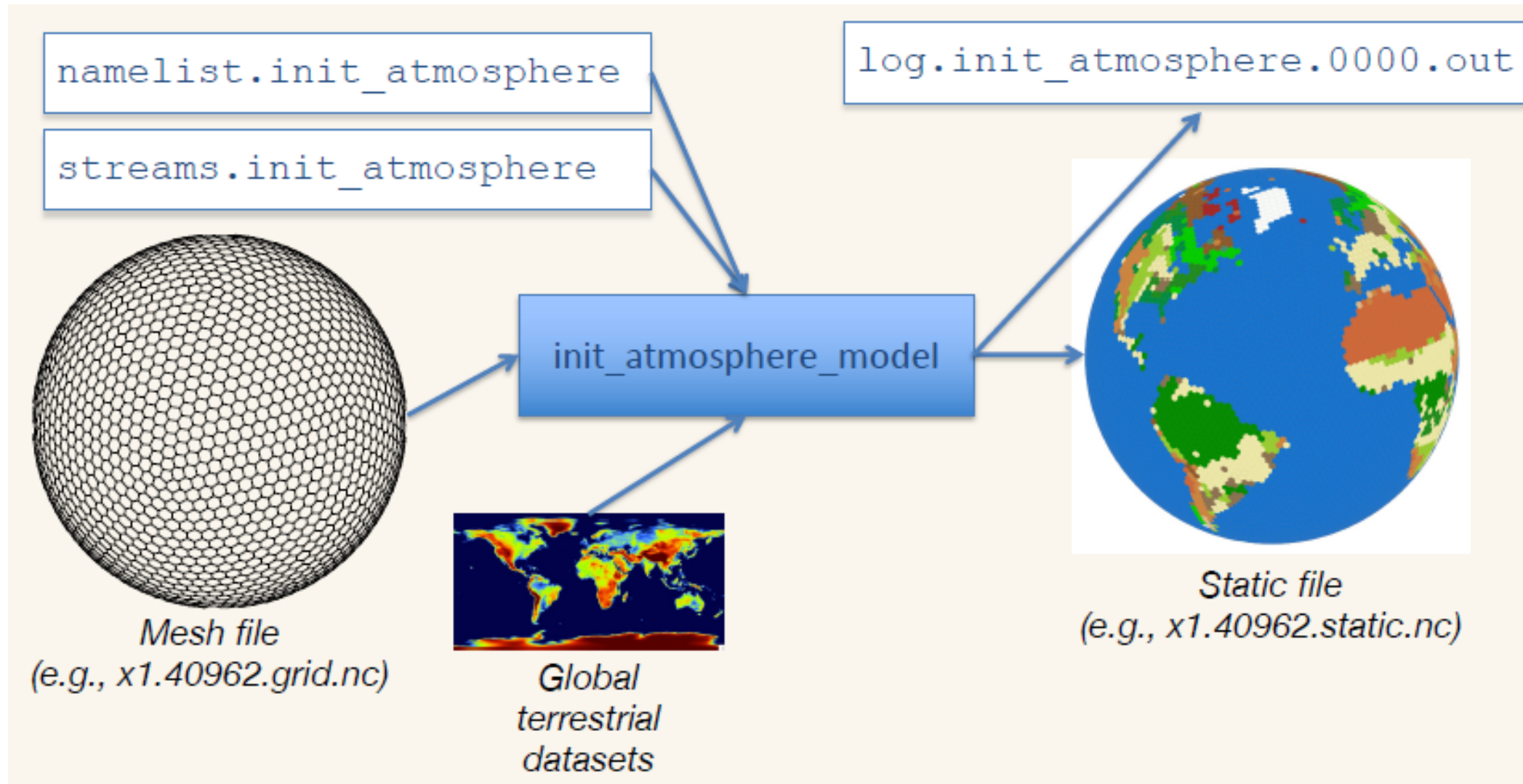
- "• Arquivo de namelist do Fortran
- Determina qual "caso" será preparado (por exemplo, casos idealizados, caso de dados reais)
- Determina sub-opções para a caso da inicialização selecionada"

`streams.init_atmosphere`

- "• Arquivo XML
- Especifica quais arquivos netCDF serão lidos e escritos pelo programa `init_atmosphere_model`"



"Arquivos de entrada e saída ao produzir um arquivo "estático":"





"Configurações-chave no arquivo
namelist.init_atmosphere:"

```
&nhyd_model
config_init_case = 7
/
&data_sources
config_geog_data_path = '/glade/work/wrfhelp/WPS_GEOG/'
config_landuse_data = 'MODIFIED_IGBP_MODIS_NOAH'
config_topo_data = 'GMTED2010'
config_vegfrac_data = 'MODIS'
config_albedo_data = 'MODIS'
config_maxsnowalbedo_data = 'MODIS'
/
&preproc_stages
config_static_interp = true
config_native_gwd_static = true
config_vertical_grid = false
config_met_interp = false
config_input_sst = false
config_frac_seaice = false
/
```

Key settings in the **streams.init_atmosphere** file:

```
<immutable_stream name="input"
    type="input"
    filename_template="x1.40962.grid.nc"
    input_interval="initial_only" />

<immutable_stream name="output"
    type="output"
    filename_template="x1.40962.static.nc"
    packages="initial_conds"
    output_interval="initial_only" />
```



The result should be a “static” netCDF file with

- terrain
- land use category
- soil category
- climatological albedo
- climatological vegetation fraction
- sub-grid-scale orography statistics for the GWDO scheme

Also, the radius of the SCVT mesh should be 6371229.0 m!



"Procure por mensagens como as seguintes no arquivo log.init_atmosphere.0000.out:"

```
--- enter subroutine init_atm_static:  
Using GMTED2010 terrain dataset  
/shared/WPS_GEOG/topo_gmted2010_30s/00001-01200.00001-01200  
/shared/WPS_GEOG/topo_gmted2010_30s/01201-02400.00001-01200  
/shared/WPS_GEOG/topo_gmted2010_30s/02401-03600.00001-01200
```

```
Computing GWDO static fields on the native MPAS mesh  
--- Using GMTED2010 terrain dataset for GWDO static fields
```

```
-----  
Total log messages printed:  
Output messages = 3067  
Warning messages = 10  
Error messages = 0  
Critical error messages = 0  
-----
```



"Condições iniciais com dados reais

- Processamento de campos invariáveis no tempo ("geração de arquivo estático")
- Interpolação de campos atmosféricos e de superfície terrestre
- Produção de arquivos de atualização de SST e gelo marinho



"Campos meteorológicos e de superfície terrestre variáveis no tempo no MPAS-Atmosphere são interpolados a partir de arquivos intermediários produzidos pelo componente ungrib do Sistema de Pré-processamento do WRF.

Vamos assumir neste tutorial que esses arquivos já foram preparados!

- Detalhes adicionais podem ser encontrados nos links abaixo."

WRF Model web page: <http://www2.mmm.ucar.edu/wrf/users/>

WRF Users' guide: https://www2.mmm.ucar.edu/wrf/users/wrf_users_guide/build/html/index.html

WPS source code: <https://github.com/wrf-model/WPS>

Tutorial slides for running ungrib: http://www2.mmm.ucar.edu/wrf/users/tutorial/201801/wps_general.pdf



"Configurações-chave no arquivo :"

namelist.init_atmosphere

```
&nhyd_model
config_init_case = 7
config_start_time = '2014-09-10_00:00:00'
/
&dimensions
config_nvertlevels = 55
config_nsoillevels = 4
config_nfglevels = 38
config_nfgsoillevels = 4
/
&data_sources
config_met_prefix = 'GFS'
/
```

```
&vertical_grid
config_ztop = 30000.0
config_nsmterrain = 1
config_smooth_surfaces = true
config_dzmin = 0.3
config_nsm = 30
config_tc_vertical_grid = true
config_blend_bdy_terrain = false
/&
preproc_stages
config_static_interp = false
config_native_gwd_static = false
config_vertical_grid = true
config_met_interp = true
config_input_sst = false
config_frac_seaice = true
/
```

streams.init_atmosphere

```
<immutable_stream name="input"
  type="input"
  filename_template="x1.40962.static.nc"
  input_interval="initial_only" />

<immutable_stream name="output"
  type="output"
  filename_template="x1.40962.init.nc"
  packages="initial_conds"
  output_interval="initial_only" />
```




"O resultado deve ser um arquivo netCDF "init" com

- tudo do arquivo "estático"
- informações da grade vertical 3D
- temperatura potencial 3D (theta)
- ventos 3D (u e w)
- razão de mistura de vapor de água 3D (qv)
- umidade do solo 2D
- temperatura do solo 2D"



"Procure por mensagens como as seguintes no arquivo log.init_atmosphere.0000.out:"

```
real-data GFS test case
Using option 'linear' for vertical extrapolation of temperature
max ter = 5393.19321458650
Setting up vertical levels as in 2014 TC experiments
--- config_tc_vertical_grid = T
--- als = 0.7500000000000000E-01
--- alt = 1.7000000000000000
--- zetal = 0.7500000000000000
```

```
Interpolating TT at 27 1000.000000000000
Interpolating U at 27 1000.000000000000
Interpolating V at 27 1000.000000000000
Interpolating RH at 27 1000.000000000000
Interpolating GHT at 27 1000.000000000000
*****
Found 27 levels in the first-guess data
*****
```

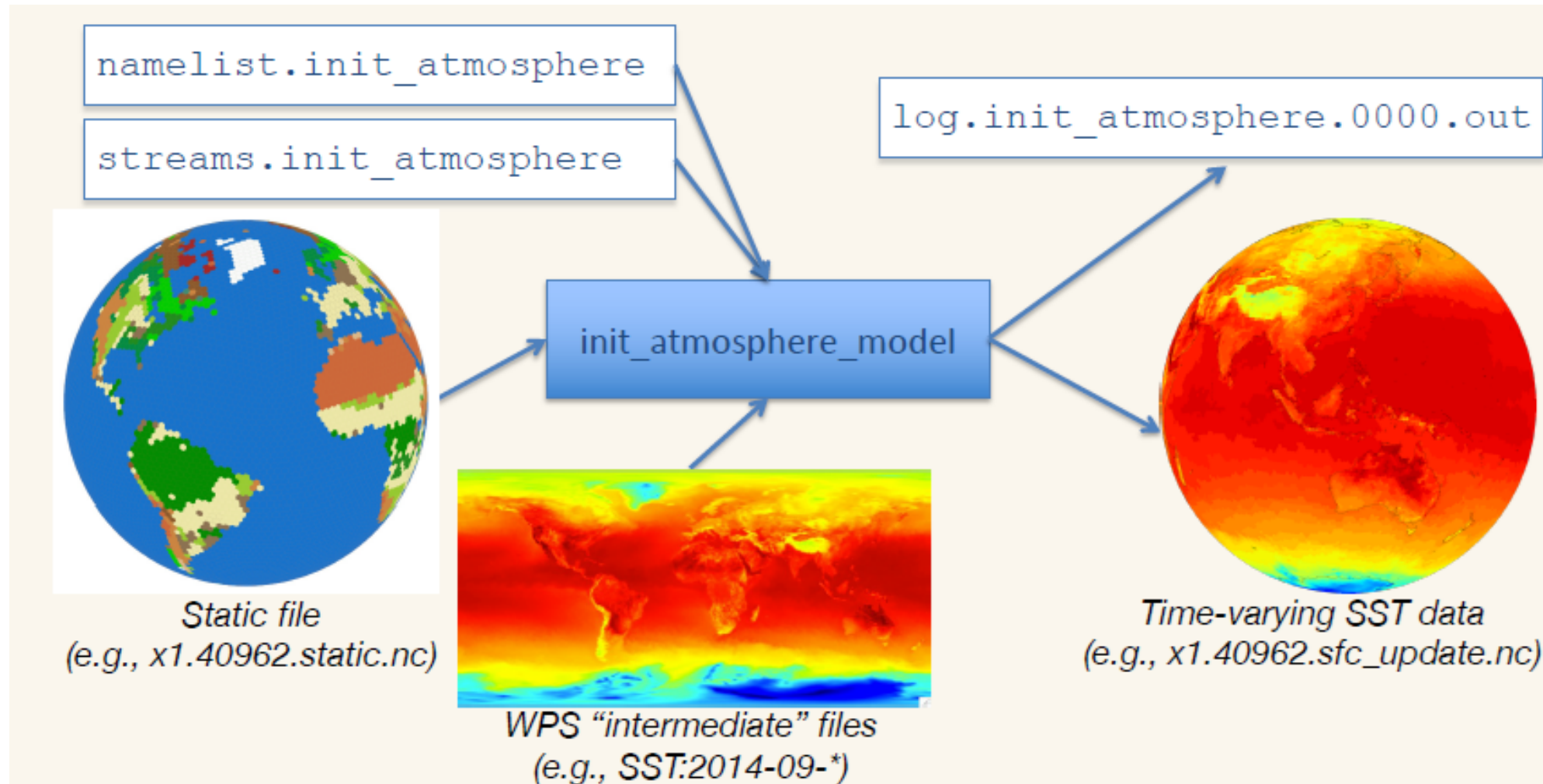


"Condições iniciais com dados reais

- Processamento de campos invariáveis no tempo ("geração de arquivo estático")
- Interpolação de campos atmosféricos e de superfície terrestre
- Produção de arquivos de atualização de SST e gelo marinho



"Arquivos de entrada e saída ao produzir um arquivo de atualização de SST:"





"Configurações-chave no arquivo
`namelist.init_atmosphere:`"

```
&nhyd_model
config_init_case = 8
config_start_time = '2014-09-10_00:00:00'
config_stop_time = '2014-09-20_00:00:00'
/
&data_sources
config_sfc_prefix = 'SST'
config_fg_interval = 86400
/
&preproc_stages
config_static_interp = false
config_native_gwd_static = false
config_vertical_grid = false
config_met_interp = false
config_input_sst = true
config_frac_seaice = true
/
```

"Configurações-chave no arquivo
`streams.init_atmosphere:`"

```
<immutable_stream name="input"
    type="input"
    filename_template="x1.40962.static.nc"
    input_interval="initial_only" />

<immutable_stream name="surface"
    type="output"
    filename_template="x1.40962.sfc_update.nc"
    filename_interval="none"
    packages="sfc_update"
    output_interval="86400" />
```



"Procure por mensagens como as seguintes no arquivo log.init_atmosphere.0000.out:"

```
real-data surface (SST) update test case
Processing file SST:2014-09-10_00
Processing file SST:2014-09-11_00
Processing file SST:2014-09-12_00
Processing file SST:2014-09-13_00
Processing file SST:2014-09-14_00
Processing file SST:2014-09-15_00
Processing file SST:2014-09-16_00
Processing file SST:2014-09-17_00
Processing file SST:2014-09-18_00
Processing file SST:2014-09-19_00
Processing file SST:2014-09-20_00
```

```
-----
Total log messages printed:
Output messages = 144
Warning messages = 0
Error messages = 0
Critical error messages = 0
-----
```

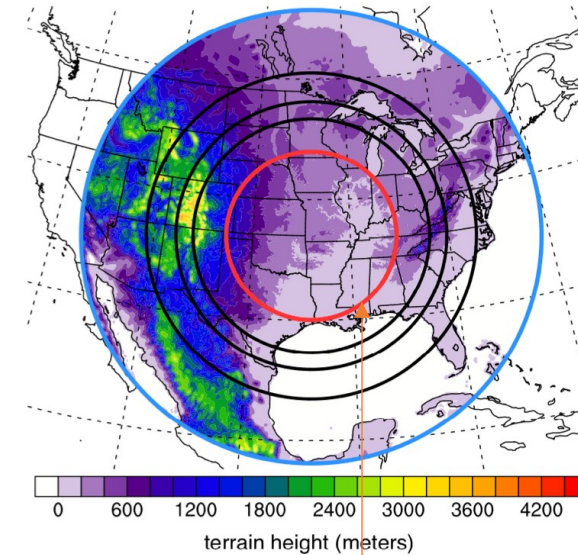


1.2 Regionalização com fronteira aberta

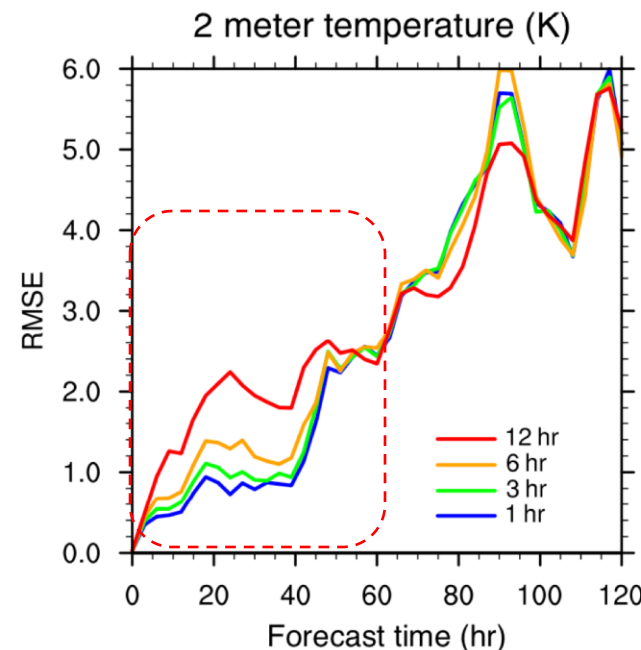
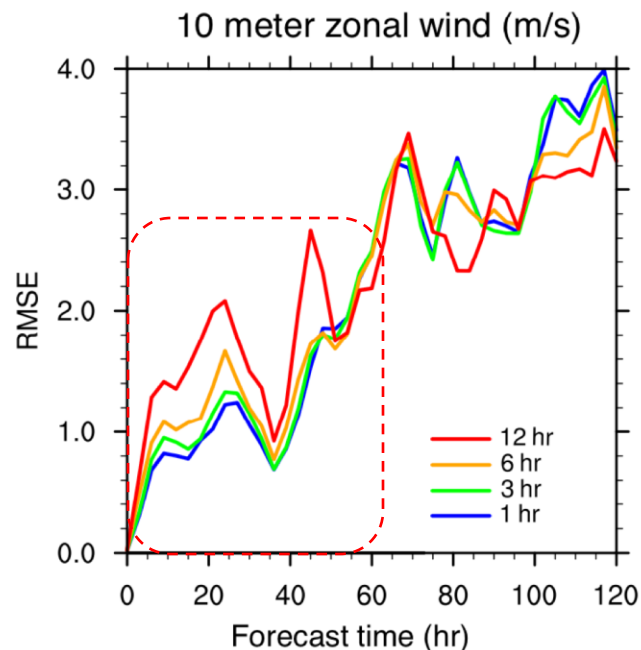
Regional MPAS



Considerações de configuração Frequência de atualização de contorno: integrações MPAS regionais de 5 dias, região do círculo vermelho (3 km de espaçamento entre células), diferentes frequências de atualização de LBC. Integração global de 3 km é verdade. Atualizações com mais frequência são melhores.



26 April 2017
00 UTC init



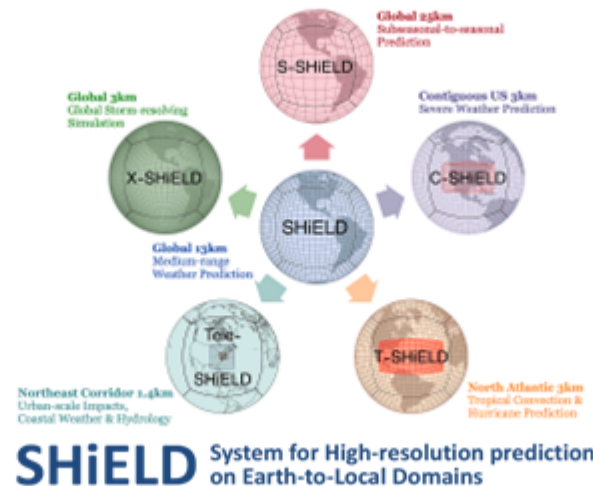
*Verificação sobre a
pequena malha de 3 km
(círculo vermelho)

**Frequência
das
condições
de
contorno**





Dinâmica Não-Hidrostática



GEF
Global Eta FrameWork



1.3 Dinâmica não-hidrostática,

Hydrostatic Equation

Hydrostatic approximation:

$$\frac{1}{\rho_0} \frac{\partial p}{\partial z} = b \quad H/L \ll 1$$

Difference in height between upper & lower isobaric surfaces

$$\frac{\partial z}{\partial p} = - \frac{RT}{p g}$$

Mean temperature within a layer

Non-hydrostatic Equation

Vertical momentum equation for a nonhydrostatic atmosphere

Nonhydrostatic vertical pressure gradient force in a grid box, where ρ_0 is the environmental density, p' is the departure of pressure from the large-scale environmental hydrostatic balance

Change in vertical motion with time $\frac{\partial w}{\partial t} = - \left\{ u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right\} - \frac{1}{\rho_0} \frac{\partial p'}{\partial z} + gB - gq$

Advection of vertical momentum by the wind field

Buoyancy (B) differences from large-scale average

Precipitation loading, where q is specific humidity

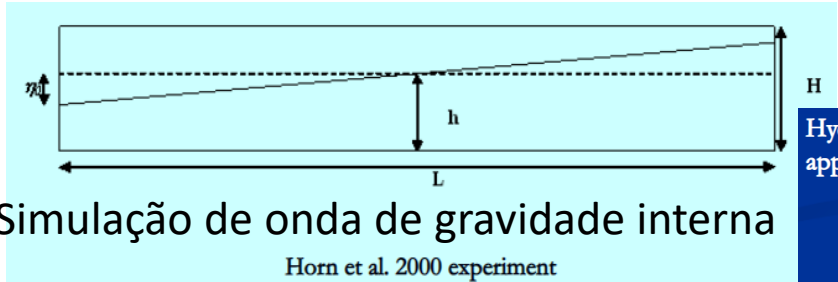
	Hydrostatic	Non-hydrostatic
Buoyancy	Indirect (first, horizontal pressure gradients are created, then convergence, finally vertical motion)	Direct (buoyancy - vertical motion)
Perturbation pressure acting against buoyancy	No	Yes, important for regulating convective updraft velocity and convective cloud structure and for gravity wave energy propagation
Previous vertical motion	No	Yes, vertical motion is advected
Water loading	No	Yes

Aproximação hidrostática

$$-\frac{1}{\rho_0} \frac{\partial p'}{\partial x} = gB - gq$$

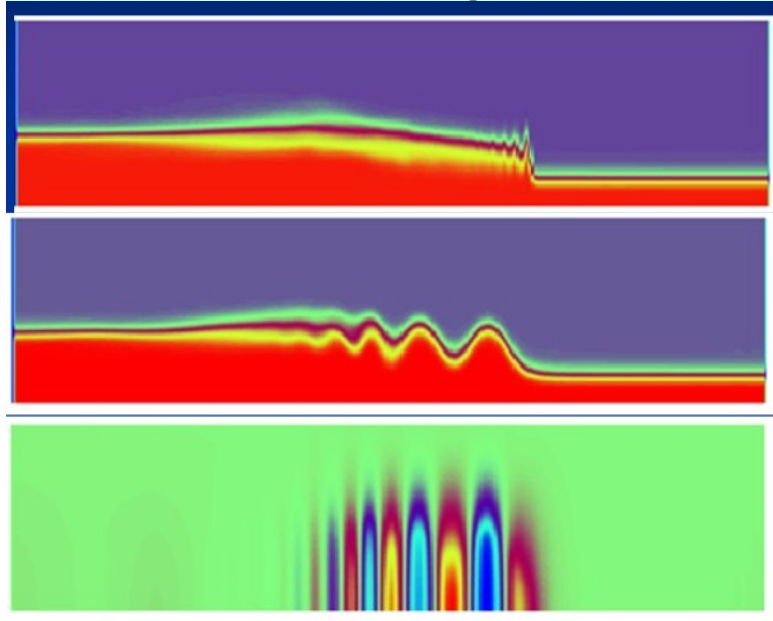


1.3 Dinâmica não-hidrostática,



Hydrostatic approximation:

$$\frac{1}{\rho_0} \frac{\partial p}{\partial z} = b \quad H/L \ll 1$$

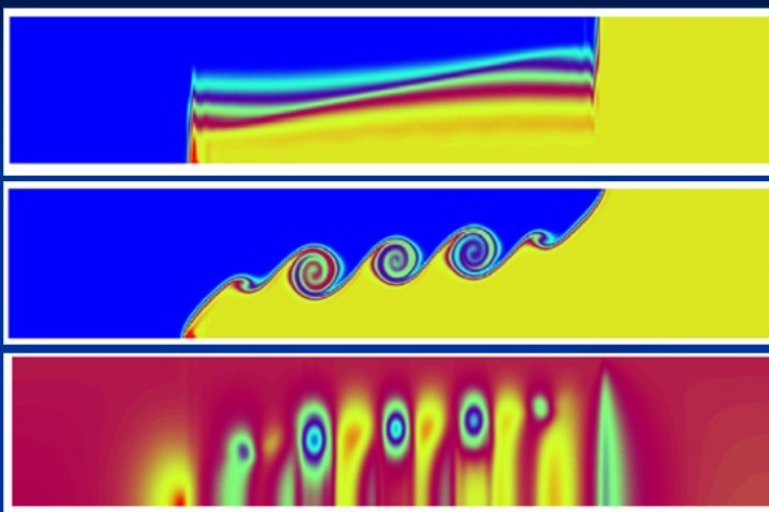


Hydrostatic

Non-hydrostatic

Non-hydrostatic pressure distribution

Intabilidade baroclinica de Kelvin Helmolztz



Density distribution in hydrostatic simulations

Density distribution in non-hydrostatic simulations

Nh pressure correction

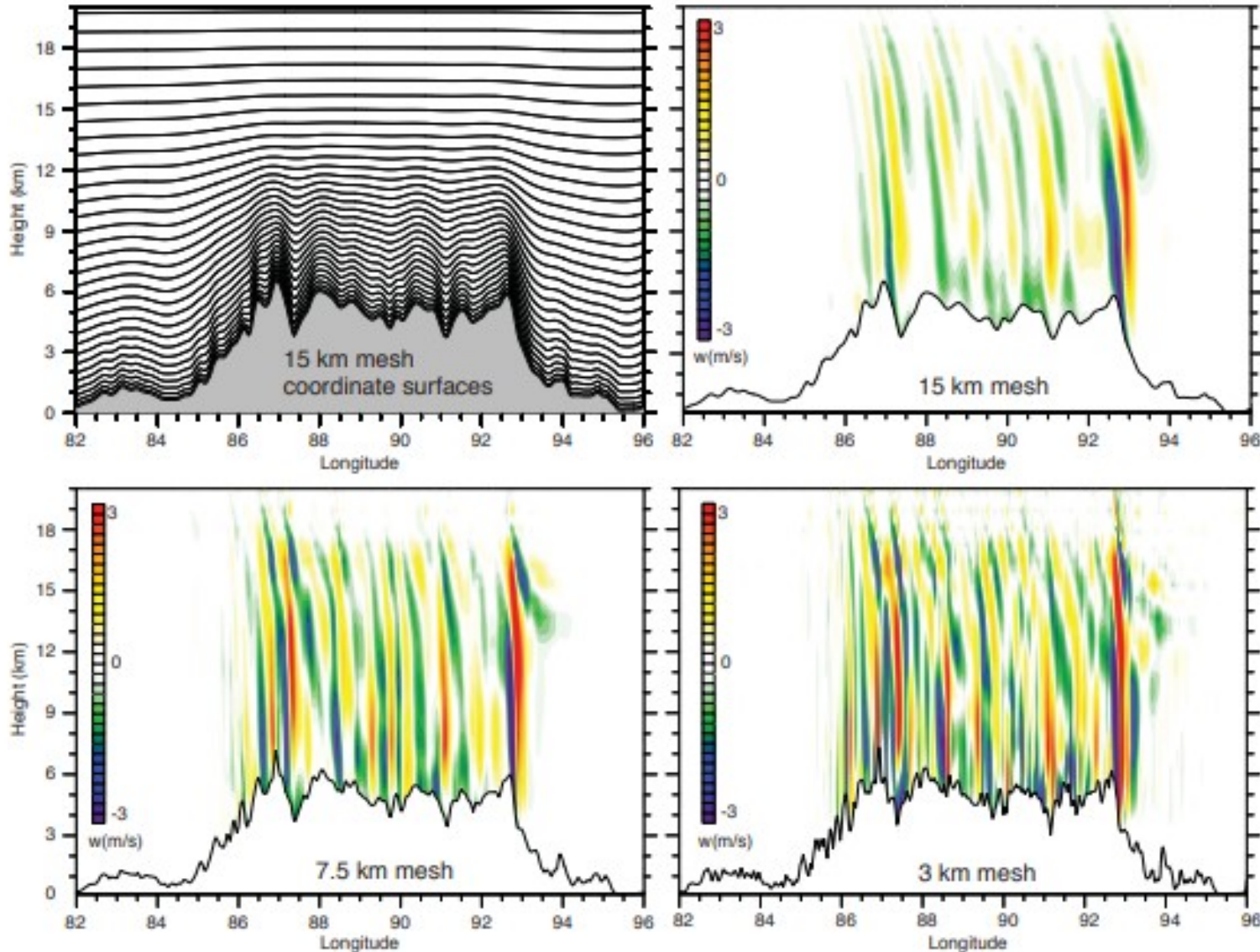
Hydrostatic stable time step two times smaller then non-hydrostatic!

Formulação /Setup	MPAS	FV3-Shied	GEF
Aproximação Vertical	nonhydrostatic	nonhydrostatic	hydrostatic



1.3 Dinâmica não-hidrostática,

coordenada vertical híbrida MPAS-A (Klemp, 2011).



Uma das principais razões para aumentar a resolução nas escalas não hidrostáticas:

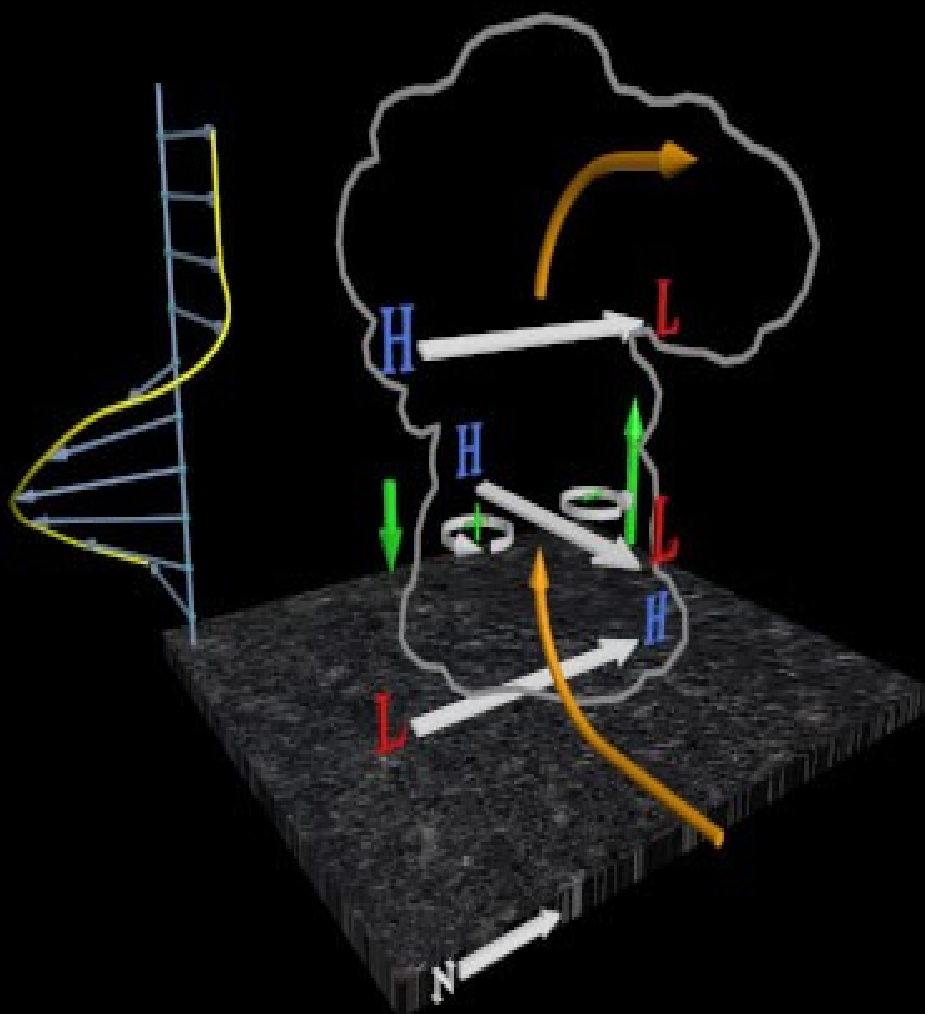
a) resolver melhor a topografia

b) começar a resolver explicitamente a estrutura e a evolução do sistema convectivo.



1.3 Dinâmica não-hidrostática,

Supercell Thunderstorm within Directionally Sheared Environment



©The COMET Program

Non-hydrostatic Equation

Vertical momentum equation for a nonhydrostatic atmosphere

Nonhydrostatic vertical pressure gradient force in a grid box, where ρ_0 is the environmental density, p' is the departure of pressure from the large-scale environmental hydrostatic balance

$$\text{Change in vertical motion with time } \frac{\partial w}{\partial t} = - \left\{ \underbrace{u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z}}_{\text{I Advection of vertical momentum by the wind field}} - \underbrace{\frac{1}{\rho_0} \frac{\partial p'}{\partial z}}_{\text{II Buoyancy (B) differences from large-scale average}} + \underbrace{gB}_{\text{III}} - \underbrace{gq}_{\text{VI Precipitation loading, where q is specific humidity}} \right.$$

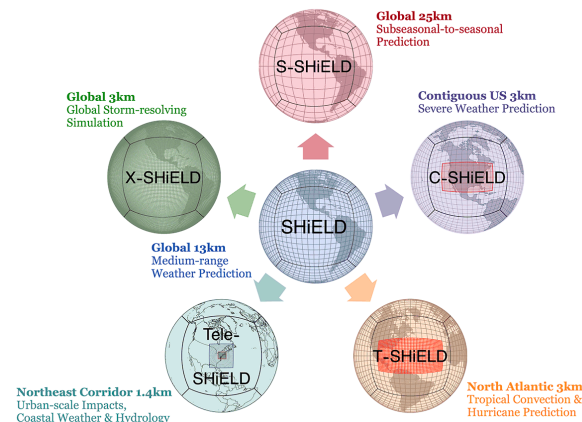
- A figura mostra um exemplo em que um modelo não hidrostático é necessário para **capturar importantes efeitos da dinâmica e da física**.
- O **cisalhamento do vento** dentro do ambiente da tempestade **dá origem a perturbação de pressões** dentro da célula da tempestade, produzindo **subida e descida** conforme mostrado pelas **setas verdes**.
- Essas **circulações induzidas pela perturbação de pressão** dão origem à divisão da tempestade.



1.4 Transporte de escalares com manutenção de positividade e monotonicidade,



transporte de escalares com manutenção de positividade e monotonicidade



GEF



1.4 Transporte de escalares com manutenção de monotonicidade, positividade



3.4.1. Conceito de Monotonicidade

Quando esquemas numéricos que são usados para **advectar** uma **função monotônica**, por exemplo, **uma função monotonicamente que decrescente com x**



As **soluções numéricas não necessariamente preservam a propriedade mononótica** – na verdade, **na maioria das vezes não o fazem**, e os erros tendem a ser grandes perto de gradiente acentuado





1.4 Transporte de escalares com manutenção de monotonicidade, positividade

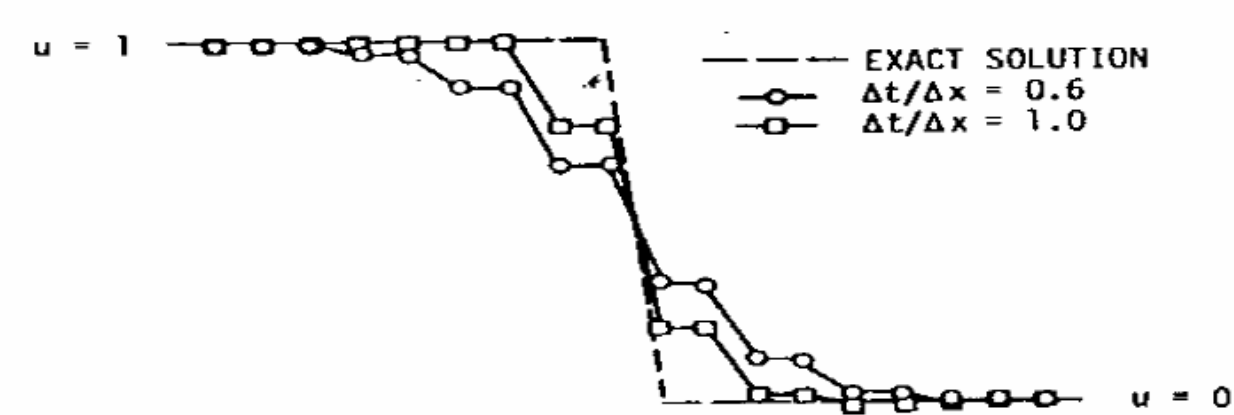


Figure 4-27 Numerical solution of Burgers' equation using Lax method.

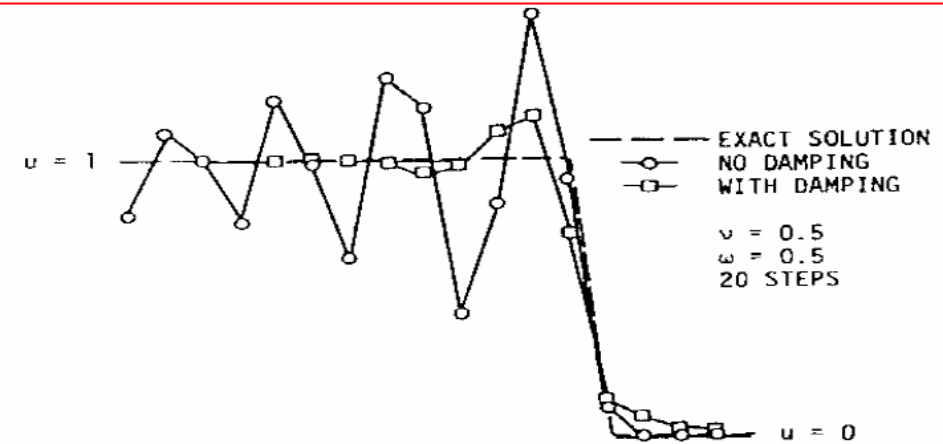


Figure 4-35 Solution of Burgers' equation using Beam-Warming (trapezoidal) method.

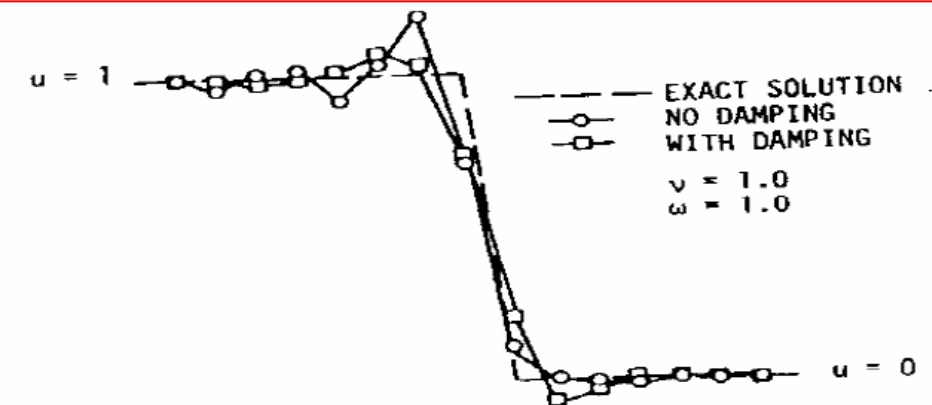


Figure 4-36 Solution for right-moving discontinuity time-centered implicit method, delta form.

Soluções Numéricas tendem a não preservar a monotonicidade



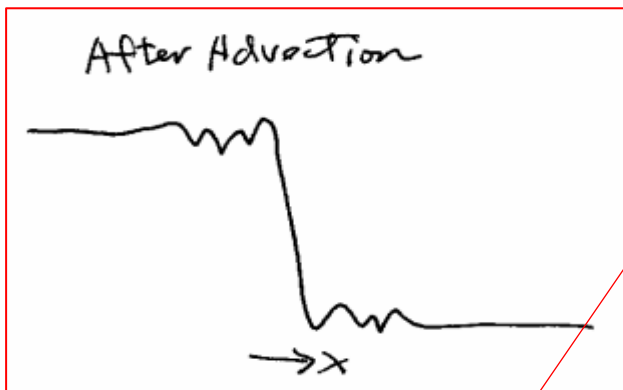


1.4 Transporte de escalares com manutenção de monotonicidade, positividade



Método de Discretização Numérica

Volume Finito
MPAS



Como Minimizar
este problema?
(Volume
Finito de
alta ordem)

$$\frac{\partial(\rho\psi)}{\partial t} = -\nabla \cdot \mathbf{V}\rho\psi,$$

$$\frac{\partial(\psi_i)}{\partial t} = L(\mathbf{V}, \psi_j),$$

$$\frac{\partial(\overline{\rho\psi})_i}{\partial t} = -\frac{1}{A_i} \int_{\Gamma_i} \rho\psi \mathbf{V} \cdot \mathbf{n}_i d\Gamma.$$

$$\frac{\partial(\overline{\rho\psi}_i)}{\partial t} = -\frac{1}{A_i} \sum_{n_{e_i}} d_{e_i} \mathbf{F}_{e_i}(\mathbf{V}, \psi),$$

$$F(u, \psi)_{i+1/2} = u_{i+1/2} \left[\frac{1}{2} (\psi_{i+1} + \psi_i) - \Delta x_e^2 \frac{1}{12} \left\{ \left(\frac{\partial^2 \psi}{\partial x^2} \right)_{i+1} + \left(\frac{\partial^2 \psi}{\partial x^2} \right)_i \right\} \right. \\ \left. + \text{sign}(u) \Delta x_e^2 \frac{\beta}{12} \left\{ \left(\frac{\partial^2 \psi}{\partial x^2} \right)_{i+1} - \left(\frac{\partial^2 \psi}{\partial x^2} \right)_i \right\} \right],$$

A equação representa algoritmos de **precisão de 3ª ordem** ($\beta \neq 0$) ou **4ª ordem** ($\beta = 0$) para malhas com espaçamento de malha constante Dx e velocidade constante.

Na **malha MPAS Voronoi**, Skamarock e Gassmann demonstraram a **convergência de 3ª ordem** para este esquema na esfera com malhas suaves é o ideal.

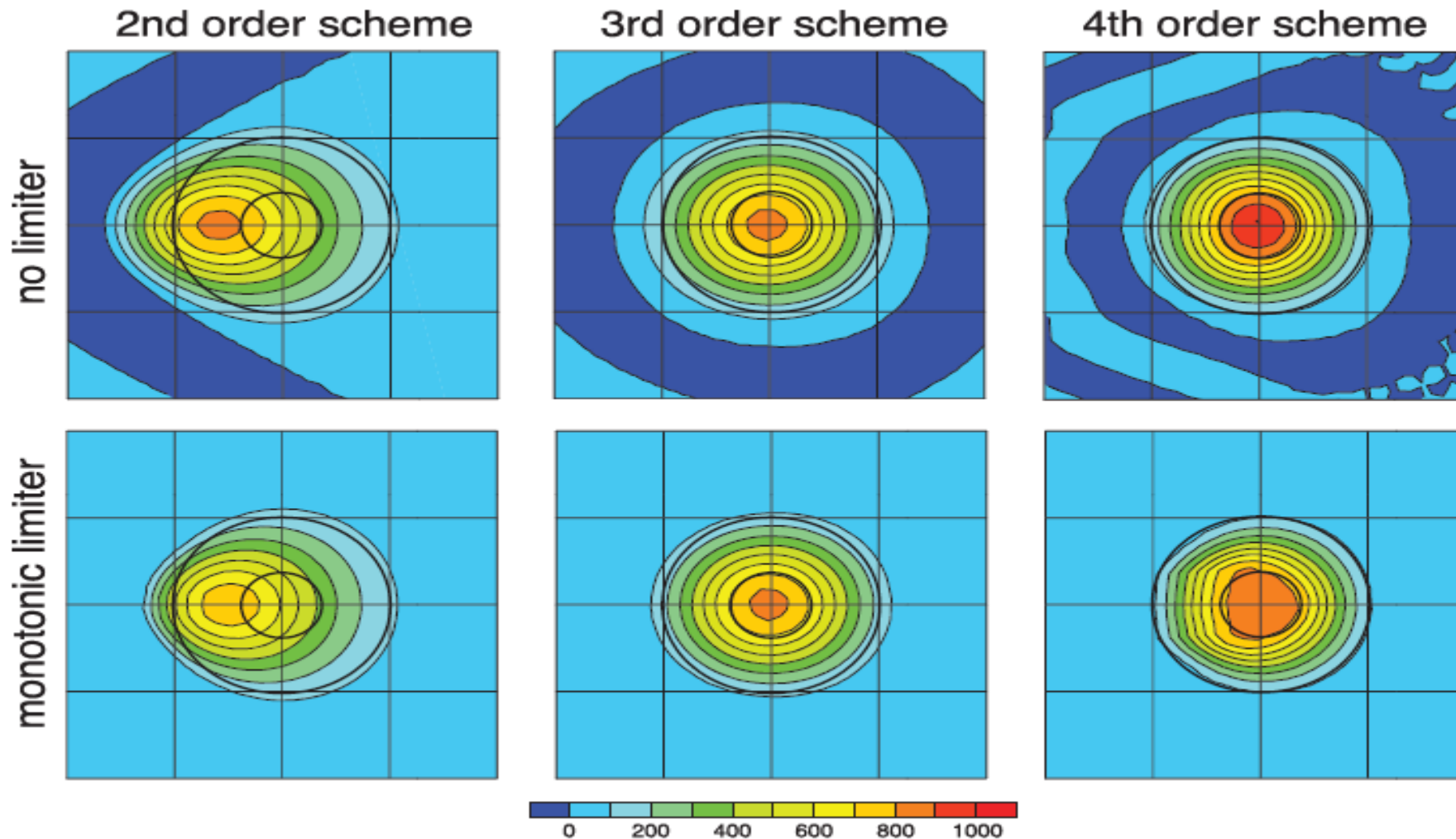
Onde **recomendaram** $\beta = 0.25$ quando usado com um limitador de preservação na forma baseada em FCT.



1.4 Transporte de escalares com manutenção de monotonicidade, positividade



Método de Discretização Numérica



*Como Minimizar
este problema?*
**(Volume
Finito de
alta ordem)**

FIG. 3. Cosine bell after one revolution on the sphere. The contour interval is 100 and the thick contours show the exact solution for $\psi = 100$ and 800. The simulations were performed on the 10 242-cell mesh.



1.4 Transporte de escalares com manutenção de monotonicidade, positividade



Método de Discretização Numérica

Interpolação usado no método Semi-Lagrangiano

*Como Minimizar
este problema?*
(Metodo
Semi-
Lagrangiano)





1.4 Transporte de escalares com manutenção de monotonicidade, positividade



Importância da qualidade da interpolação

2.2 Interpolation Procedures

Inverse Distance Weighted (**IDW**) scheme
(Esquema de Distância Inversa Ponderada (IDW))

Least Squares (**LS**) scheme
Esquema de mínimos quadrados (LS)

Thin-Plate Spline (**TPS**) scheme
Esquema Thin-Plate Spline (TPS)

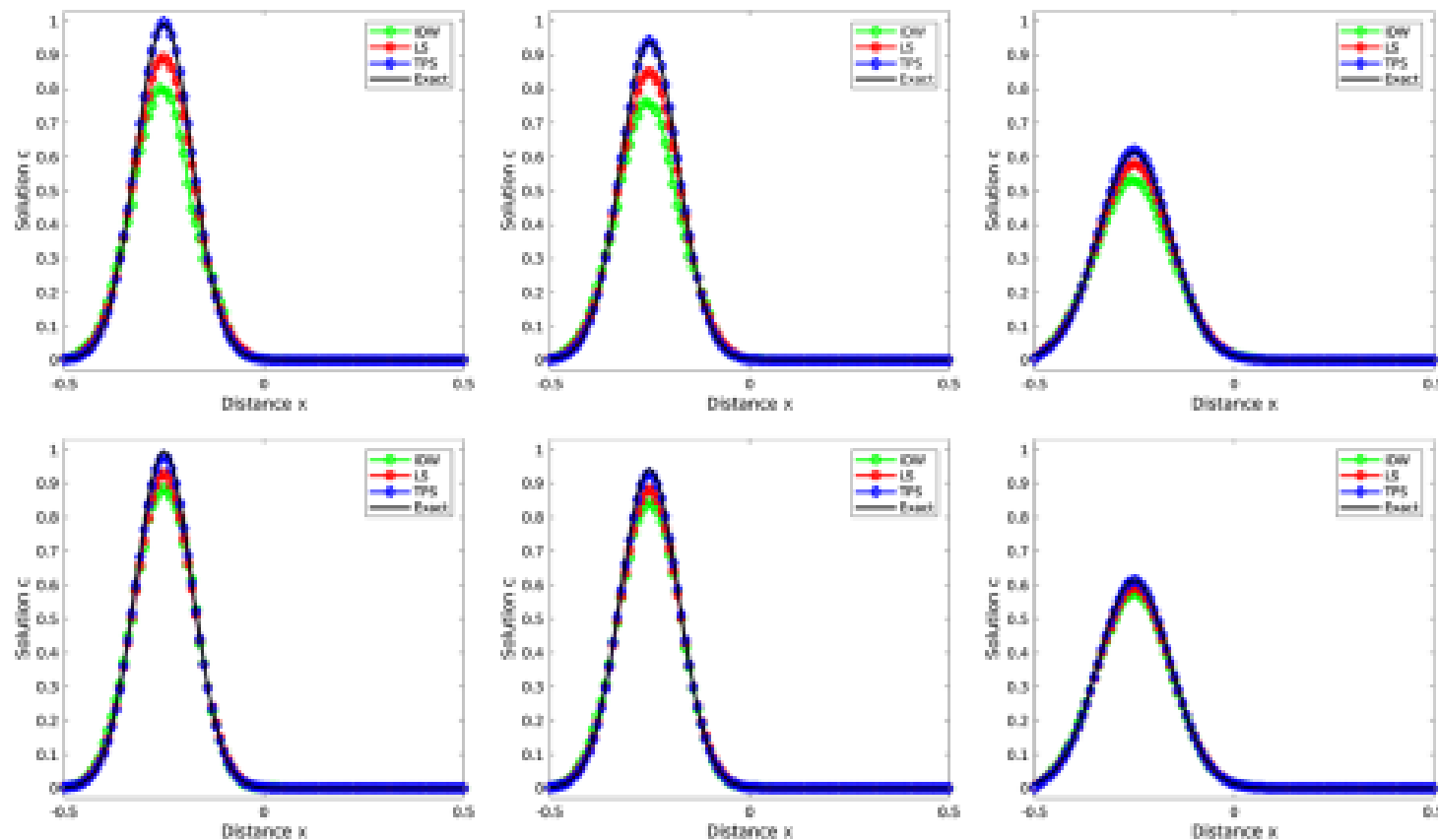


Fig. 6 Cross-sections of the results for the advection-diffusion problem with $\nu = 10^{-5}$ (first column), $\nu = 10^{-4}$ (second column) and $\nu = 10^{-3}$ (third column) using a mesh with 64×64 control volumes (first row) and 128×128 control volumes (second row)

$\nu = \text{coeficiente de difusão}$



1.4 Transporte de escalares com manutenção de monotonicidade, positividade



Método de Discretização Numérica

2.2 Interpolation Procedures

Inverse Distance Weighted (IDW) scheme
(Esquema de Distância Inversa Ponderada (IDW))

Least Squares (LS) scheme
Esquema de mínimos quadrados (LS)

Thin-Plate Spline (TPS) scheme
Esquema Thin-Plate Spline (TPS)

Importância da qualidade da interpolação

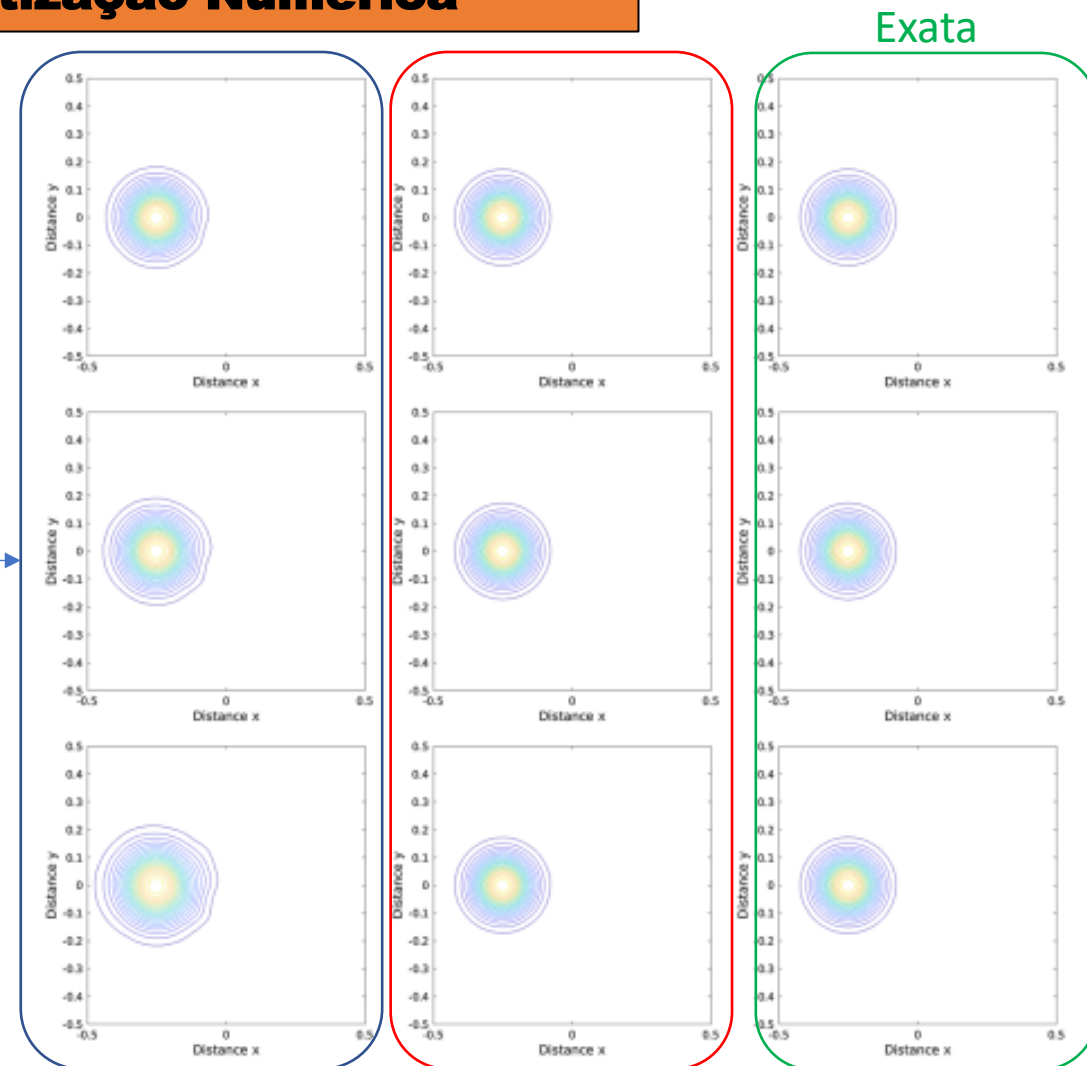


Fig. 4 Numerical solutions obtained using LS (first column), TPS (second column) and exact solution (third column) for $\nu = 0$ on a mesh with 64×64 control volumes after one rotation (first row) and 2 rotations (second row) and 5 rotations (third row)



1.4 Transporte de escalares com manutenção de monotonicidade, positividade



Interpolação não conservativa X conservativa

2.2 Interpolation Procedures

Inverse Distance Weighted (IDW) scheme
(Esquema de Distância Inversa Ponderada (IDW))

Least Squares (LS) scheme
Esquema de mínimos quadrados (LS)

Thin-Plate Spline (TPS) scheme
Esquema Thin-Plate Spline (TPS)

Importância da
qualidade da
interpolação
conservativa

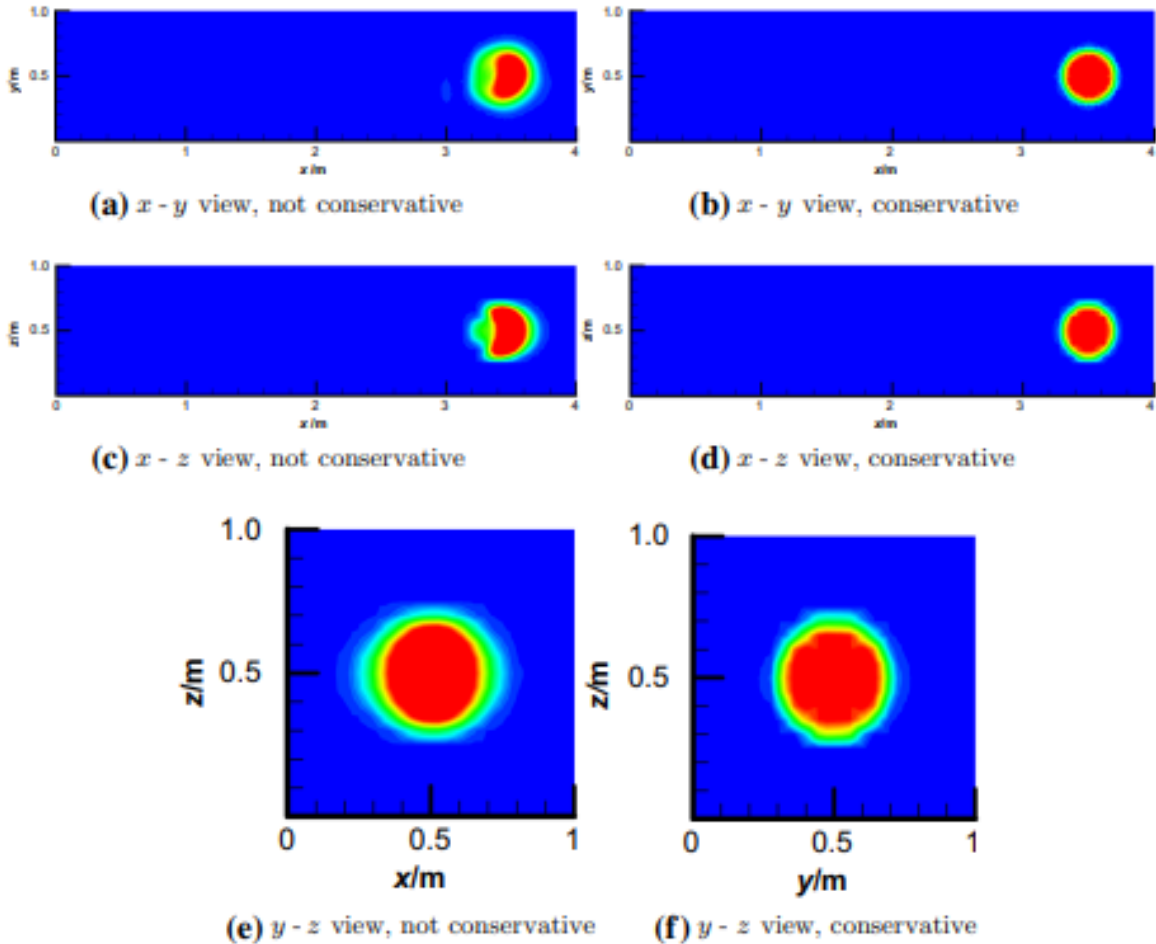


Fig. 14 Solution at time $t = 3$ for the advection of a sphere of contaminant problem



1.4 Transporte de escalares com manutenção de positividade e monotonicidade,



Formulação /Setup	MPAS	FV3-Shied	GEF
Transporte de escalares	Volume finito polinômio de 3rd and 4th-order	Esquema Semi-Lagrangiano	Esquema Semi-Lagrangiano

Semi-Lagrangian Advection on a Cubic Gnomonic Projection of the Sphere

John L. McGregor (1997) Semi-Lagrangian Advection on a Cubic Gnomonic Projection of the Sphere, Atmosphere-Ocean, 35:sup1, 153-169, DOI:10.1080/07055900.1997.9687346

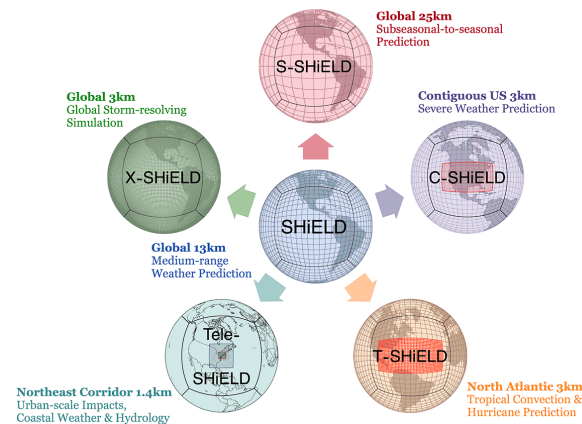
Higher-order finite-volume flux operators for transport algorithms:

Skamarock, W. C., and A. Gassmann, 2011: Conservative Transport Schemes for Spherical Geodesic Grids: High-Order Flux Operators for ODE-Based Time Integration. *Mon. Wea. Rev.*, **139**, 2962–2975, <https://doi.org/10.1175/MWR-D-10-05056.1>.



1.5 Suítes físicas avançadas, etc.

físicas avançadas

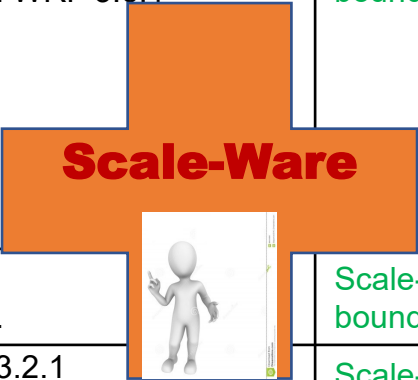


GEF



1.5 Suítes físicas avançadas, etc.

Formulação /Setup	MPAS	FV3-Shied	GEF
Gravity Wave Drag	YSU GWDO as in WRF 3.6.1.	GFS (orographic and convective) gravity wave drag scheme	??
Land Surface Model	Noah (4-layers) as in WRF 3.3.1	Noah land surface model	Noah land surface model
Surface Layer	module_sf_sfclay.F (Monin-Obukhov) as in WRF 3.8.1; MYNN as in WRF 3.6.1	Scale-aware TKE-EDMF planetary boundary layer scheme	Monin–Obukhov similarity theory is combined with Paulson stability functions (Paulson, 1970) and applied to the surface layer which includes a molecular viscous sublayer over land according to Zilitinkevich (1995), and over water according to Janjic (1994).
PBL	YSU as in WRF 3.8.1; MYNN as in WRF 3.6.1.	Scale-aware TKE-EDMF planetary boundary layer scheme	Mellor-Yamada 2.5 closure (Mellor and Yamada, 1982;Janjic, 1990).
Convection	Kain-Fritsch as in WRF 3.2.1 Tiedtke as in WRF 3.3.1 New Tiedtke as in WRF 3.8.1 version of scale-aware Grell-Freitas as in WRF 3.6.1	Scale-aware SAS (shallow and deep) convection scheme	Betts-Miller-Janjic (Bettsand Miller, 1986; Janjic, 1994) and Kain-Fritsch (Kain,2004),
Microphysics	WSM6 as in WRF 3.8.1 Thompson (non-aerosol aware) as in WRF 3.8.1	GFDL cloud microphysics v3	Zhao (Zhao et al., 1997) and Ferrier (Ferrier et al., 2002).
Radiation	RRTMG sw as in WRF 3.8.1 RRTMG lw as in WRF 3.8.1 CAM radiation as in WRF 3.3.1	RRTM radiation with GFDL cloud-radiation interaction scheme	radiation package (GFDL), which includes the short-wave radiation scheme of Lacis and Hansen (1974) and the long-wave radiation scheme of Schwarzkopf and Fels (1991), Paulo Yoshio Kubota





1.5 Suítes físicas avançadas, etc.

**O Modelo MONAN
Necessita de
Parametrizações Físicas
Com o Tratamento
de **Scale-Ware****





2. Dados Estáticos [Nedilson Junior-PGMET /Dra. Aline -UFPA]



Pre_process_geogrid.x



2. Dados Estáticos [Nedilson Junior-PGMET /Dra. Aline -UFPA]



/mnt/beegfs/paulo.kubota/monan_project/MPAS_Model_Regional//pre/databcs/WPS_GEOG

modis_landuse_20class_30s

```
00001-01200.00001-01200
08401-09600.10801-12000
18001-19200.00001-01200
26401-27600.10801-12000
36001-37200.00001-01200
. . . .
```

modis_landuse_new_20class_30s

```
00001-01200.00001-01200.new
08401-09600.10801-12000.new
18001-19200.00001-01200.new
26401-27600.10801-12000.new
36001-37200.00001-01200.new
. . . .
```

Arquivos intermediários

```
00001-01200.00001-01200.bin
08401-09600.10801-12000.bin
18001-19200.00001-01200.bin
26401-27600.10801-12000.bin
36001-37200.00001-01200.bin
. . . .
```

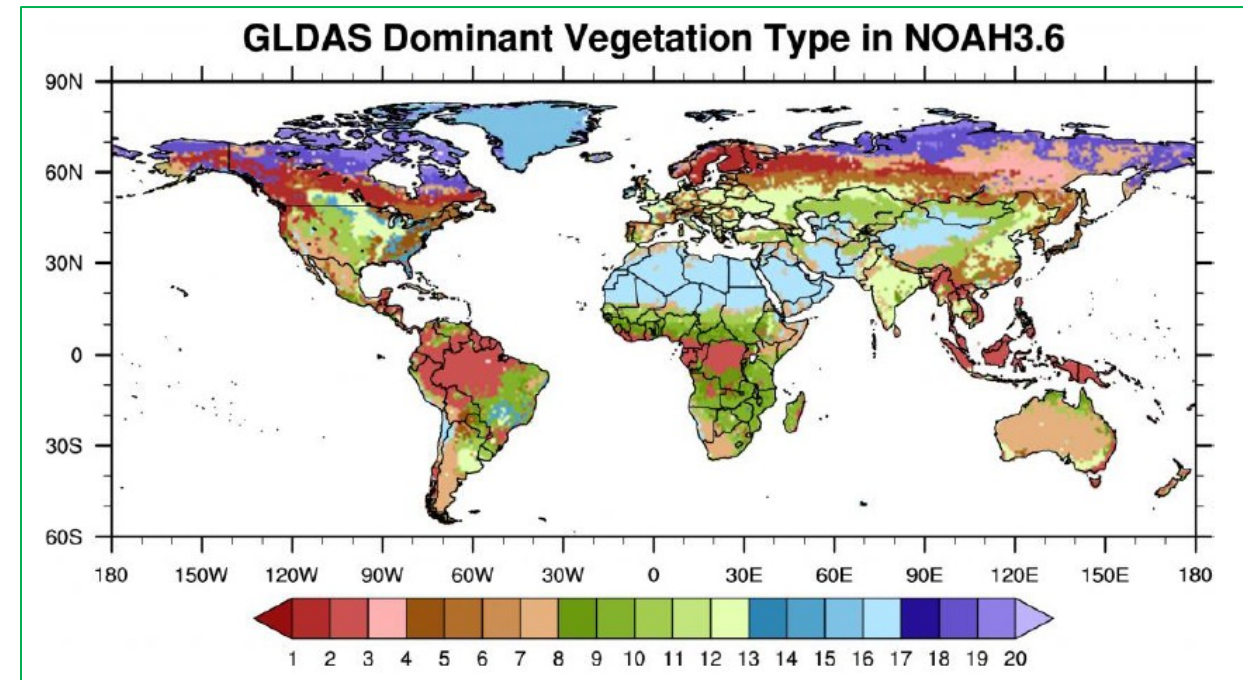


2. Dados Estáticos [Nedilson Junior-PGMET/Dra. Aline-UFPA]



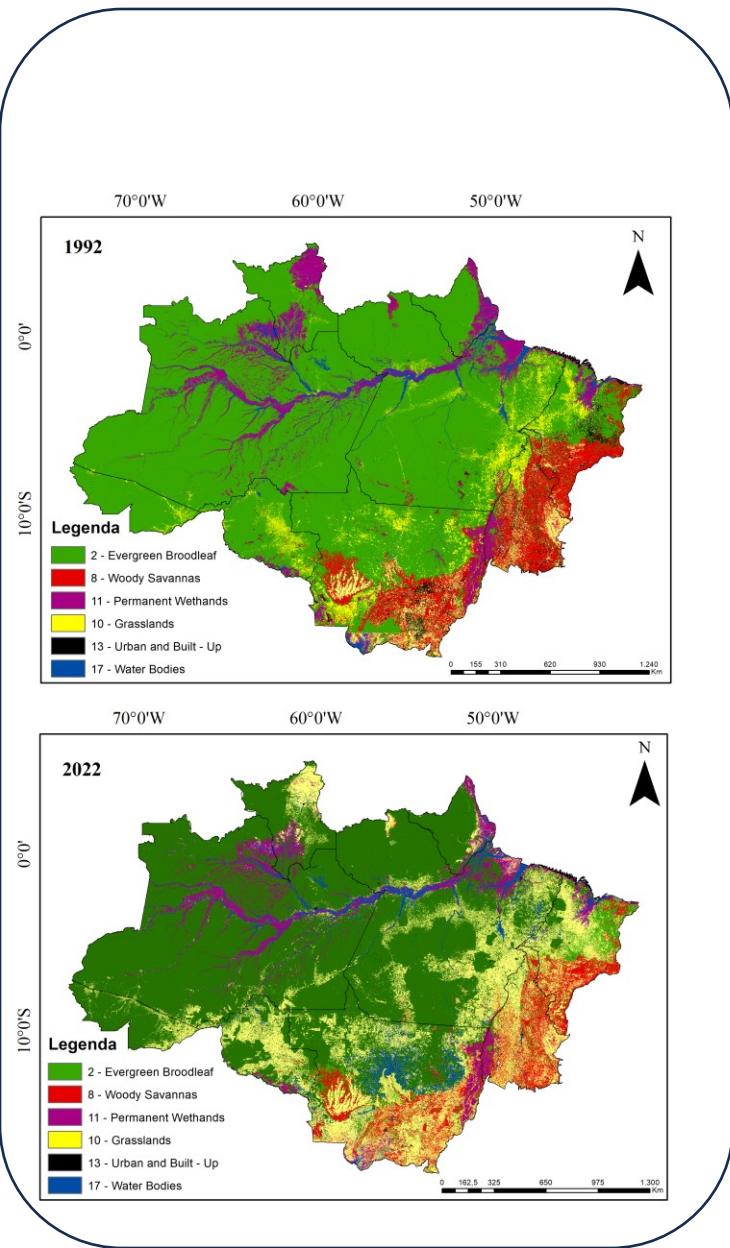
Model	Noah	Catchment	VIC, Catchment (daily)
Index	Modified MODIS IGBP Vegetation Type	Mosaic Vegetation Type	UMD Vegetation Type
0	missing value	missing value	missing value
1	Evergreen Needleleaf Forest	Broadleaf Evergreen Trees	Evergreen Needleleaf Forest
2	Evergreen Broadleaf Forest	Broadleaf Deciduous Trees	Evergreen Broadleaf Forest
3	Deciduous Needleleaf Forest	Needleleaf Trees	Deciduous Needleleaf Forest
4	Deciduous Broadleaf Forest	Grassland	Deciduous Broadleaf Forest
5	Mixed Forest	Broadleaf Shrubs, Bare Soil, Desert Soil	Mixed Cover
6	Closed Shrublands	Dwarf Trees	Woodland
7	Open Shrublands		Wooded Grasslands
8	Woody Savannas		Closed Shrublands
9	Savannas		Open Shrublands
10	Grassland		Grasslands
11	Permanent Wetland		Cropland
12	Cropland		Barren
13	Urban and Built-Up		Urban and Built-Up
14	Cropland/Natural Vegetation Mosaic		
15	Snow and Ice		
16	Barren or Sparsely Vegetated		
17	Ocean		
18	Wooded Tundra		
19	Mixed Tundra		
20	Bare Ground Tundra		

- GLDAS2/Noah Dominant Vegetation Type Data (NetCDF): [0.25 degree](#), [1 degree](#)
- GLDAS2.0 & GLDAS2.1/Catchment Dominant Vegetation Type Data (NetCDF): [1 degree](#)
- GLDAS2.0 & GLDAS2.2/Catchment Dominant Vegetation Type Data (NetCDF): [0.25 degree](#)
- GLDAS2/VIC Dominant Vegetation Type Data (NetCDF): [1 degree](#)





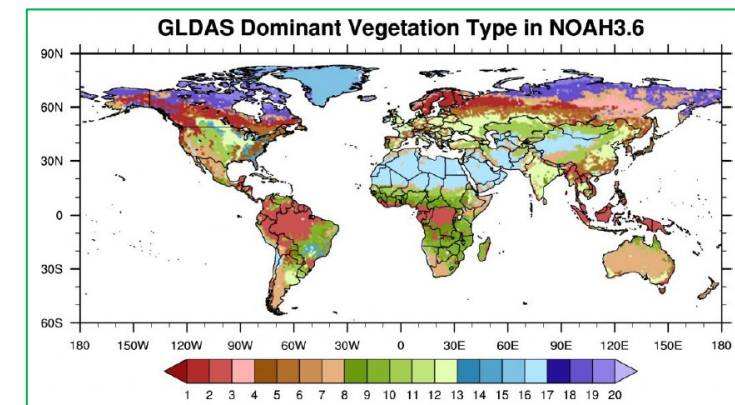
2. Dados Estáticos [Nedilson Junior-PGMET/Dra. Aline-UFPA]



1992

2022

QGIS



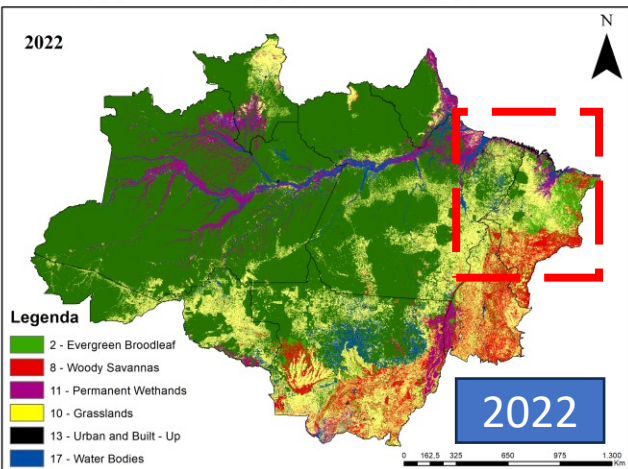
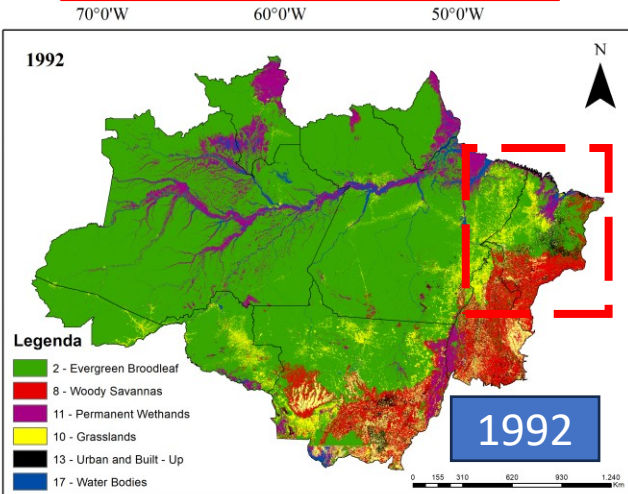
Base de
Dados
MapBiomias



2. Dados Estáticos [Nedilson Junior-PGMET /Dra. Aline -UFPA]



Classificação MapBiomias



modis_landuse_20class_30s

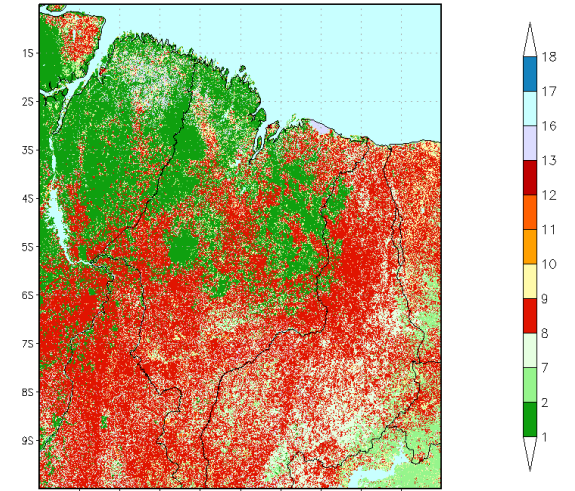
Floresta Amazonica
Wood Savannas

Arquivos intermediários

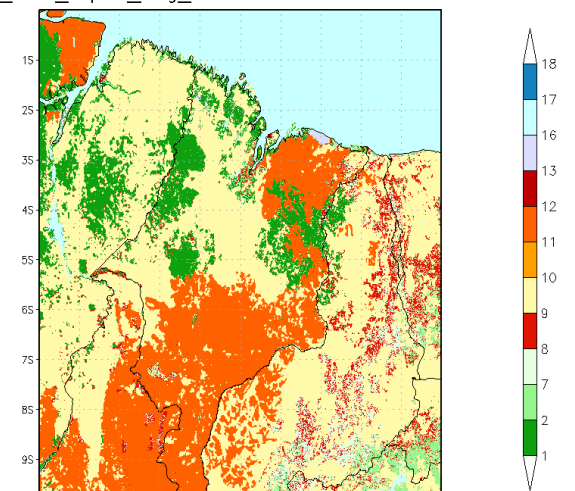
Area pastagem
Area agricola

modis_landuse_new_20class_30s

NE_Old_mpas_veg_15601-16800.09601-10800



NE_New_mpas_veg_15601-16800.09601-10800





3. Dados Oceânicos



pre_mpas_sst



3. Dados Oceânicos



sst.day.mean.1997.nc

ice.day.mean.1997.nc

```
dimensions:
  time = UNLIMITED ; // (365 currently)
  lat = 720 ;
  lon = 1440 ;
variables:
  double time(time) ;
    time:long_name = "Time" ;
    time:units = "days since 1800-01-01 00:00:00" ;
    time:delta_t = "0000-00-01 00:00:00" ;
    time:avg_period = "0000-00-01 00:00:00" ;
    time:axis = "T" ;
    time:actual_range = 71953., 72317. ;
  float lat(lat) ;
    lat:long_name = "Latitude" ;
    lat:standard_name = "latitude" ;
    lat:units = "degrees_north" ;
    lat:actual_range = -89.875f, 89.875f ;
    lat:axis = "Y" ;
  float lon(lon) ;
    lon:long_name = "Longitude" ;
    lon:standard_name = "longitude" ;
    lon:units = "degrees_east" ;
    lon:actual_range = 0.125f, 359.875f ;
    lon:axis = "X" ;
  float sst(time, lat, lon) ;
    sst:long_name = "Daily Sea Surface Temperature" ;
    sst:units = "degC" ;
    sst:valid_range = -3.f, 45.f ;
    sst:missing_value = -9.96921e+36f ;
    sst:precision = 2.f ;
    sst:dataset = "NOAA High-resolution Blended Analysis" ;
    sst:var_desc = "Sea Surface Temperature" ;
    sst:level_desc = "Surface" ;
    sst:statistic = "Mean" ;
    sst:parent_stat = "Individual Observations" ;
    sst:actual_range = -1.8f, 34.56f ;
```



3. Dados Oceânicos



sst.day.mean.1997.nc

ice.day.mean.1997.nc

```
dimensions:
  time = UNLIMITED ; // (365 currently)
  lat = 720 ;
  lon = 1440 ;
variables:
  double time(time) ;
    time:long_name = "Time" ;
    time:units = "days since 1800-01-01 00:00:00" ;
    time:delta_t = "0000-00-01 00:00:00" ;
    time:avg_period = "0000-00-01 00:00:00" ;
    time:axis = "T" ;
    time:actual_range = 71953., 72317. ;
  float lat(lat) ;
    lat:long_name = "Latitude" ;
    lat:standard_name = "latitude" ;
    lat:units = "degrees_north" ;
    lat:actual_range = -89.875f, 89.875f ;
    lat:axis = "Y" ;
  float lon(lon) ;
    lon:long_name = "Longitude" ;
    lon:standard_name = "longitude" ;
    lon:units = "degrees_east" ;
    lon:actual_range = 0.125f, 359.875f ;
    lon:axis = "X" ;
  float icec(time, lat, lon) ;
    icec:long_name = "Daily Sea Ice Concentration" ;
    icec:units = "percent" ;
    icec:valid_range = 0.f, 1.f ;
    icec:missing_value = -9.96921e+36f ;
    icec:precision = 2.f ;
    icec:dataset = "NOAA High-resolution Blended Analysis" ;
    icec:var_desc = "Sea Ice Concentration" ;
    icec:level_desc = "Surface" ;
    icec:statistic = "Mean" ;
```



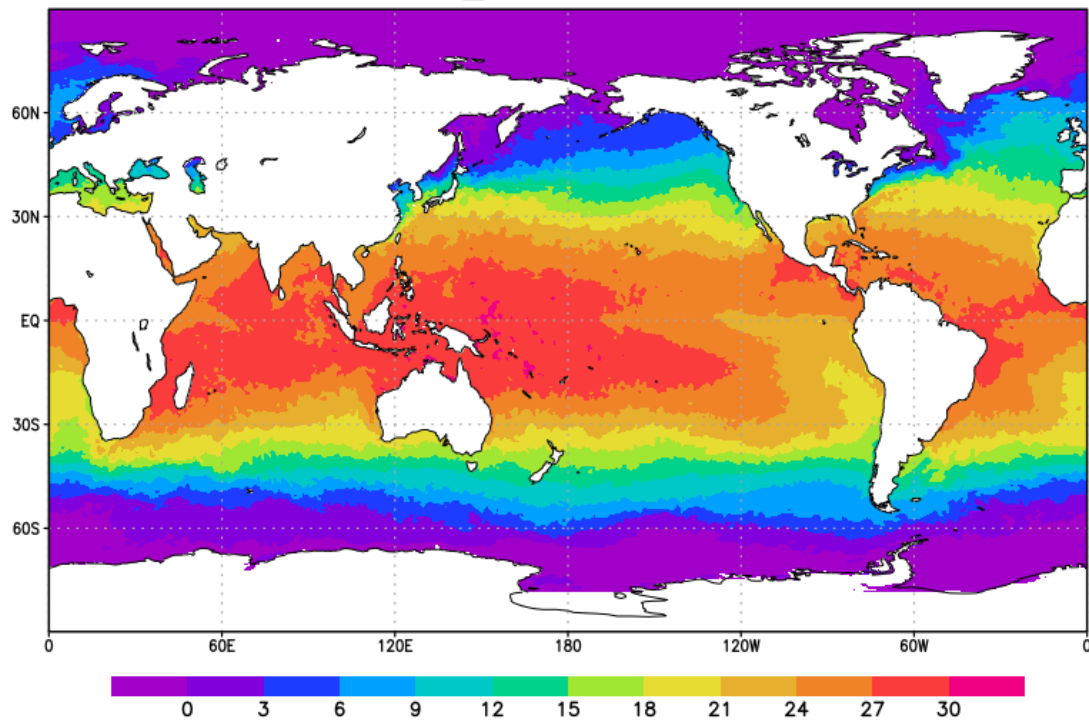
3. Dados Oceânicos



sst.day.mean.1997.nc

ice.day.mean.1997.nc

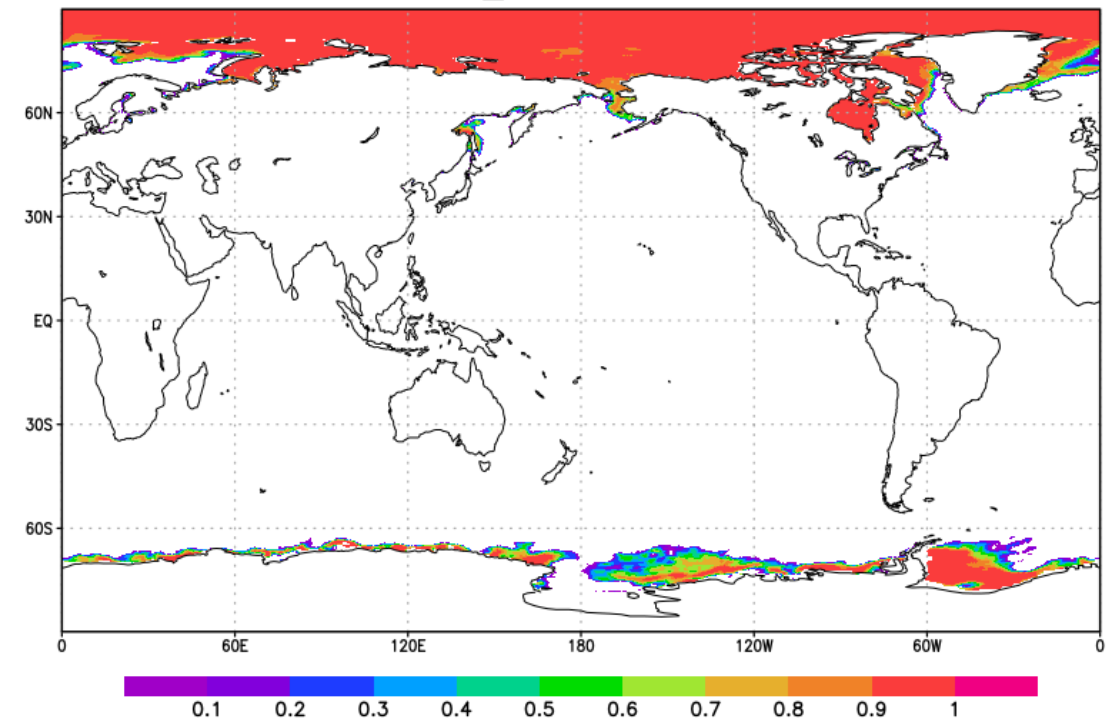
SST_00Z01JAN1997



GrADS/COLA

2024-04-29-14:17

ICEC_00Z01JAN1997



GrADS/COLA

2024-04-29-14:19



3. Dados Oceânicos



sst.day.mean.1997.nc

ice.day.mean.1997.nc

pre_mpas_sst

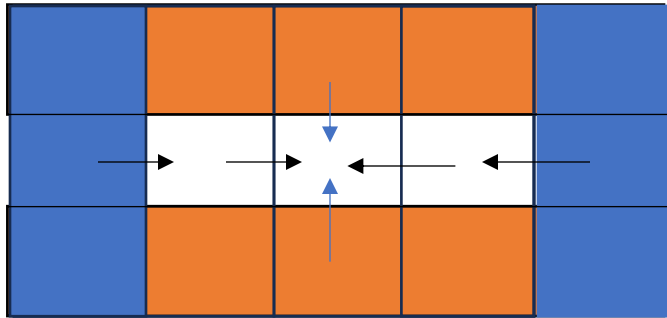
SST:1997-01-01_00
...
SST:1997-07-20_00
....
SST:1998-12-19_00

init_atmosphere_model

x1.40962.sfc_update.nc



3. Dados Oceânicos



SST:1997-01-01_00

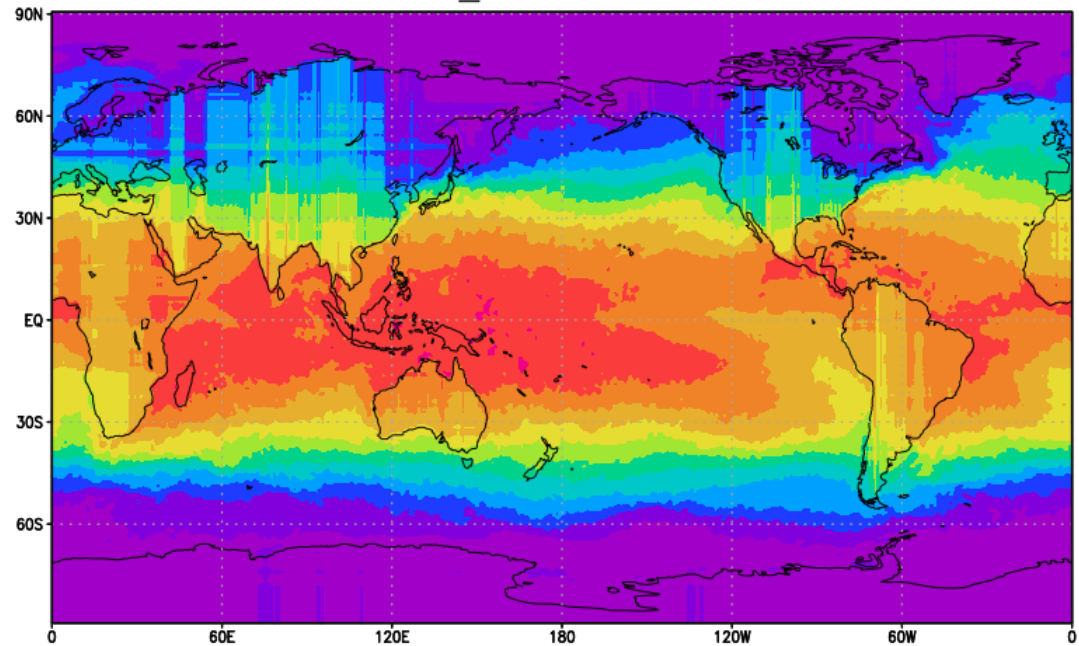
...

SST:1997-07-20_00

.....

SST:1998-12-19_00

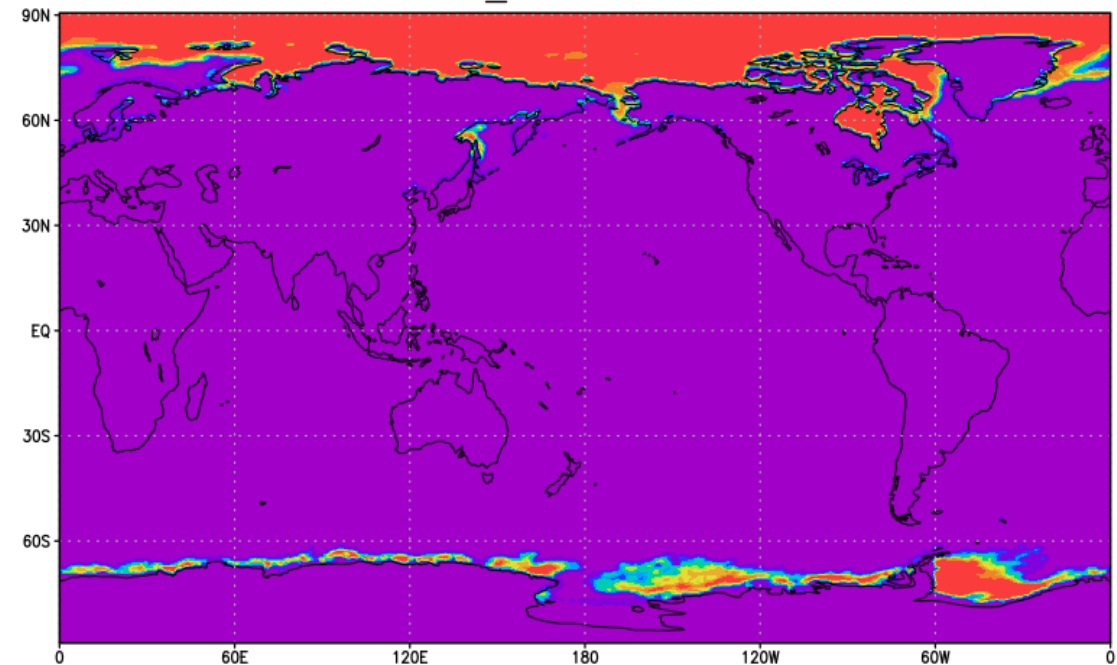
SST_00Z01JAN1997



GrADS/COLA

2024-04-29-14:30

ICEC_00Z01JAN1997



GrADS/COLA

2024-04-29-14:31



3. Dados Oceânicos



sst.day.mean.1997.nc

ice.day.mean.1997.nc

pre_mpas_sst

SST:1997-01-01_00
...
SST:1997-07-20_00
.....
SST:1998-12-19_00

init_atmosphere_model

x1.40962.sfc_update.nc



3. Dados Oceânicos



x1.40962.sfc_update.nc

```
dimensions:
  StrLen = 64 ;
  Time = UNLIMITED ; // (1 currently)
  nCells = 40962 ;
variables:
  char xtime(Time, StrLen) ;
    xtime:units = "YYYY-MM-DD_hh:mm:ss" ;
    xtime:long_name = "Model valid time" ;
  float Time(Time) ;
    Time:units = "seconds since 1997-01-01 00:00:00" ;
    Time:long_name = "CF-compliant valid time" ;
    Time:standard_name = "time" ;
  float sst(Time, nCells) ;
    sst:units = "K" ;
    sst:long_name = "sea-surface temperature" ;
  float xice(Time, nCells) ;
    xice:units = "unitless" ;
    xice:long_name = "fractional area coverage of sea-ice" ;
```

`ncdump -v sst x1.40962.sfc_update.nc`

`ncdump -v icec x1.40962.sfc_update.nc`



3. Dados Oceânicos



```
ncdump -v sst x1.40962.sfc_update.nc
```

data:

```
sst =  
  295.0165, 294.2157, 271.43, 295.2659,  
297.7201, 290.1828, 295.6052,  
  297.5935, 271.35, 296.3419,  
297.4984, 293.4944, 290.1667, 301.98,  
  302.1353, 289.4696, 275.1285,  
300.3037, 296.7001, 292.1894, 273.2,  
  274.5934, 278.1795, 280.2252,  
300.3073, 302.9193, 295.4616, 273.3606,  
  296.52, 302.99, 273.8472, 295.4381,  
303.5397, 294.2111, 276.9515,  
  302.0513, 291.3456, 273.8145,  
295.7006, 302.9274, 276.1938, 294.1543,
```

```
ncdump -v icec x1.40962.sfc_update.nc
```

data:

```
xice =  
  0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
  0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
  0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
  0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

Check se realmente tem
valores de icec.
Não pode ser somente
zeros



3. Dados Meteorológicos



Chopping_Monan



3. Dados Meteorológicos

gdas1.T00Z.sfcan1.netcdf.2022080300

gdas1.T00Z.atman1.netcdf.2022080300

Chopping_Monan

FILE3:2024-04-03_00

init_atmosphere_model

x1.40962.init.nc



3. Dados Meteorológicos

gdas1.T00Z.sfcan1.netcdf.2022080300

gdas1.T00Z.atman1.netcdf.2022080300

Adaptação para a
grade gaussiana

Surf_Netcdf_Parameter.f90

Atmos_Netcdf_Parameter.f90

Chopping_Monan

Níveis isobáricos

mpas_VerticalInterpolation.F

Níveis em coordenada hibrida sigma/pressão

VerticalInterpolation.f90

FILE3:2024-04-03_00



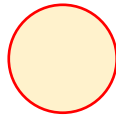
3. Dados Meteorológicos



gdas1.T00Z.sfcan1.netcdf.2022080300

gdas1.T00Z.atman1.netcdf.2022080300

Chopping_Monan



Status Atual
Checando a decomposição
de domínio MPI.

FILE3:2024-04-03_00



3. Dados Meteorológicos



gdas1.T00Z.sfcan1.netcdf.2022080300

gdas1.T00Z.atman1.netcdf.2022080300

Chopping_Monan

FILE3:2024-04-03_00

```
module met_data_module

  ! Derived types

  type met_data
    integer :: version
    integer :: nx
    integer :: ny
    integer :: iproj
    real :: xfcst
    real :: xlv1
    real :: startlat
    real :: startlon
    real :: starti
    real :: startj
    real :: deltalat
    real :: deltalon
    real :: dx
    real :: dy
    real :: xlonc
    real :: centerlat
    real :: centerlon
    real :: pole_lat
    real :: pole_lon
    real :: truelat1
    real :: truelat2
    real :: earth_radius
    real, pointer, dimension(:,,:) :: slab
    logical :: is_wind_grid_rel
    character (len=9) :: field
    character (len=24) :: hdate
    character (len=25) :: units
    character (len=32) :: map_source
    character (len=46) :: desc
  end type met_data

end module met_data_module
```




4. Padronização das saídas no modelo MONAN

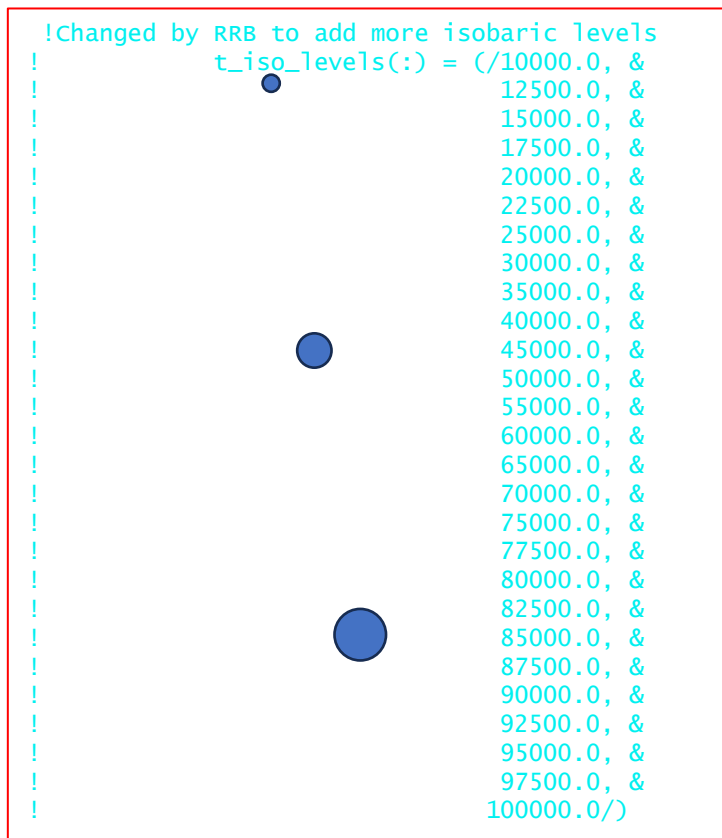


Diagnostics/

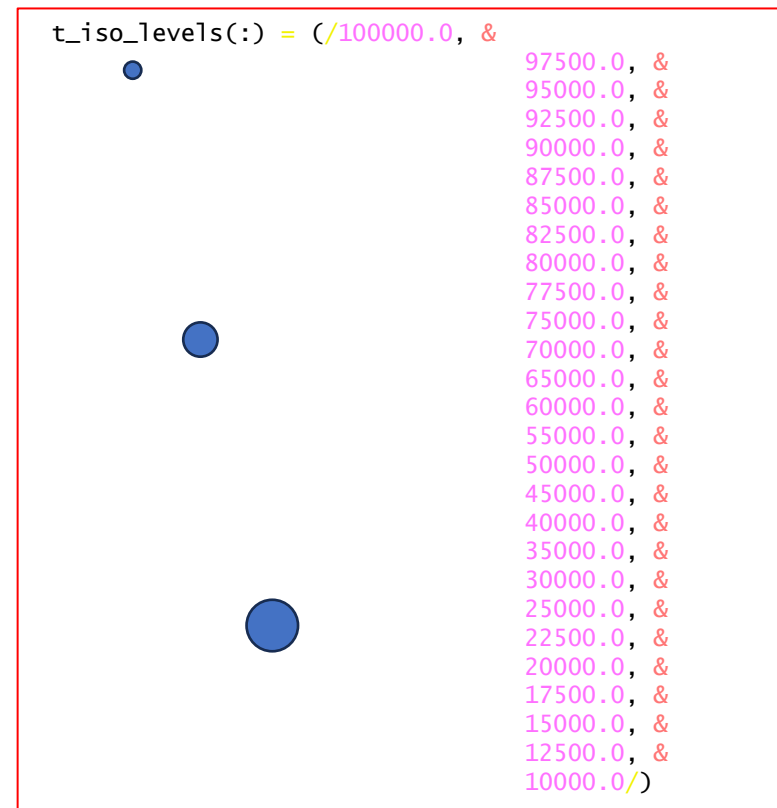
Makefile	README
mpas_atm_diagnostics_manager.F	Registry_diagnostics.xml
mpas_atm_diagnostic_template.F	Registry_template.xml
mpas_isobaric_diagnostics.F	Registry_isobaric.xml
mpas_lsm_diagnostics.F	Registry_lsm.xml
mpas_sfclayer_diagnostics.F	Registry_sfclayer.xml
mpas_pbl_diagnostics.F	Registry_pbl.xml
mpas_cloud_diagnostics.F	Registry_cloud.xml
mpas_radiation_diagnostics.F	Registry_radiation.xml
mpas_convective_diagnostics.F	Registry_convective.xml
mpas_microphysics_diagnostics.F	Registry_microphysics.xml
mpas_pv_diagnostics.F	Registry_pv.xml
mpas_soundings.F	Registry_soundings.xml
mpas_VerticalInterpolation.F	
mpas_atm_diagnostics_utils.F	



4. Padronização das saídas no modelo MONAN



Atrapalha o pós-processamento. É invertido em relação aos níveis do modelo



Ideal para o pós-processamento. Os níveis verticais (modelo e isobárico) tem a mesma distribuição



4. Padronização das saídas no modelo MONAN



```
do icell=1,nCells
  do k=1,nVertLevels
    kk = nVertLevels+1-k
    press_in(icell,kk) = pressure(k,icell) * 100.0
  end do
end do

if (need_t_isobaric .or. need_qv_isobaric .or. need_rh_isobaric) then
  ! Additional temperature levels for vortex tracking
  do icell=1,nCells
    do k=1,nVertLevels
      kk = nVertLevels+1-k
      field_in(icell,kk) = temperature(k,icell)
    end do
  end do

  call interp_tofixed_pressure(nCells, nVertLevels, nIsoLevelst,&
    press_in, field_in, press_interp, field_interp)

  do k=1,nIsoLevelst
    t_isobaric(k,1:nCells) = field_interp(1:nCells,k)
  end do
```

```
do icell=1,nCells
  do k=1,nVertLevels

    press_in(icell,k) = pressure(k,icell) * 100.0
  end do
end do

if (need_t_isobaric .or. need_qv_isobaric .or. need_rh_isobaric) then
  ! Additional temperature levels for vortex tracking
  do icell=1,nCells
    do k=1,nVertLevels

      field_in(icell,k) = temperature(k,icell)
    end do
  end do

  call interp_tofixed_pressure(nCells, nVertLevels, nIsoLevelst,&
    press_in, field_in, press_interp, field_interp)

  do k=1,nIsoLevelst
    t_isobaric(k,1:nCells) = field_interp(1:nCells,k)
  end do
```

Lembrando que nos arquivos history.xxxxxx e diag.xxxxxx podem conter variáveis em níveis do modelo e níveis isobáricos.

Então é importante que a distribuição dos níveis (sfc-top) seja igual para os níveis isobáricos e os níveis do modelo



5. Impacto das saídas do modelo no convert_mpas

convert_mpas



5. Impacto das saídas do modelo no convert_mpas

- **Padronização do convert_mpas**
- **Níveis isobáricos e níveis do modelo com a mesma distribuição (sfc-top)**



Pauta da 3 Reunião Geral do Modelo MONAN



- **Avanços na Análise Modal no MONAN [Bonatti].**
- **Avanços na modelagem de Superfície [A. Manzi].**
- **Parametrização de Turbulência [Paulo Kubota]**
- **Parametrização de Radiação [Paulo Kubota]**
- **Parametrização de Microfísica [Enver/Jorge Gomes]**
- **LES_SAM/Convecção Rasa [Silvio Nilo/Jhonatan]**
- **Convecção Grell-Freitas [Saulo Freitas]**



- Avanços na Análise Modal no MONAN [Bonatti].



Slides Grupo Atmos



- Avanços na modelagem de Superfície [A. Manzi].



Slides Grupo Superfície



Implementação da parametrização YSU BL

Implementation of a revised SBL scheme in the YSU BL package

“Propõem-se a adição da contribuição do fluxo de calor devido o entranhamento no topo da Camada Limite Planetária (PBL) ”

Noh et al 2003, modificaram o esquema original

$$\frac{\partial C}{\partial t} = \frac{\partial [-\overline{w'c'}]}{\partial z}$$

$$-\overline{w'c'} = K_c \left(\frac{\partial C}{\partial z} - \gamma_c \right) - \overline{(w'c')}_h \left(\frac{z}{h} \right)^n$$

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial z} \left[K_c \left(\frac{\partial C}{\partial z} - \gamma_c \right) \right]$$

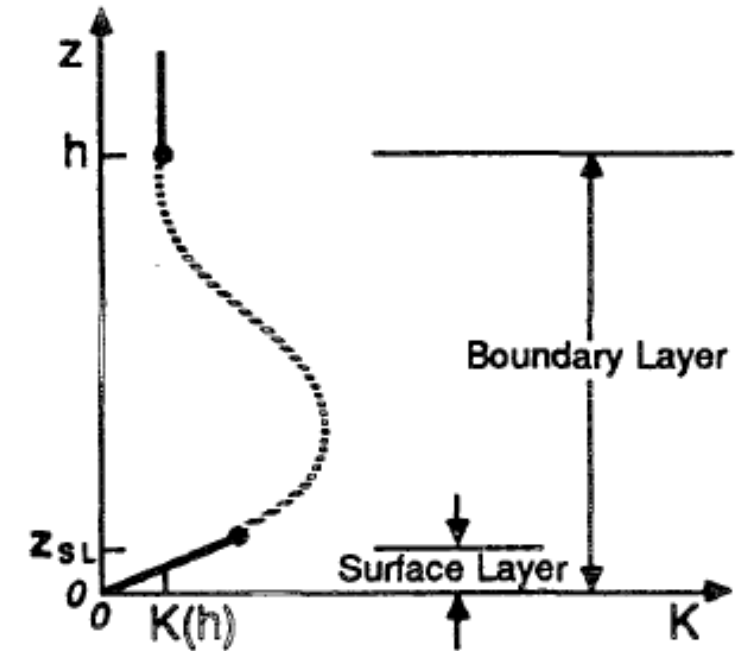


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

"Aqui obtivemos a constante empírica n como $n = 3$ com base na comparação entre os resultados do modelo e dados de LES, mas descobriu-se que os resultados são altamente insensíveis à escolha de n . Por exemplo, simulações com valores substancialmente diferentes de n , como $n = 1$ ou 5 , também apresentam resultados semelhantes, pois a contribuição do novo termo é desprezado $\left(\frac{z}{h} \right)^n$, exceto próximo ao topo da Camada Limite Planetária (PBL $z = h \Rightarrow \frac{z}{h} = 1$)."

(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite

Implementation of a revised SBL scheme in the YSU BL package

"O **esquema YSU antigo** usava uma teoria K modificada, com um termo de contra-gradiente adicional que incorpora a contribuição de vórtices de grande escala ao fluxo total."

Hong SY et al 2006. **Noh et al 2003**, modificaram o esquema original introduzindo o **efeito de entranhamento**

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial z} \left\{ K_c \left(\frac{\partial C}{\partial z} - \gamma_c \right) - \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right\}$$

$$\frac{\partial C}{\partial t} = \left\{ \frac{\partial}{\partial z} K_c \frac{\partial C}{\partial z} - \frac{\partial}{\partial z} \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right) \right\}$$

$\overline{(w'c')}_h$ é o fluxo na camada de inversão

"A fórmula mantém o conceito básico de Hong SY et al 1996, mas inclui um **termo assintótico de fluxo de entrada na camada de inversão** $\overline{(w'c')}_h \left(\frac{z}{h} \right)^3$."

Neste caso a altura da (PBL) h é definida como o nível em que o fluxo mínimo ocorre na camada de inversão, enquanto em Hong SY et al 1996 é definida como o nível em que a mistura turbulenta da camada limite diminui.

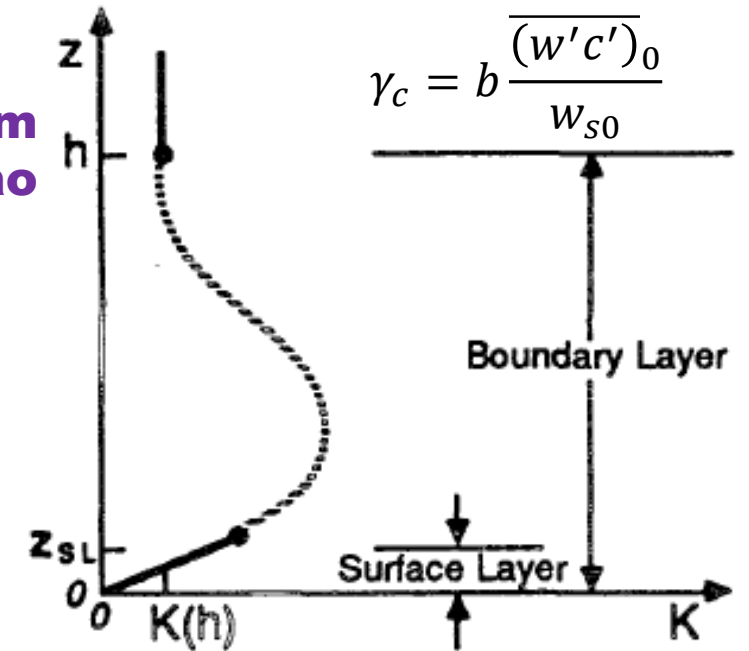


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

Implementation of a revised SBL scheme in the YSU BL package

$$\frac{\partial C}{\partial t} = \left\{ \frac{\partial}{\partial z} K_c \frac{\partial C}{\partial z} - \frac{\partial}{\partial z} \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right) \right\}$$

$$\int_k^{k+1} \frac{\partial C}{\partial t} dz = \left\{ \int_k^{k+1} \frac{\partial}{\partial z} K_c \frac{\partial C}{\partial z} dz - \int_k^{k+1} \frac{\partial}{\partial z} \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right) dz \right\}$$

$$\frac{\partial}{\partial t} \int_k^{k+1} C dz = \left\{ K_c \frac{\partial C}{\partial z} \Big|_k^{k+1} - \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right) \Big|_k^{k+1} \right\}$$

$$\frac{\partial \bar{C}}{\partial t} = \left\{ \left(K_c \frac{\partial C}{\partial z} \right)^{k+1} - \left(K_c \frac{\partial C}{\partial z} \right)^k - \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^{k+1} + \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^k \right\}$$

$$\frac{\partial \bar{C}}{\partial t} - \left(K_c \frac{\partial C}{\partial z} \right)^{k+1} + \left(K_c \frac{\partial C}{\partial z} \right)^k = \left\{ - \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^{k+1} + \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^k \right\}$$

Implementation of a revised SBL scheme in the YSU BL package

$$\frac{\partial \bar{c}}{\partial t} - \left(K_c \frac{\partial \bar{c}}{\partial z} \right)^{k+1} + \left(K_c \frac{\partial \bar{c}}{\partial z} \right)^k = - \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^{k+1} + \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^k$$

$$\frac{C_k^{n+1} - C_k^{n-1}}{2\Delta t} - \left(K_{c,k+1}^n \frac{C_{k+1}^{n+1} - C_k^{n+1}}{\Delta z} \right)^{k+1} + \left(K_{c,k}^n \frac{C_k^{n+1} - C_{k-1}^{n+1}}{\Delta z} \right)^k = - \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^{k+1} + \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^k$$

$$\frac{C_k^{n+1} - C_k^{n-1}}{2\Delta t} - K_{c,k+1}^n \frac{C_{k+1}^{n+1} - C_k^{n+1}}{\Delta z} + K_{c,k}^n \frac{C_k^{n+1} - C_{k-1}^{n+1}}{\Delta z} = - \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^{k+1} + \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^k$$

$$C_k^{n+1} - C_k^{n-1} - \frac{K_{c,k+1}^n 2\Delta t}{\Delta z} (C_{k+1}^{n+1} - C_k^{n+1}) + \frac{K_{c,k}^n 2\Delta t}{\Delta z} (C_k^{n+1} - C_{k-1}^{n+1}) = -2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^{k+1} + 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^k$$

$$C_k^{n+1} - C_k^{n-1} + \left(-\frac{K_{c,k+1}^n 2\Delta t}{\Delta z} C_{k+1}^{n+1} + \frac{K_{c,k+1}^n 2\Delta t}{\Delta z} C_k^{n+1} \right) + \left(\frac{K_{c,k}^n 2\Delta t}{\Delta z} C_k^{n+1} - \frac{K_{c,k}^n 2\Delta t}{\Delta z} C_{k-1}^{n+1} \right) = -2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^{k+1} + 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^k$$

$$-\frac{K_{c,k+1}^n 2\Delta t}{\Delta z} C_{k+1}^{n+1} + \frac{K_{c,k+1}^n 2\Delta t}{\Delta z} C_k^{n+1} + \frac{K_{c,k}^n 2\Delta t}{\Delta z} C_k^{n+1} + C_k^{n+1} + \left(-\frac{K_{c,k}^n 2\Delta t}{\Delta z} C_{k-1}^{n+1} \right) = C_k^{n-1} - 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^{k+1} + 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^k$$



(1)Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite



Implementation of a revised SBL scheme in the YSU BL package

$$-\frac{K_{c_{k+1}}^n 2\Delta t}{\Delta z} C_{k+1}^{n+1} + \frac{K_{c_{k+1}}^n 2\Delta t}{\Delta z} C_k^{n+1} + \frac{K_{c_k}^n 2\Delta t}{\Delta z} C_k^{n+1} + C_k^{n+1} + \left(-\frac{K_{c_k}^n 2\Delta t}{\Delta z} C_{k-1}^{n+1}\right) = C_k^{n-1} - 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h}\right)^3\right)^{k+1} + 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h}\right)^3\right)^k$$

$$-\frac{K_{c_{k+1}}^n 2\Delta t}{\Delta z} C_{k+1}^{n+1} + \frac{K_{c_{k+1}}^n 2\Delta t}{\Delta z} C_k^{n+1} + \frac{K_{c_k}^n 2\Delta t}{\Delta z} C_k^{n+1} + C_k^{n+1} + \left(-\frac{K_{c_k}^n 2\Delta t}{\Delta z} C_{k-1}^{n+1}\right) = C_k^{n-1} - 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h}\right)^3\right)^{k+1} + 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h}\right)^3\right)^k$$

$$\left(-\frac{K_{c_k}^n 2\Delta t}{\Delta z}\right) C_{k-1}^{n+1} + \left(\frac{K_{c_{k+1}}^n 2\Delta t}{\Delta z} + \frac{K_{c_k}^n 2\Delta t}{\Delta z} + 1\right) C_k^{n+1} - \frac{K_{c_{k+1}}^n 2\Delta t}{\Delta z} C_{k+1}^{n+1} = C_k^{n-1} - 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h}\right)^3\right)^{k+1} + 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h}\right)^3\right)^k$$

$$(AA)C_{k-1}^{n+1} + BB C_k^{n+1} - DD C_{k+1}^{n+1} = C_k^{n-1} + f1_k - f1_{k+1}$$

$$(AA)C_{k-1}^{n+1} + BB C_k^{n+1} - DD C_{k+1}^{n+1} = FF_k^{n-1}$$

$$\begin{bmatrix} BB & DD & 0 & 0 & 0 & 0 \\ AA & BB & DD & 0 & 0 & 0 \\ 0 & AA & BB & DD & 0 & 0 \\ 0 & 0 & AA & BB & DD & 0 \\ 0 & 0 & 0 & AA & BB & DD \\ 0 & 0 & 0 & 0 & AA & BB \end{bmatrix} * \begin{bmatrix} C_1^{n+1} \\ C_2^{n+1} \\ C_3^{n+1} \\ C_4^{n+1} \\ C_5^{n+1} \\ C_6^{n+1} \end{bmatrix} = \begin{bmatrix} FF_1^{n-1} \\ FF_2^{n-1} \\ FF_3^{n-1} \\ FF_4^{n-1} \\ FF_5^{n-1} \\ FF_6^{n-1} \end{bmatrix} \leftarrow \frac{\partial C}{\partial t} = \frac{\partial [-\overline{w'c'}]}{\partial z}$$



Implementação da parametrização de Melor Yamada Nino Nakanishi com fluxo de Massa

$$\overline{w'\phi'} = -K \frac{\partial \phi}{\partial z} + M_{\phi,u}(\phi_u - \phi) - M_{\phi,d}(\phi_d - \phi)$$

$$K_\phi = c_\phi L_k \sqrt{\bar{e}}$$

$$K_\phi = S_\phi L_k \sqrt{2\bar{e}}$$



Linearização da Equações de Navier Stokes

5.2 TURBULENT KINETIC ENERGY

A equação de Navier Stokes: conservação de momentum

$$\begin{matrix} i=1,2,3 \\ j=1,2,3 \\ K=1,2,3 \end{matrix} \quad \frac{Du_i}{Dt} = \frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho_0} \frac{\partial P}{\partial x_i} - \frac{\rho}{\rho_0} g \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j u_k + \nu \left(\frac{\partial^2 u_i}{\partial x_j^2} \right)$$

$$u_i = \bar{u}_i + u_i'$$

Aplique a Média de Reynolds na Variáveis

$$\frac{\partial(\bar{u}_i + u_i')}{\partial t} + (\bar{u}_j + u_j') \frac{\partial(\bar{u}_i + u_i')}{\partial x_j} = -\frac{1}{\rho_0} \frac{\partial(\bar{P} + P')}{\partial x_i} - g \frac{(\rho + \rho')}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k + u_k') + \nu \left(\frac{\partial^2(\bar{u}_i + u_i')}{\partial x_j^2} \right)$$

Expanda os termos

$$\begin{aligned} & \frac{\partial(\bar{u}_i)}{\partial t} + \frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + (u_j') \frac{\partial(\bar{u}_i)}{\partial x_j} + (u_j') \frac{\partial(u_i')}{\partial x_j} = \\ & -\frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - \frac{1}{\rho_0} \frac{\partial(P')}{\partial x_i} - g \frac{\rho}{\rho_0} \delta_{i3} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2} + \nu \frac{\partial^2(u_i')}{\partial x_j^2} \end{aligned}$$



A equação de Navier Stokes: conservação de momentum

$$\begin{aligned} \frac{\partial(\bar{u}_i)}{\partial t} + \frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + (u_j') \frac{\partial(\bar{u}_i)}{\partial x_j} + (u_j') \frac{\partial(u_i')}{\partial x_j} = \\ - \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - \frac{1}{\rho_0} \frac{\partial(P')}{\partial x_i} - g \frac{\rho}{\rho_0} \delta_{i3} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2} + \nu \frac{\partial^2(u_i')}{\partial x_j^2} \end{aligned}$$

Separa os termos na equação acima

$$\begin{aligned} \frac{\partial(\bar{u}_i)}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + (u_j') \frac{\partial(\bar{u}_i)}{\partial x_j} + (u_j') \frac{\partial(u_i')}{\partial x_j} = \\ - \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - g \frac{\rho}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2} \end{aligned}$$



Linearização da Equações de Navier Stokes

A equação de Navier Stokes: conservação de momentum

$$\frac{\partial(\bar{u}_i)}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} + (u_j') \frac{\partial(\bar{u}_i)}{\partial x_j} + \boxed{(u_j') \frac{\partial(u_i')}{\partial x_j}} =$$

$$-\frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - g \frac{\rho}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2}$$

$$\frac{\partial(\bar{u}_i)}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} + (u_j') \frac{\partial(\bar{u}_i)}{\partial x_j} + \boxed{\frac{\partial(u_j' u_i')}{\partial x_j} - (u_i') \frac{\partial(u_j')}{\partial x_j}}$$

$$= -\frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - g \frac{\rho}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2}$$

Aplique as media de Reynolds

$$\frac{\partial(\bar{\bar{u}}_i)}{\partial t} + (\bar{\bar{u}}_j) \frac{\partial(\bar{\bar{u}}_i)}{\partial x_j} + \frac{\partial(\overline{u_j' u_i'})}{\partial x_j} - \overline{(u_i')} \frac{\partial(\overline{u_j'})}{\partial x_j} + (\overline{u_j'}) \frac{\partial(\bar{\bar{u}}_i)}{\partial x_j}$$

$$= -\frac{1}{\rho_0} \frac{\partial(\bar{\bar{P}})}{\partial x_i} - g \frac{\bar{\rho}}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{\bar{u}}_k) + \nu \frac{\partial^2(\bar{\bar{u}}_i)}{\partial x_j^2}$$



Linearização da Equações de Navier Stokes



A equação de Navier Stokes: conservação de momentum

$$\begin{aligned} \frac{\partial(\bar{u}_i)}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} + \frac{\partial(\overline{u_j' u_i'})}{\partial x_j} - \overline{(u_i')} \frac{\partial(\overline{u_j'})}{\partial x_j} + (\overline{u_j'}) \frac{\partial(\bar{u}_i)}{\partial x_j} \\ = - \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - g \frac{\bar{\rho}}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2} \end{aligned}$$

Aplique as considerações da media de Reynolds

$$\bar{\bar{u}}_i = \bar{u}_i$$

$$\overline{u_j' u_i'} \neq 0$$

$$\overline{u_j'} = 0$$

$$\overline{u_i'} = 0$$

$$\frac{\partial(\bar{u}_i)}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} = - \frac{\partial(\overline{u_j' u_i'})}{\partial x_j} - \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - g \frac{\bar{\rho}}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2}$$



- **Equação primitiva não linear para o escoamento da Atmosfera**

$$\frac{Du_i}{Dt} = \frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho_0} \frac{\partial P}{\partial x_i} - \frac{\rho}{\rho_0} g \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j u_k + \nu \left(\frac{\partial^2 u_i}{\partial x_j^2} \right)$$

- **Equação Governante Linearizada do Estado Médio do escoamento da Atmosfera**

$$\frac{\partial(\bar{u}_i)}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} = -\frac{\partial(\overline{u_j' u_i'})}{\partial x_j} - \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - g \frac{\bar{\rho}}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2}$$

5.2 TURBULENT KINETIC ENERGY A equação de Navier Stokes: conservação de momentum

$$\begin{aligned} \frac{\partial(\bar{u}_i)}{\partial t} + \frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + (u_j') \frac{\partial(\bar{u}_i)}{\partial x_j} + (u_j') \frac{\partial(u_i')}{\partial x_j} = \\ - \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - \frac{1}{\rho_0} \frac{\partial(P')}{\partial x_i} - g \frac{\rho}{\rho_0} \delta_{i3} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2} + \nu \frac{\partial^2(u_i')}{\partial x_j^2} \end{aligned}$$

Separe os termos com perturbação que se cancelariam com a media de Reynolds

$$\begin{aligned} \frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + \boxed{(u_j') \frac{\partial(\bar{u}_i)}{\partial x_j}} + \boxed{(u_j') \frac{\partial(u_i')}{\partial x_j}} \\ = - \frac{1}{\rho_0} \frac{\partial(P')}{\partial x_i} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(u_i')}{\partial x_j^2} \end{aligned}$$

$$\begin{aligned} \frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + \boxed{\frac{\partial(u_j' \bar{u}_i)}{\partial x_j} - (\bar{u}_i) \frac{\partial(u_j')}{\partial x_j}} + \boxed{\frac{\partial(u_j' u_i')}{\partial x_j} - (u_i') \frac{\partial(u_j')}{\partial x_j}} \\ = - \frac{1}{\rho} \frac{\partial(P')}{\partial x_i} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(u_i')}{\partial x_j^2} \end{aligned}$$



Linearização da Equações de Navier Stokes



5.2 TURBULENT KINETIC ENERGY A equação de Navier Stokes: conservação de momentum

$$\frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + \boxed{(u_j') \frac{\partial(\bar{u}_i)}{\partial x_j}} + \boxed{(u_j') \frac{\partial(u_i')}{\partial x_j}} = -\frac{1}{\rho_0} \frac{\partial(P')}{\partial x_i} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(u_i')}{\partial x_j^2}$$

Aplique a derivada do produto nos termos em destaque:

$$\begin{aligned} \frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + \boxed{\frac{\partial(u_j' \bar{u}_i)}{\partial x_j} - (\bar{u}_i) \frac{\partial(u_j')}{\partial x_j}} + \boxed{\frac{\partial(u_j' u_i')}{\partial x_j} - (u_i') \frac{\partial(u_j')}{\partial x_j}} \\ = -\frac{1}{\rho} \frac{\partial(P')}{\partial x_i} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(u_i')}{\partial x_j^2} \end{aligned}$$



Linearização da Equações de Navier Stokes



5.2 TURBULENT KINETIC ENERGY A equação de Navier Stokes: conservação de momentum

$$\begin{aligned} \frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + \frac{\partial(u_j' \bar{u}_i)}{\partial x_j} - (\bar{u}_i) \frac{\partial(u_j')}{\partial x_j} + \frac{\partial(u_j' u_i')}{\partial x_j} - (u_i') \frac{\partial(u_j')}{\partial x_j} \\ = -\frac{1}{\rho} \frac{\partial(P')}{\partial x_i} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(u_i')}{\partial x_j^2} \end{aligned}$$

Considere que a propriedade da equação da continuidade para o fluxo $\nabla \cdot \vec{V} = 0$

$$\nabla \cdot \vec{V}' = \frac{\partial(u_j')}{\partial x_j} = 0, \quad j = 1, 2, 3$$

$$\frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + \frac{\partial(u_j' \bar{u}_i)}{\partial x_j} + \frac{\partial(u_j' u_i')}{\partial x_j} = -\frac{1}{\rho_0} \frac{\partial(P')}{\partial x_i} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(u_i')}{\partial x_j^2}$$



Linearização da Equações de Navier Stokes



5.2 TURBULENT KINETIC ENERGY A equação de Navier Stokes: conservação de momentum

$$\frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + \frac{\partial(u_j' \bar{u}_i)}{\partial x_j} + \frac{\partial(u_j' u_i')}{\partial x_j} = -\frac{1}{\rho_0} \frac{\partial(P')}{\partial x_i} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(u_i')}{\partial x_j^2}$$



**Gorge_mellor_Analytic Prediction of the
Properties of Stratified Planetary Surface
Layers_atsc-1520-0469_1973**



1.1 A Teoria de Similaridade de Monin-Obukhov



As equações são apresentadas de uma forma geral considerável para que, em princípio, possam ser integradas para simular, por exemplo, uma camada limite planetária completa.

Aqui, no entanto, restringimos a atenção à região da superfície de fluxo constante, evitando assim, por enquanto, o considerável esforço computacional necessário para a camada completa.

No entanto, este é um primeiro passo lógico, uma vez que é possível comparar diretamente com os dados de fluxo constante de Businger et al. (1971) na forma de variáveis de similaridade de Monin-Obukhoff.



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

As equações de movimento para a velocidade média $\bar{u}_j$; e a temperatura potencial média $\bar{\theta}$, As barras superiores representam as médias do conjunto e os termos minúsculos, u'_k e θ' , são os componentes flutuantes da velocidade e da temperatura e são governados por

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0 \quad (1)$$

$$\frac{\partial \bar{u}_j}{\partial t} + \bar{u}_k \frac{\partial (\bar{u}_j)}{\partial x_k} + \bar{u}_j \left[\frac{\partial (\bar{u}_k)}{\partial x_k} \right] + \frac{\partial (\overline{u'_k u'_j})}{\partial x_k} + \epsilon_{j,k,l} f_k \bar{u}_l = - \frac{\partial P}{\partial x_j} - g_j \beta \bar{\theta} + \nu \nabla^2 \bar{u}_j \quad (2)$$

$$\frac{\partial \bar{u}_j}{\partial t} + \frac{\partial (\bar{u}_k \bar{u}_j)}{\partial x_k} + \frac{\partial (\overline{u'_k u'_j})}{\partial x_k} + \epsilon_{j,k,l} f_k \bar{u}_l = - \frac{\partial P}{\partial x_j} - g_j \beta \bar{\theta} + \nu \nabla^2 \bar{u}_j \quad (2)$$

$$\frac{\partial \bar{u}_j}{\partial t} + \frac{\partial}{\partial x_k} (\bar{u}_k \bar{u}_j + \overline{u'_k u'_j}) + \epsilon_{j,k,l} f_k \bar{u}_l = - \frac{\partial P}{\partial x_j} - g_j \beta \bar{\theta} + \nu \nabla^2 \bar{u}_j \quad (2)$$

$$\frac{\partial \bar{\theta}}{\partial t} + \frac{\partial}{\partial x_k} (\bar{u}_k \bar{\theta} + \overline{u'_k \theta'}) = \alpha \nabla^2 \bar{\theta} \quad (3)$$

onde P é a pressão cinemática média, $g_j = (0, 0, -g)$ vetor de gravidade, $f_j = (0, f_y, f)$ o parâmetro de Coriolis (o componente vertical de não terá subscrito), $\beta = \left(\frac{\partial \rho}{\partial T} \right)_p / \rho$ o coeficiente de expansão térmica, ν é a viscosidade cinemática e α é a condutividade térmica cinemática (ou difusividade térmica).



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\frac{\partial u'_i}{\partial x_i} = 0 \quad (4)$$

$$\frac{\partial(u'_j)}{\partial t} + (\overline{u_k}) \frac{\partial(u'_j)}{\partial x_k} + \frac{\partial(u'_k \overline{u_j})}{\partial x_k} + \frac{\partial(u'_k u'_j)}{\partial x_k} = -\frac{1}{\rho_0} \frac{\partial(P')}{\partial x_j} - g \frac{\rho'}{\rho_0} \delta_{j3} - 2\Omega \varepsilon_{jki} \eta_k(u'_k) + \nu \frac{\partial^2(u'_j)}{\partial x_k^2}$$

Aplique a media de reynolds

$$\frac{\partial(\overline{u'_j})}{\partial t} + (\overline{u_k}) \frac{\partial(\overline{u'_j})}{\partial x_k} + \frac{\partial(\overline{u'_k \overline{u_j}})}{\partial x_k} + \frac{\partial(\overline{u'_k u'_j})}{\partial x_k} = -\frac{1}{\rho_0} \frac{\partial(\overline{P'})}{\partial x_j} - g \frac{\overline{\rho'}}{\rho_0} \delta_{j3} - 2\Omega \varepsilon_{jki} \eta_k(\overline{u'_k}) + \nu \frac{\partial^2(\overline{u'_j})}{\partial x_k^2}$$

$$\frac{\partial(\overline{u'_k u'_j})}{\partial x_k} = 0$$

$$\frac{\partial u'_j}{\partial t} + \left(\overline{u_k} \frac{\partial u'_j}{\partial x_k} + u'_j \frac{\partial \overline{u_k}}{\partial x_k} + \frac{\partial(\overline{u_j} u'_k)}{\partial x_k} + \frac{\partial(u'_k u'_j)}{\partial x_k} - \frac{\partial(\overline{u'_k u'_j})}{\partial x_k} \right) + \epsilon_{j,k,l} f_k u'_l = -\frac{\partial P'}{\partial x_j} - g_j \beta \theta' + \nu \nabla^2 u'_j \quad (5)$$

Como as equações médias (1), (2) e (3) envolvem a tensão de Reynolds $\overline{u'_i u'_j}$ e os momentos de condução de calor, $u'_i \theta'$, obtemos suas equações governantes de (5) e (6) usando a (4)



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2. As equações básicas

$$\frac{\partial u'_i}{\partial x_i} = 0 \quad (4)$$

$$\frac{\partial(u'_j)}{\partial t} + (\overline{u_k}) \frac{\partial(u'_j)}{\partial x_k} + \frac{\partial(u'_k \overline{u_j})}{\partial x_k} + \frac{\partial(u'_k u'_j)}{\partial x_k} = -\frac{1}{\rho_0} \frac{\partial(P')}{\partial x_j} - g \frac{\rho'}{\rho_0} \delta_{j3} - 2\Omega \varepsilon_{jki} \eta_k(u'_k) + \nu \frac{\partial^2(u'_j)}{\partial x_k^2}$$

$$\frac{\partial u'_j}{\partial t} + \left(\overline{u_k} \frac{\partial u'_j}{\partial x_k} + u'_j \frac{\partial \overline{u_k}}{\partial x_k} + \frac{\partial(\overline{u_j} u'_k)}{\partial x_k} + \frac{\partial(u'_k u'_j)}{\partial x_k} - \frac{\partial(\overline{u'_k u'_j})}{\partial x_k} \right) + \epsilon_{j,k,l} f_k u'_l = -\frac{\partial P'}{\partial x_j} - g_j \beta \theta' + \nu \nabla^2 u'_j$$

$$\frac{\partial u'_j}{\partial t} + \left(\frac{\partial \overline{u_k} u'_j}{\partial x_k} + \frac{\partial \overline{u_j} u'_k}{\partial x_k} + \frac{\partial u'_k u'_j}{\partial x_k} - \frac{\partial \overline{u'_k u'_j}}{\partial x_k} \right) + \epsilon_{j,k,l} f_k u'_l = -\frac{\partial P'}{\partial x_j} - g_j \beta \theta' + \nu \nabla^2 u'_j \quad (5)$$

$$\frac{\partial u'_j}{\partial t} + \frac{\partial}{\partial x_k} (\overline{u_k} u'_j + \overline{u_j} u'_k + u'_k u'_j - \overline{u'_k u'_j}) + \epsilon_{j,k,l} f_k u'_l = -\frac{\partial P'}{\partial x_j} - g_j \beta \theta' + \nu \nabla^2 u'_j \quad (5)$$

$$\frac{\partial \theta'}{\partial t} + \frac{\partial}{\partial x_k} (\overline{\theta} u'_k + \overline{u_k} \theta' + u'_k \theta' + \overline{u'_k \theta'}) = \alpha \nabla^2 \theta' \quad (6)$$

Como as equações médias (1), (2) e (3) envolvem a tensão de Reynolds $\overline{u'_i u'_j}$ e os momentos de condução de calor, $u'_i \theta'$, obtemos suas equações governantes de (5) e (6) usando a (4)



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2. As equações básicas

$$\frac{\partial u'_j}{\partial t} + \frac{\partial}{\partial x_k} (\bar{u}_k u'_j + \bar{u}_j u'_k + u'_k u'_j - \overline{u'_k u'_j}) + \epsilon_{j,k,l} f_k u'_l = -\frac{\partial P'}{\partial x_j} - g_j \beta \theta' + \nu \nabla^2 u'_j \quad (5)$$

$$u'_i \frac{\partial u'_j}{\partial t} + u'_i \frac{\partial}{\partial x_k} (\bar{u}_k u'_j + \bar{u}_j u'_k + u'_k u'_j - \overline{u'_k u'_j}) + \epsilon_{j,k,l} f_k u'_l u'_i = -u'_i \frac{\partial P'}{\partial x_j} - g_j \beta u'_i \theta' + \nu u'_i \nabla^2 u'_j \quad (5a)$$

$$u'_j \frac{\partial u'_i}{\partial t} + u'_j \frac{\partial}{\partial x_k} (\bar{u}_k u'_i + \bar{u}_i u'_k + u'_k u'_i - \overline{u'_k u'_i}) + \epsilon_{i,k,l} f_k u'_l u'_j = -u'_j \frac{\partial P'}{\partial x_i} - g_i \beta u'_j \theta' + \nu u'_j \nabla^2 u'_i \quad (5b)$$

$$\begin{aligned} & \left(u'_i \frac{\partial u'_j}{\partial t} + u'_j \frac{\partial u'_i}{\partial t} \right) + \left(u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} + u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} \right) + \left(u'_i \frac{\partial \bar{u}_j u'_k}{\partial x_k} + u'_j \frac{\partial \bar{u}_i u'_k}{\partial x_k} \right) + \left(u'_i \frac{\partial u'_k u'_j}{\partial x_k} + u'_j \frac{\partial u'_k u'_i}{\partial x_k} \right) - \left(u'_i \frac{\partial \overline{u'_k u'_j}}{\partial x_k} + u'_j \frac{\partial \overline{u'_k u'_i}}{\partial x_k} \right) \\ & + \epsilon_{j,k,l} f_k u'_l u'_i + \epsilon_{i,k,l} f_k u'_l u'_j = -u'_i \frac{\partial P'}{\partial x_j} - u'_j \frac{\partial P'}{\partial x_i} - g_j \beta u'_i \theta' - g_i \beta u'_j \theta' + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i \quad (5c) \end{aligned}$$



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2. As equações básicas

$$\left(u'_i \frac{\partial u'_j}{\partial t} + u'_j \frac{\partial u'_i}{\partial t} \right) + \left(u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} + u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} \right) + \left(u'_i \frac{\partial \bar{u}_j u'_k}{\partial x_k} + u'_j \frac{\partial \bar{u}_i u'_k}{\partial x_k} \right) + \left(u'_i \frac{\partial u'_k u'_j}{\partial x_k} + u'_j \frac{\partial u'_k u'_i}{\partial x_k} \right) - \left(u'_i \frac{\partial \overline{u'_k u'_j}}{\partial x_k} + u'_j \frac{\partial \overline{u'_k u'_i}}{\partial x_k} \right) \\ + \epsilon_{j,k,l} f_k u'_l u'_i + \epsilon_{i,k,l} f_k u'_l u'_j = -u'_i \frac{\partial P'}{\partial x_j} - u'_j \frac{\partial P'}{\partial x_i} - g_j \beta u'_i \theta' - g_i \beta u'_j \theta' + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i$$

$$\left(u'_i \frac{\partial u'_j}{\partial t} + u'_j \frac{\partial u'_i}{\partial t} \right) + \left(u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} + u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} \right) + \left(u'_i \frac{\partial u'_k u'_j}{\partial x_k} + u'_j \frac{\partial u'_k u'_i}{\partial x_k} \right) + \left(u'_i \frac{\partial \bar{u}_j u'_k}{\partial x_k} + u'_j \frac{\partial \bar{u}_i u'_k}{\partial x_k} \right) - \left(u'_i \frac{\partial \overline{u'_k u'_j}}{\partial x_k} + u'_j \frac{\partial \overline{u'_k u'_i}}{\partial x_k} \right) \\ + \epsilon_{j,k,l} f_k u'_l u'_i + \epsilon_{i,k,l} f_k u'_l u'_j = -u'_i \frac{\partial P'}{\partial x_j} - u'_j \frac{\partial P'}{\partial x_i} - g_j \beta u'_i \theta' - g_i \beta u'_j \theta' + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i$$

$$\left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} - u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} \right] + \left[\frac{\partial \bar{u}_k u'_j u'_i}{\partial x_k} - u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} - u'_j \frac{\partial u'_k u'_i}{\partial x_k} \right] + \left[\frac{\partial u'_k u'_j u'_i}{\partial x_k} - u'_i \frac{\partial u'_k u'_j}{\partial x_k} \right] \right) \\ + \left(\left[\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} - u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} \right] + \left[\frac{\partial \bar{u}_i u'_j u'_k}{\partial x_k} - u'_k \frac{\partial \bar{u}_i u'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i \overline{u'_k u'_j}}{\partial x_k} - \overline{u'_k u'_j} \frac{\partial u'_i}{\partial x_k} \right] + \left[\frac{\partial u'_j \overline{u'_k u'_i}}{\partial x_k} - \overline{u'_k u'_i} \frac{\partial u'_j}{\partial x_k} \right] \right) \\ + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) = - \left[\frac{\partial u'_i P'}{\partial x_j} - P' \frac{\partial u'_i}{\partial x_j} \right] - \left[\frac{\partial u'_j P'}{\partial x_i} - P' \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i$$



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2. As equações básicas

$$\begin{aligned} & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} - u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} \right] + \left[\frac{\partial \bar{u}_k u'_j u'_i}{\partial x_k} - u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} - u'_j \frac{\partial u'_k u'_i}{\partial x_k} \right] + \left[\frac{\partial u'_k u'_j u'_i}{\partial x_k} - u'_i \frac{\partial u'_k u'_j}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} - u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} \right] + \left[\frac{\partial \bar{u}_i u'_j u'_k}{\partial x_k} - u'_k \frac{\partial \bar{u}_i u'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i \bar{u}'_k u'_j}{\partial x_k} - \overline{u'_k u'_j} \frac{\partial u'_i}{\partial x_k} \right] + \left[\frac{\partial u'_j \bar{u}'_k u'_i}{\partial x_k} - \overline{u'_k u'_i} \frac{\partial u'_j}{\partial x_k} \right] \right) \\ & + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) = - \left[\frac{\partial u'_i P'}{\partial x_j} - P' \frac{\partial u'_i}{\partial x_j} \right] - \left[\frac{\partial u'_j P'}{\partial x_i} - P' \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i \end{aligned}$$

$$\begin{aligned} & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} + \frac{\partial \bar{u}_k u'_j u'_i}{\partial x_k} \right] + \left[-u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} - u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} + \frac{\partial u'_k u'_j u'_i}{\partial x_k} \right] + \left[-u'_j \frac{\partial u'_k u'_i}{\partial x_k} - u'_i \frac{\partial u'_k u'_j}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} + \frac{\partial \bar{u}_i u'_j u'_k}{\partial x_k} \right] + \left[-u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} - u'_k \frac{\partial \bar{u}_i u'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i \bar{u}'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j \bar{u}'_k u'_i}{\partial x_k} \right] \right) - \left(\left[-\overline{u'_k u'_j} \frac{\partial u'_i}{\partial x_k} \right] + \left[-\overline{u'_k u'_i} \frac{\partial u'_j}{\partial x_k} \right] \right) \\ & + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + \left[P' \frac{\partial u'_i}{\partial x_j} + P' \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i \end{aligned}$$



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2. As equações básicas

$$\begin{aligned} & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} + \frac{\partial \bar{u}_k u'_j u'_i}{\partial x_k} \right] + \left[-u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} - u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} + \frac{\partial u'_k u'_j u'_i}{\partial x_k} \right] + \left[-u'_j \frac{\partial u'_k u'_i}{\partial x_k} - u'_i \frac{\partial u'_k u'_j}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} + \frac{\partial \bar{u}_i u'_j u'_k}{\partial x_k} \right] + \left[-u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} - u'_k \frac{\partial \bar{u}_i u'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i u'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j u'_k u'_i}{\partial x_k} \right] \right) - \left(\left[-\overline{u'_k u'_j} \frac{\partial u'_i}{\partial x_k} \right] + \left[-\overline{u'_k u'_i} \frac{\partial u'_j}{\partial x_k} \right] \right) \\ & + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + \left[P' \frac{\partial u'_i}{\partial x_j} + P' \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i \end{aligned}$$

$$\begin{aligned} & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} + \frac{\partial \bar{u}_k u'_j u'_i}{\partial x_k} \right] + \left[-u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} - u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} + \frac{\partial u'_k u'_j u'_i}{\partial x_k} \right] + \left[-u'_j \frac{\partial u'_k u'_i}{\partial x_k} - u'_i \frac{\partial u'_k u'_j}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} + \frac{\partial \bar{u}_i u'_j u'_k}{\partial x_k} \right] + \left[-u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} - u'_k \frac{\partial \bar{u}_i u'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i u'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j u'_k u'_i}{\partial x_k} \right] \right) - \left(\left[-\overline{u'_k u'_j} \frac{\partial u'_i}{\partial x_k} \right] + \left[-\overline{u'_k u'_i} \frac{\partial u'_j}{\partial x_k} \right] \right) \\ & + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + \left[P' \frac{\partial u'_i}{\partial x_j} + P' \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i \end{aligned}$$



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2. As equações básicas

$$\begin{aligned}
 & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} + \frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] + \left[-u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} - u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} + \frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] + \left[-u'_j \frac{\partial u'_k u'_i}{\partial x_k} - u'_i \frac{\partial u'_k u'_j}{\partial x_k} \right] \right) \\
 & + \left(\left[\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} + \frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} \right] + \left[-u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} - u'_k \frac{\partial \bar{u}_j u'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i u'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j u'_k u'_i}{\partial x_k} \right] \right) - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} \right] + \left[-\bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) \\
 & + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + \left[P' \frac{\partial u'_i}{\partial x_j} + P' \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i
 \end{aligned}$$

$$\begin{aligned}
 & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(2 \left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] - \left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] \right) + \left(2 \left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] - \left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} + \frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} \right] + \left[-u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} - u'_k \frac{\partial \bar{u}_j u'_j}{\partial x_k} \right] \right) \\
 & - \left(\left[\frac{\partial u'_i u'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j u'_k u'_i}{\partial x_k} \right] \right) - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} - \bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) \\
 & = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu u'_i \frac{\partial}{\partial x_k} \frac{\partial u'_j}{\partial x_k} + \nu u'_j \frac{\partial}{\partial x_k} \frac{\partial u'_i}{\partial x_k}
 \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned}
 & \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(2 \left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] - \left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(2 \left[\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] - \left[\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial \bar{u}_j \mathbf{u}'_i \mathbf{u}'_k}{\partial x_k} + \frac{\partial \bar{u}_i \mathbf{u}'_j \mathbf{u}'_k}{\partial x_k} \right] + \left[-u'_k \frac{\partial \bar{u}_j \mathbf{u}'_i}{\partial x_k} - u'_k \frac{\partial \bar{u}_i \mathbf{u}'_j}{\partial x_k} \right] \right. \\
 & \left. - \left(\left[\frac{\partial \mathbf{u}'_i \mathbf{u}'_k \mathbf{u}'_j}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \mathbf{u}'_k \mathbf{u}'_i}{\partial x_k} \right] \right) - \left(\left[-\bar{u}'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} - \bar{u}'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l \mathbf{u}'_i + \epsilon_{i,k,l} u'_l \mathbf{u}'_j) \\
 & = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} + \frac{\partial \mathbf{u}'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial \mathbf{u}'_i}{\partial x_j} + \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \mathbf{u}'_i \frac{\partial}{\partial x_k} \frac{\partial \mathbf{u}'_j}{\partial x_k} + \nu \mathbf{u}'_j \frac{\partial}{\partial x_k} \frac{\partial \mathbf{u}'_i}{\partial x_k}
 \end{aligned}$$

$$\begin{aligned}
 & \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[u'_k \frac{\partial \bar{u}_j \mathbf{u}'_i}{\partial x_k} + \mathbf{u}'_i \frac{\partial \bar{u}_j \mathbf{u}'_k}{\partial x_k} + u'_k \frac{\partial \bar{u}_i \mathbf{u}'_j}{\partial x_k} + \mathbf{u}'_j \frac{\partial \bar{u}_i \mathbf{u}'_k}{\partial x_k} \right] + \left[-u'_k \frac{\partial \bar{u}_j \mathbf{u}'_i}{\partial x_k} - u'_k \frac{\partial \bar{u}_i \mathbf{u}'_j}{\partial x_k} \right] \right) \\
 & - \left(\left[\frac{\partial \mathbf{u}'_i \mathbf{u}'_k \mathbf{u}'_j}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \mathbf{u}'_k \mathbf{u}'_i}{\partial x_k} \right] \right) - \left(\left[-\bar{u}'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} - \bar{u}'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l \mathbf{u}'_i + \epsilon_{i,k,l} u'_l \mathbf{u}'_j) \\
 & = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} + \frac{\partial \mathbf{u}'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial \mathbf{u}'_i}{\partial x_j} + \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \left[\frac{\partial}{\partial x_k} \left(\mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right) - \left(\frac{\partial \mathbf{u}'_i}{\partial x_k} \frac{\partial \mathbf{u}'_j}{\partial x_k} \right) \right] \\
 & + \nu \left[\frac{\partial}{\partial x_k} \left(\mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} \right) - \left(\frac{\partial \mathbf{u}'_j}{\partial x_k} \frac{\partial \mathbf{u}'_i}{\partial x_k} \right) \right]
 \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned}
 & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[+u'_i \frac{\partial \bar{u}_j u'_k}{\partial x_k} + u'_j \frac{\partial \bar{u}_i u'_k}{\partial x_k} \right] + \left[u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} - u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} + u'_k \frac{\partial \bar{u}_i u'_j}{\partial x_k} - u'_k \frac{\partial \bar{u}_i u'_j}{\partial x_k} \right] \right) \\
 & - \left(\left[\frac{\partial u'_i u'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j u'_k u'_i}{\partial x_k} \right] \right) - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} - \bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) \\
 & = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu \left[\frac{\partial}{\partial x_k} \left(u'_i \frac{\partial u'_j}{\partial x_k} \right) - \left(\frac{\partial u'_i}{\partial x_k} \frac{\partial u'_j}{\partial x_k} \right) \right] \\
 & + \nu \left[\frac{\partial}{\partial x_k} \left(u'_j \frac{\partial u'_i}{\partial x_k} \right) - \left(\frac{\partial u'_j}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \right) \right]
 \end{aligned}$$

$$\begin{aligned}
 & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[u'_i \frac{\partial \bar{u}_j u'_k}{\partial x_k} + u'_j \frac{\partial \bar{u}_i u'_k}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i u'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j u'_k u'_i}{\partial x_k} \right] \right) \\
 & - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} - \bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) \\
 & = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu \frac{\partial}{\partial x_k} \left[\left(u'_i \frac{\partial u'_j}{\partial x_k} \right) + \left(u'_j \frac{\partial u'_i}{\partial x_k} \right) \right] - 2\nu \left(\frac{\partial u'_j}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \right)
 \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned}
 & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[u'_i \frac{\partial \bar{u}_j u'_k}{\partial x_k} + u'_j \frac{\partial \bar{u}_i u'_k}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i \bar{u}'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j \bar{u}'_k u'_i}{\partial x_k} \right] \right) \\
 & - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} - \bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) \\
 & = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu \frac{\partial}{\partial x_k} \left[\left(u'_i \frac{\partial u'_j}{\partial x_k} \right) + \left(u'_j \frac{\partial u'_i}{\partial x_k} \right) \right] - 2\nu \left(\frac{\partial u'_j}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \right)
 \end{aligned}$$

$$\begin{aligned}
 & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[u'_i u'_k \frac{\partial \bar{u}_j}{\partial x_k} + u'_i \bar{u}_j \frac{\partial u'_k}{\partial x_k} + u'_j u'_k \frac{\partial \bar{u}_i}{\partial x_k} + u'_j \bar{u}_i \frac{\partial u'_k}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i \bar{u}'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j \bar{u}'_k u'_i}{\partial x_k} \right] \right) \\
 & - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} - \bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) \\
 & = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu \frac{\partial}{\partial x_k} \left[\left(u'_i \frac{\partial u'_j}{\partial x_k} \right) + \left(u'_j \frac{\partial u'_i}{\partial x_k} \right) \right] - 2\nu \left(\frac{\partial u'_j}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \right)
 \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned} & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[u'_i u'_k \frac{\partial \bar{u}_j}{\partial x_k} + u'_i \bar{u}_j \frac{\partial u'_k}{\partial x_k} + u'_j u'_k \frac{\partial \bar{u}_i}{\partial x_k} + u'_j \bar{u}_i \frac{\partial u'_k}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i \bar{u}'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j \bar{u}'_k u'_i}{\partial x_k} \right] \right) \\ & - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} - \bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) \\ & = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu \frac{\partial}{\partial x_k} \left[\left(u'_i \frac{\partial u'_j}{\partial x_k} \right) + \left(u'_j \frac{\partial u'_i}{\partial x_k} \right) \right] - 2\nu \left(\frac{\partial u'_j}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \right) \end{aligned}$$

$$\begin{aligned} & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[u'_i u'_k \frac{\partial \bar{u}_j}{\partial x_k} + u'_j u'_k \frac{\partial \bar{u}_i}{\partial x_k} + u'_i \bar{u}_j \frac{\partial u'_k}{\partial x_k} + u'_j \bar{u}_i \frac{\partial u'_k}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i \bar{u}'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j \bar{u}'_k u'_i}{\partial x_k} \right] \right) \\ & - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} - \bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) \\ & = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu \frac{\partial}{\partial x_k} \left[\left(u'_i \frac{\partial u'_j}{\partial x_k} \right) + \left(u'_j \frac{\partial u'_i}{\partial x_k} \right) \right] - 2\nu \left(\frac{\partial u'_j}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \right) \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned}
 & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[u'_i u'_k \frac{\partial \bar{u}_j}{\partial x_k} + u'_j u'_k \frac{\partial \bar{u}_i}{\partial x_k} + u'_i \bar{u}_j \frac{\partial u'_k}{\partial x_k} + u'_j \bar{u}_i \frac{\partial u'_k}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i \bar{u}'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j \bar{u}'_k u'_i}{\partial x_k} \right] \right) \\
 & - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} - \bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) \\
 & = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu \frac{\partial}{\partial x_k} \left[\left(u'_i \frac{\partial u'_j}{\partial x_k} \right) + \left(u'_j \frac{\partial u'_i}{\partial x_k} \right) \right] - 2\nu \left(\frac{\partial u'_j}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \right)
 \end{aligned}$$

$$\begin{aligned}
 & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[u'_i u'_k \frac{\partial \bar{u}_j}{\partial x_k} + u'_j u'_k \frac{\partial \bar{u}_i}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i \bar{u}'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j \bar{u}'_k u'_i}{\partial x_k} \right] \right) \\
 & - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} - \bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) \\
 & = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu \frac{\partial}{\partial x_k} \left[\left(u'_i \frac{\partial u'_j}{\partial x_k} \right) + \left(u'_j \frac{\partial u'_i}{\partial x_k} \right) \right] - 2\nu \left(\frac{\partial u'_j}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \right)
 \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned}
 & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[u'_i u'_k \frac{\partial \bar{u}_j}{\partial x_k} + u'_j u'_k \frac{\partial \bar{u}_i}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i \overline{u'_k u'_j}}{\partial x_k} \right] + \left[\frac{\partial u'_j \overline{u'_k u'_i}}{\partial x_k} \right] \right) \\
 & - \left(\left[-\overline{u'_k u'_j} \frac{\partial u'_i}{\partial x_k} - \overline{u'_k u'_i} \frac{\partial u'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) \\
 & = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu \frac{\partial}{\partial x_k} \left[\left(u'_i \frac{\partial u'_j}{\partial x_k} \right) + \left(u'_j \frac{\partial u'_i}{\partial x_k} \right) \right] - 2\nu \left(\frac{\partial u'_j}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \right)
 \end{aligned}$$

$$\begin{aligned}
 & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[u'_i u'_k \frac{\partial \bar{u}_j}{\partial x_k} + u'_j u'_k \frac{\partial \bar{u}_i}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i \overline{u'_k u'_j}}{\partial x_k} \right] + \left[\frac{\partial u'_j \overline{u'_k u'_i}}{\partial x_k} \right] \right) \\
 & - \left(\left[-\overline{u'_k u'_j} \frac{\partial u'_i}{\partial x_k} - \overline{u'_k u'_i} \frac{\partial u'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) \\
 & = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu \frac{\partial}{\partial x_k} \left(\frac{\partial u'_i u'_j}{\partial x_k} \right) - 2\nu \left(\frac{\partial u'_j}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \right)
 \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned}
 & \left(\frac{\partial \overline{u'_i u'_j}}{\partial t} \right) + \left(\left[\frac{\partial \overline{u_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\frac{\partial \overline{u'_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\overline{u'_i u'_k} \frac{\partial \overline{u_j}}{\partial x_k} + \overline{u'_j u'_k} \frac{\partial \overline{u_i}}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \overline{u'_i u'_k u'_j}}{\partial x_k} \right] + \left[\frac{\partial \overline{u'_j u'_k u'_i}}{\partial x_k} \right] \right) \\
 & - \left(\left[-\overline{u'_k u'_j} \frac{\partial \overline{u'_i}}{\partial x_k} - \overline{u'_k u'_i} \frac{\partial \overline{u'_j}}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} \overline{u'_l u'_i} + \epsilon_{i,k,l} \overline{u'_l u'_j}) \\
 & = - \left[\frac{\partial \overline{u'_i P'}}{\partial x_j} + \frac{\partial \overline{u'_j P'}}{\partial x_i} \right] + \overline{P'} \left[\frac{\partial \overline{u'_i}}{\partial x_j} + \frac{\partial \overline{u'_j}}{\partial x_i} \right] - \beta (g_j \overline{u'_i \theta'} - g_i \overline{u'_j \theta'}) + \nu \frac{\partial}{\partial x_k} \left(\frac{\partial \overline{u'_i u'_j}}{\partial x_k} \right) - 2\nu \left(\frac{\partial \overline{u'_j}}{\partial x_k} \frac{\partial \overline{u'_i}}{\partial x_k} \right)
 \end{aligned}$$

$$\begin{aligned}
 & \left(\frac{\partial \overline{u'_i u'_j}}{\partial t} \right) + \left(\left[\frac{\partial \overline{u_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\frac{\partial \overline{u'_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\overline{u'_i u'_k} \frac{\partial \overline{u_j}}{\partial x_k} + \overline{u'_j u'_k} \frac{\partial \overline{u_i}}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \overline{u'_i u'_k u'_j}}{\partial x_k} \right] + \left[\frac{\partial \overline{u'_j u'_k u'_i}}{\partial x_k} \right] \right) \\
 & + \left(\left[\overline{u'_k u'_j} \frac{\partial \overline{u'_i}}{\partial x_k} + \overline{u'_k u'_i} \frac{\partial \overline{u'_j}}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} \overline{u'_l u'_i} + \epsilon_{i,k,l} \overline{u'_l u'_j}) \\
 & = - \left[\frac{\partial \overline{u'_i P'}}{\partial x_j} + \frac{\partial \overline{u'_j P'}}{\partial x_i} \right] + \overline{P'} \left[\frac{\partial \overline{u'_i}}{\partial x_j} + \frac{\partial \overline{u'_j}}{\partial x_i} \right] - \beta (g_j \overline{u'_i \theta'} - g_i \overline{u'_j \theta'}) + \nu \frac{\partial}{\partial x_k} \left(\frac{\partial \overline{u'_i u'_j}}{\partial x_k} \right) - 2\nu \left(\frac{\partial \overline{u'_j}}{\partial x_k} \frac{\partial \overline{u'_i}}{\partial x_k} \right)
 \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned}
 & \left(\frac{\partial \overline{u'_i u'_j}}{\partial t} \right) + \left(\left[\frac{\partial \overline{u_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\frac{\partial \overline{u'_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\overline{u'_i u'_k} \frac{\partial \overline{u_j}}{\partial x_k} + \overline{u'_j u'_k} \frac{\partial \overline{u_i}}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \overline{u'_i u'_k u'_j}}{\partial x_k} \right] + \left[\frac{\partial \overline{u'_j u'_k u'_i}}{\partial x_k} \right] \right) \\
 & + \left(\left[\overline{u'_k u'_j} \frac{\partial \overline{u'_i}}{\partial x_k} + \overline{u'_k u'_i} \frac{\partial \overline{u'_j}}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} \overline{u'_l u'_i} + \epsilon_{i,k,l} \overline{u'_l u'_j}) \\
 & = - \left[\frac{\partial \overline{u'_i P'}}{\partial x_j} + \frac{\partial \overline{u'_j P'}}{\partial x_i} \right] + \overline{P'} \left[\frac{\partial \overline{u'_i}}{\partial x_j} + \frac{\partial \overline{u'_j}}{\partial x_i} \right] - \beta (g_j \overline{u'_i \theta'} - g_i \overline{u'_j \theta'}) + \nu \frac{\partial}{\partial x_k} \left(\frac{\partial \overline{u'_i u'_j}}{\partial x_k} \right) - 2\nu \left(\frac{\partial \overline{u'_j}}{\partial x_k} \frac{\partial \overline{u'_i}}{\partial x_k} \right)
 \end{aligned}$$

$$\begin{aligned}
 & \left(\frac{\partial \overline{u'_i u'_j}}{\partial t} \right) + \left(\left[\frac{\partial \overline{u_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\frac{\partial \overline{u'_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\overline{u'_i u'_k} \frac{\partial \overline{u_j}}{\partial x_k} + \overline{u'_j u'_k} \frac{\partial \overline{u_i}}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} \overline{u'_l u'_i} + \epsilon_{i,k,l} \overline{u'_l u'_j}) \\
 & = - \left[\frac{\partial \overline{u'_i P'}}{\partial x_j} + \frac{\partial \overline{u'_j P'}}{\partial x_i} \right] + \overline{P'} \left[\frac{\partial \overline{u'_i}}{\partial x_j} + \frac{\partial \overline{u'_j}}{\partial x_i} \right] - \beta (g_j \overline{u'_i \theta'} - g_i \overline{u'_j \theta'}) + \nu \frac{\partial}{\partial x_k} \left(\frac{\partial \overline{u'_i u'_j}}{\partial x_k} \right) - 2\nu \left(\frac{\partial \overline{u'_j}}{\partial x_k} \frac{\partial \overline{u'_i}}{\partial x_k} \right)
 \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned} & \left(\frac{\partial \overline{u'_i u'_j}}{\partial t} \right) + \left(\left[\frac{\partial \overline{u_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\overline{u'_i u'_k} \frac{\partial \bar{u}_j}{\partial x_k} + \overline{u'_j u'_k} \frac{\partial \bar{u}_i}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} \overline{u'_l u'_i} + \epsilon_{i,k,l} \overline{u'_l u'_j}) \\ &= - \left[\frac{\partial \overline{u'_i P'}}{\partial x_j} + \frac{\partial \overline{u'_j P'}}{\partial x_i} \right] + \bar{P}' \left[\frac{\partial \overline{u'_i}}{\partial x_j} + \frac{\partial \overline{u'_j}}{\partial x_i} \right] - \beta (g_j \overline{u'_i \theta'} - g_i \overline{u'_j \theta'}) + \nu \frac{\partial}{\partial x_k} \left(\frac{\partial \overline{u'_i u'_j}}{\partial x_k} \right) - 2\nu \left(\frac{\partial \overline{u'_j}}{\partial x_k} \frac{\partial \overline{u'_i}}{\partial x_k} \right) \end{aligned}$$

$$\begin{aligned} & \frac{\partial \overline{u'_i u'_j}}{\partial t} + \frac{\partial}{\partial x_k} \left[\bar{u}_k \overline{u'_i u'_j} + \overline{u'_k u'_i u'_j} - \nu \frac{\partial \overline{u'_i u'_j}}{\partial x_k} \right] + \frac{\partial \overline{P' u'_i}}{\partial x_j} + \frac{\partial \overline{P' u'_j}}{\partial x_i} + f_k (\epsilon_{j,k,l} \overline{u'_l u'_i} + \epsilon_{i,k,l} \overline{u'_l u'_j}) \\ &= - \overline{u'_k u'_i} \frac{\partial \bar{u}_j}{\partial x_k} - \overline{u'_k u'_j} \frac{\partial \bar{u}_i}{\partial x_k} - \beta (g_j \overline{u'_i \theta'} + g_i \overline{u'_j \theta'}) + \overline{P' \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)} - 2\nu \overline{\left(\frac{\partial u'_i}{\partial x_k} \frac{\partial u'_j}{\partial x_k} \right)} \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\frac{\partial u'_j}{\partial t} + \frac{\partial}{\partial x_k} (\bar{u}_k u'_j + \bar{u}_j u'_k + u'_k u'_j - \overline{u'_k u'_j}) + \epsilon_{j,k,l} f_k u'_l = -\frac{\partial P'}{\partial x_j} - g_j \beta \theta' + \nu \nabla^2 u'_j \quad (5)$$

$$\frac{\partial \theta'}{\partial t} + \frac{\partial}{\partial x_k} (\bar{\theta} u'_k + \bar{u}_k \theta' + u'_k \theta' + \overline{u'_k \theta'}) = \alpha \nabla^2 \theta' \quad (6)$$

$$\frac{\partial u'_j \theta'}{\partial t} = u'_j \frac{\partial \theta'}{\partial t} + \theta' \frac{\partial u'_j}{\partial t}$$

$$\begin{aligned} \frac{\partial \overline{u'_j \theta'}}{\partial t} + \frac{\partial}{\partial x_k} \left[\bar{u}_k \overline{\theta' u'_j} + \overline{u'_k u'_j \theta'} - \alpha \overline{u'_j \frac{\partial \theta'}{\partial x_k}} - \nu \overline{\theta' \frac{\partial u'_j}{\partial x_k}} \right] + \frac{\partial}{\partial x_j} \overline{P' \theta'} + \epsilon_{j,k,l} f_k \overline{u'_l \theta'} \\ = -\overline{u'_j u'_k} \frac{\partial \bar{\theta}}{\partial x_k} - \overline{\theta' u'_k} \frac{\partial \bar{u}_j}{\partial x_k} - \beta g_j \overline{\theta'^2} + \overline{P' \frac{\partial \theta'}{\partial x_j}} - (\alpha + \nu) \overline{\frac{\partial u'_j}{\partial x_k} \frac{\partial \theta'}{\partial x_k}} \end{aligned} \quad (8)$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

A equação 8 envolve $\overline{\theta'^2}$, na equação . Onde é necessário obter uma equação para este termo da equação 6

$$\frac{\partial \theta'}{\partial t} + \frac{\partial}{\partial x_k} (\bar{\theta} u'_k + \bar{u}_k \theta' + u'_k \theta' + \overline{u'_k \theta'}) = \alpha \nabla^2 \theta' \quad (6)$$

$$\frac{\partial \overline{\theta'^2}}{\partial t} = + \frac{\partial}{\partial x_k} \left[\bar{u}_k \overline{\theta'^2} + \overline{k'_k \theta'^2} - \alpha \frac{\partial \overline{\theta'^2}}{\partial x_k} \right] = 2 \overline{k'_k \theta'^2} - 2 \alpha \frac{\partial \theta'}{\partial x_k} \frac{\partial \theta'}{\partial x_k} \quad (9)$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

Suposições de modelagem

. A principal contribuição de Rotta (1951) foi sugerir uma suposição para o termo $\overline{P' \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)}$, que ele chamou de "**termo de redistribuição de energia**", **uma vez que uma de suas funções é particionar a energia entre os três componentes de energia sem contribuir para o total**. contração, o termo sai da Ed.7. Com base nas relações integrais obtidas da Eq.5 no caso neutro onde $-\beta(g_j u'_i \theta')$ não é significativa, Rotta sugeriu que o termo poderia ser razoavelmente proporcional a $u'_i u'_j$ e $\frac{\partial \bar{u}_i}{\partial x_j}$. Por isso,

$$\overline{P' \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)} = C_{ijk m} \overline{u'_k u'_m} + C'_{ijk m} \frac{\partial \bar{u}_k}{\partial x_m}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

Suposições de modelagem

aqui assumimos que os coeficientes constitutivos são tensores isotrópicos, ou seja

$$C_{ijklm} = C_1 \delta_{i,j} \delta_{k,m} + C_2 \delta_{i,k} \delta_{j,m} + C_3 \delta_{j,k} \delta_{i,m}$$

Da eq. Da continuidade obtém-se $C_1 = \frac{(C_2+C_3)}{3}$ raciocínio semelhante se aplica ao C_{ijklm} . Obtém-se portanto

$$\overline{P' \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)} = C_{ijklm} \overline{u'_k u'_m} + C'_{ijklm} \frac{\partial \bar{u}_k}{\partial x_m}$$

$$\overline{P' \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)} = -\frac{q}{3l_1} \left(\overline{u'_i u'_j} - \frac{\delta_{i,j}}{3} q^2 + C q^2 \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) \right)$$

Onde $-\frac{q}{3l_1}$ e Cq^2 the sido substituído por coeficientes escalares sobreviventes $q \equiv \left(\overline{u'^2_i} \right)^{1/2}$. O comprimento l_1 e a constante C devem ser determinadas empiricamente.



Linearização da Equações de Navier Stokes



5.2 TURBULENT KINETIC ENERGY

A equação de Navier Stokes: conservação de momentum

$$\begin{aligned} & \frac{\partial(u_i' u_k')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i' u_k')}{\partial x_j} \\ &= -(u_j' u_k') \frac{\partial(\bar{u}_i)}{\partial x_j} - (u_j' \bar{u}_i) \frac{\partial(u_k')}{\partial x_j} - \bar{u}_i u_k' \frac{\partial(u_j')}{\partial x_j} - \frac{\partial(u_j' u_i' u_k')}{\partial x_j} - \frac{1}{\rho_0} \frac{\partial(P' u_k')}{\partial x_i} - g \frac{\rho' u_k'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k' u_k') + \nu \frac{\partial^2(u_i' u_k')}{\partial x_j^2} \end{aligned}$$

Considere que a propriedade da equação da continuidade para o fluxo $\nabla \cdot \vec{V} = 0$

$$\nabla \cdot \vec{V}' = \frac{\partial(u_j')}{\partial x_j} = 0, \quad j = 1, 2, 3$$

$$\frac{\partial(u_i' u_k')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i' u_k')}{\partial x_j} = -(u_j' u_k') \frac{\partial(\bar{u}_i)}{\partial x_j} - (u_j' \bar{u}_i) \frac{\partial(u_k')}{\partial x_j} - \frac{\partial(u_j' u_i' u_k')}{\partial x_j} - \frac{1}{\rho_0} \frac{\partial(P' u_k')}{\partial x_i} - g \frac{\rho' u_k'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k' u_k') + \nu \frac{\partial^2(u_i' u_k')}{\partial x_j^2}$$



Linearização da Equações de Navier Stokes



$$\begin{aligned} & \frac{\partial \overline{u'_j \theta'}}{\partial t} + \frac{\partial}{\partial x_k} \left[\bar{u}_k \overline{\theta' u'_j} + \overline{u'_k u'_j \theta'} - \alpha \overline{u'_j \frac{\partial \theta'}{\partial x_k}} - \nu \overline{\theta' \frac{\partial u'_j}{\partial x_k}} \right] + \frac{\partial}{\partial x_j} \overline{P' \theta'} + \epsilon_{j,k,l} f_k \overline{u'_l \theta'} \\ &= -\overline{u'_j u'_k} \frac{\partial \bar{\theta}}{\partial x_k} - \overline{\theta' u'_k} \frac{\partial \bar{u}_j}{\partial x_k} - \beta g_j \overline{\theta'^2} + \overline{P' \frac{\partial \theta'}{\partial x_j}} - (\alpha + \nu) \overline{\frac{\partial u'_j}{\partial x_k} \frac{\partial \theta'}{\partial x_k}} \end{aligned} \quad (8)$$

$$\begin{aligned} & \frac{\partial \overline{u'_i u'_j}}{\partial t} + \frac{\partial}{\partial x_k} \left[\bar{u}_k \overline{u'_i u'_j} + \overline{u'_k u'_i u'_j} - \nu \overline{\frac{\partial u'_i u'_j}{\partial x_k}} \right] + \frac{\partial \overline{P' u'_i}}{\partial x_j} + \frac{\partial \overline{P' u'_j}}{\partial x_i} + f_k (\epsilon_{j,k,l} u'_l \textcolor{red}{u'_i} + \epsilon_{i,k,l} u'_l \textcolor{green}{u'_j}) \\ &= -\overline{u'_k u'_i} \frac{\partial \bar{u}_j}{\partial x_k} - \overline{u'_k u'_j} \frac{\partial \bar{u}_i}{\partial x_k} - \beta (g_j \overline{u'_i \theta'} + g_i \overline{u'_j \theta'}) + \overline{P' \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)} - 2\nu \overline{\left(\frac{\partial u'_i}{\partial x_k} \frac{\partial u'_j}{\partial x_k} \right)} \end{aligned}$$



Linearização da Equações de Navier Stokes

5.2 TURBULENT KINETIC ENERGY

A equação de Navier Stokes: conservação de momentum



$$\begin{aligned} & \frac{\partial \overline{u'_i u'_j}}{\partial t} + \frac{\partial}{\partial x_k} [\bar{u}_k \overline{u'_i u'_j}] \\ &= -\overline{u'_k u'_i} \frac{\partial \bar{u}_j}{\partial x_k} - \overline{u'_k u'_j} \frac{\partial \bar{u}_i}{\partial x_k} - \frac{\partial \overline{u'_k u'_i u'_j}}{\partial x_k} - \beta (g_j \overline{u'_i \theta'} + g_i \overline{u'_j \theta'}) - f_k (\epsilon_{j,k,l} \overline{u'_l u'_i} + \epsilon_{i,k,l} \overline{u'_l u'_j}) - \left[\frac{\partial \overline{P' u'_i}}{\partial x_j} + \frac{\partial \overline{P' u'_j}}{\partial x_i} \right] - 2\nu \left(\frac{\partial u'_i}{\partial x_k} \frac{\partial u'_j}{\partial x_k} \right) \\ &+ \nu \frac{\partial}{\partial x_k} \left[\frac{\partial \overline{u'_i u'_j}}{\partial x_k} \right] + \overline{P'} \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right) \end{aligned}$$

O termo do lado esquerdo é a taxa temporal local de mudança e advecção de $\overline{(u'_i u'_k)}$

O **1 e termo** do lado direito são os termos de produção resultante da interação da turbulência e o escoamento médio

O **2 termo (terceiro momento)** correlação tripla pode ser interpretado como transporte de turbulência (segundo momento) pela flutuação turbulenta com o ganho ou perda devido a divergência do fluxo turbulento

O **3 termo** representa a produção e destruição da flutuabilidade (conversão da energia cinética turbulenta para a energia potencial turbulenta)

O **termo 4** é a rotação e pode ser desprezado para média temporal menor do que 1 hora

O **termo 5** é a interação da flutuação de pressão e do campo de velocidade

O **termo 6** é a dissipação molecular

O termo 7 é a redistribuição de energia, uma vez que uma de suas funções é particionar a energia entre os três componentes de energia sem contribuir para o total



Linearização da Equações de Navier Stokes

Para caso de homogeneidade horizontal

$$\begin{aligned} & \frac{\partial \overline{(u_i' u_k')}}{\partial t} + (\bar{u}_j) \frac{\partial \overline{(u_i' u_k')}}{\partial x_j} \\ &= -\overline{(u_j' u_k')} \frac{\partial (\bar{u}_i)}{\partial x_j} - \frac{\partial \overline{(u_j' u_i' u_k')}}{\partial x_j} - g \frac{\overline{\rho' u_k'}}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j \overline{(u_k' u_k')} - \frac{1}{\rho_0} \frac{\partial \overline{(P' u_k')}}{\partial x_i} + \nu \frac{\partial^2 \overline{(u_i' u_k')}}{\partial x_j^2} \end{aligned}$$

O termo molecular $\nu \frac{\partial^2 \overline{(u_i' u_k')}}{\partial x_j^2} \rightarrow 0$ é desprezado no caso da covariância, porque a viscosidade é dominante somente em numero de ondas grandes.

Porém, neste caso a turbulência é isotrópica e assim a covariância é zero na horizontal.

$$(\mathbf{j}=\mathbf{k}) \quad \frac{\partial \overline{u' w'}}{\partial t} = -\overline{w'^2} \frac{\partial \bar{u}}{\partial z} + \frac{g}{\theta_v} \overline{u' \theta_v'} - \frac{\partial \overline{u' w'^2}}{\partial z} - \frac{1}{\rho} \left(\overline{u' \frac{\partial P'}{\partial z}} + \overline{w' \frac{\partial P'}{\partial x}} \right)$$

Isotrópico é a caracterização de uma substância que possui as mesmas propriedades físicas, independentemente da direção considerada.



Linearização da Equações de Navier Stokes

Para caso de homogeneidade horizontal e o estado básico em condições neutra $\frac{\partial \overline{u'w'}}{\partial t} = 0$ e $\frac{g}{\theta_v} \overline{u'\theta'_v} = 0$

$$(\mathbf{j}=\mathbf{k}) \quad \frac{\partial \overline{u'w'}}{\partial t} = -\overline{w'^2} \frac{\partial \bar{u}}{\partial z} + \frac{g}{\theta_v} \overline{u'\theta'_v} - \frac{\partial \overline{u'w'^2}}{\partial z} - \frac{1}{\rho} \left(\overline{u' \frac{\partial P'}{\partial z}} + \overline{w' \frac{\partial P'}{\partial x}} \right)$$

Isto mostra que o termo de correlação de pressão-velocidade destrói o stress na mesma taxa como ela é produzida

$$0 = -\overline{w'^2} \frac{\partial \bar{u}}{\partial z} - \frac{\partial \overline{u'w'^2}}{\partial z} - \frac{1}{\rho} \left(\overline{u' \frac{\partial P'}{\partial z}} + \overline{w' \frac{\partial P'}{\partial x}} \right)$$

$$+\overline{w'^2} \frac{\partial \bar{u}}{\partial z} + \frac{\partial \overline{u'w'^2}}{\partial z} = -\frac{1}{\rho} \left(\overline{u' \frac{\partial P'}{\partial z}} + \overline{w' \frac{\partial P'}{\partial x}} \right)$$



Equação da Energia Cinética Turbulenta



Linearização da Equações de Navier Stokes



5.2 Energia Cinética Turbulenta

$$\begin{aligned} & \frac{\partial \overline{(u_i' u_k')}}{\partial t} + (\bar{u}_j) \frac{\partial \overline{(u_i' u_k')}}{\partial x_j} \\ &= -\overline{(u_j' u_k')} \frac{\partial (\bar{u}_i)}{\partial x_j} - \frac{\partial \overline{(u_j' u_i' u_k')}}{\partial x_j} - g \frac{\overline{\rho' u_k'}}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j \overline{(u_k' u_k')} - \frac{1}{\rho_0} \frac{\partial \overline{(P' u_k')}}{\partial x_i} + \nu \frac{\partial^2 \overline{(u_i' u_k')}}{\partial x_j^2} \end{aligned}$$

$$\overline{u_k' u_k'} = \overline{u_k'^2} = \overline{u_k'^2} = 0$$

$$\bar{e} = \frac{\overline{u_i'^2}}{2} = \frac{\overline{(u'^2 + v'^2 + w'^2)}}{2} \quad i = k = 1, 2, 3$$

$$\frac{\partial \bar{e}}{\partial t} + (\bar{u}_j) \frac{\partial \bar{e}}{\partial x_j} = -\overline{(u_j' u_i')} \frac{\partial (\bar{u}_i)}{\partial x_j} - g \frac{\overline{u_i' \rho'}}{\rho_0} \delta_{i3} - \frac{\partial \overline{(e u_j')}}{\partial x_j} - \frac{1}{\rho_0} \frac{\partial \overline{(P' u_i')}}{\partial x_i} + \nu \frac{\partial^2 \overline{(u_i' u_k')}}{\partial x_j^2}$$

$$\frac{\partial \bar{e}}{\partial t} + (\bar{u}_j) \frac{\partial \bar{e}}{\partial x_j} = -\frac{\overline{(u_j' u_i')}}{2} \frac{\partial (\bar{u}_i)}{\partial x_j} - g \frac{\overline{u_i' \rho'}}{2\rho_0} \delta_{i3} - \frac{\partial \overline{(e u_j')}}{\partial x_j} - \frac{1}{2\rho_0} \frac{\partial \overline{(P' u_i')}}{\partial x_i} - \epsilon$$



Linearização da Equações de Navier Stokes

5.2 Energia Cinética Turbulenta

$$\frac{\partial \bar{e}}{\partial t} + (\bar{u}_j) \frac{\partial \bar{e}}{\partial x_j} = - \frac{\overline{(u_j' u_i')}}{2} \frac{\partial (\bar{u}_i)}{\partial x_j} - g \frac{\overline{u_i' \rho'}}{2 \rho_0} \delta_{i3} - \frac{\partial (\overline{e u_j'})}{\partial x_j} - \frac{1}{2 \rho_0} \frac{\partial (\overline{P' u_i'})}{\partial x_i} - \epsilon$$

A quantidade ϵ é um parâmetro significativo para a atmosfera desde que seja relacionado a dissipação da energia cinética turbulenta de todos os movimentos atmosféricos



5.2 Energia Cinética Turbulenta

A essência da equação da energia cinética turbulenta pode ser expressa pela equação:

$$\frac{D\bar{e}}{Dt} = -\frac{\overline{(u_j' u_i')}}{2} \frac{\partial(\bar{u}_i)}{\partial x_j} - g \frac{\overline{u_i' \rho'}}{2\rho_0} \delta_{i3} - \frac{\partial(\overline{e u_j'})}{\partial x_j} - \frac{1}{2\rho_0} \frac{\partial(\overline{P' u_i'})}{\partial x_i} - \epsilon$$

$$\frac{\bar{D}(TKE)}{Dt} = MP + BPL + TR - \epsilon$$

MP é a produção mecânica

BPL é a produção e perda por flutuabilidade

TR redistribuição de tke por transporte e força de pressão

ϵ dissipação por atrito

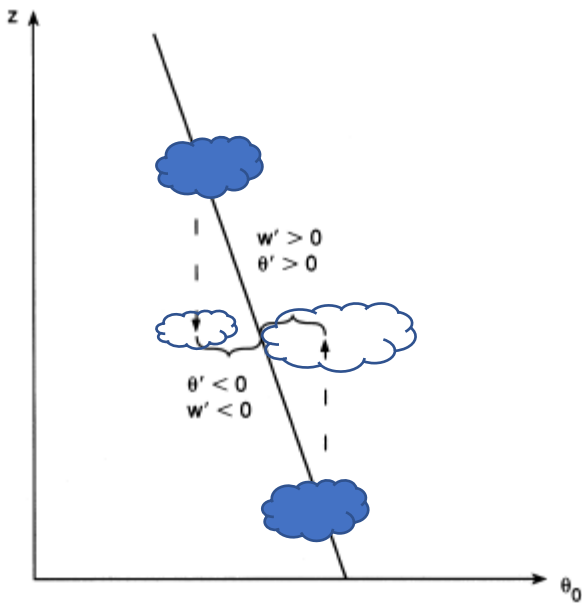


5.2 Energia Cinética Turbulenta

$$BPL \equiv \overline{w' \theta'} \frac{g}{\theta_0}$$

É a conversão da energia potencial do escoamento médio e a energia cinética turbulenta:
É positivo para movimentos que baixa o centro de massa da atmosfera
É negativo para movimentos que aumenta o centro de massa da atmosfera

Correlação positiva(fonte tke)
Atms. instável



Correlação negativa (destroi tke)
Atms. estável

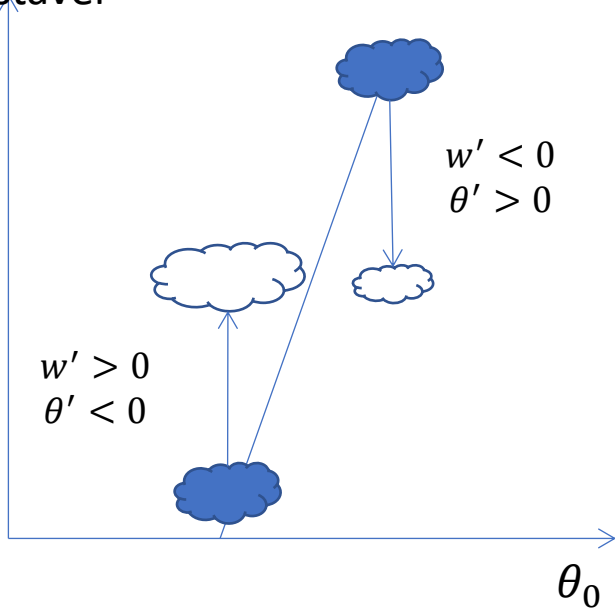


Fig. 5.1 Correlation between vertical velocity and potential temperature perturbations for upward or downward parcel displacements when the mean potential temperature $\theta_0(z)$ decreases with height.



5.2 Energia Cinética Turbulenta

Para ambos as condições estáveis e instáveis da CLP a turbulência pode ser produzida mecanicamente pela instabilidade dinâmica através do cisalhamento. Conversão de energia entre o escoamento médio e a flutuação turbulenta.

$$MP \equiv -\frac{\overline{u'w'}}{2} \frac{\partial \bar{u}}{\partial z} - \frac{\overline{v'w'}}{2} \frac{\partial \bar{v}}{\partial z}$$

$MP > 0$ quando o fluxo de momentum ($\overline{u'w'}$) é direcionado para baixo e o gradiente vertical é positivo



5.2 Energia Cinética Turbulenta

Estatisticamente na camada limite estável a turbulência pode existir somente se a produção mecânica for grande o suficiente para superar o efeito de supressão da estabilidade e da viscosidade

Esta condição é medida pelo numero de Richardson de fluxo

$$Rf \equiv -\frac{BPL}{MP} \equiv \frac{\overline{w'\theta' \frac{g}{\theta_0}}}{-\overline{u'w'} \frac{\partial \bar{u}}{\partial z} - \overline{v'w'} \frac{\partial \bar{v}}{\partial z}}$$



5.2 Energia Cinética Turbulenta

$$Rf \equiv -\frac{BPL}{MP} \equiv \frac{\overline{w'\theta' \frac{g}{\theta_0}}}{-\overline{u'w'} \frac{\partial \bar{u}}{\partial z} - \overline{v'w'} \frac{\partial \bar{v}}{\partial z}}$$

- **Se $Rf < 0$ a CLP é estatisticamente instável (a turbulências é sustentada pela convecção)**
- **$Rf > 0$ a CLP é estatisticamente estável**
- **$Rf < 0.25$ (a produção mecânica excede a produção por flutuabilidade por um fato de 4)**



5.2 Energia Cinética Turbulenta

5.3 Equações de momentum da camada limite Planetária

$$\frac{\partial(\bar{u}_i)}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} = - \frac{\partial(\overline{u_j' u_i'})}{\partial x_j} - \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - g \frac{\bar{\rho}}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2}$$

$$\begin{aligned} & \frac{\partial(\overline{u_i' u_k'})}{\partial t} + (\bar{u}_j) \frac{\partial(\overline{u_i' u_k'})}{\partial x_j} \\ &= - \overline{(u_j' u_k')} \frac{\partial(\bar{u}_i)}{\partial x_j} - g \frac{\overline{\rho' u_k'}}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j \overline{(u_k' u_k')} - \frac{\partial(\overline{u_j' u_i' u_k'})}{\partial x_j} - \frac{1}{\rho_0} \frac{\partial(\overline{P' u_k'})}{\partial x_i} + \nu \frac{\partial^2(\overline{u_i' u_k'})}{\partial x_j^2} \end{aligned}$$

$$\frac{\partial \bar{e}}{\partial t} + (\bar{u}_j) \frac{\partial \bar{e}}{\partial x_j} = - \frac{\overline{(u_j' u_i')}}{2} \frac{\partial(\bar{u}_i)}{\partial x_j} - g \frac{\overline{u_i' \rho'}}{2\rho_0} \delta_{i3} - \frac{\partial(\overline{e u_j'})}{\partial x_j} - \frac{1}{2\rho_0} \frac{\partial(\overline{P' u_i'})}{\partial x_i} - \epsilon$$



5.2 Energia Cinética Turbulenta

5.3 Equações de momentum da camada limite Planetária

$$\frac{\partial(\bar{u}_i)}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} = - \frac{\partial(u_j' u_i')}{\partial x_j} - \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - g \frac{\rho}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2}$$

Para o caso especial de turbulência horizontalmente homogênea:

-> A camada viscosa, a viscosidade molecular e o termo da divergência horizontal do fluxo de momentum turbulento podem ser desprezados.

$$\frac{D\bar{u}}{Dt} = - \frac{1}{\rho_0} \frac{\partial \bar{P}}{\partial x} + f\bar{v} - \frac{\partial \overline{u'w'}}{\partial z}$$

$$\frac{D\bar{v}}{Dt} = - \frac{1}{\rho_0} \frac{\partial \bar{P}}{\partial y} - f\bar{u} - \frac{\partial \overline{v'w'}}{\partial z}$$

Só pode ser resolvida se conhecermos a distribuição vertical do fluxo de momentum



- Parametrização de Radiação

[Paulo Kubota]



Implementação da parametrização de Radiação ECRAD do ECMWF



Slides Grupo Atmos



Slides Grupo Atmos



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1.5 Suítes físicas avançadas, etc.



**Obrigado Pela Atenção
de Todos**

